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(54) **DELIVERY OF CELLS AND TISSUES WITH SELF-ASSEMBLING PEPTIDE HYDROGEL MATERIALS**

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A61L 27/54 (2006.01)

C08L 89/00 (2006.01)

(52) **U.S. Cl.**

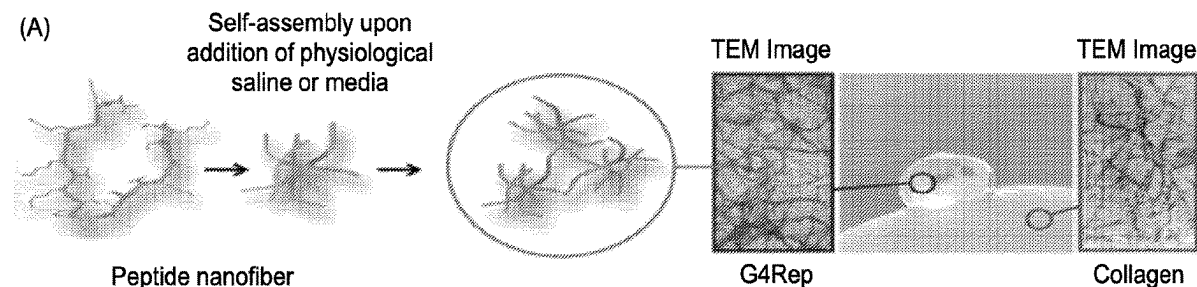
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(57)

ABSTRACT

Methods of administering biological material to a subject are provided. Methods of culturing biological material are also provided. The methods include combining biological material with a thermally stable preparation having a purified amphiphilic peptide in an aqueous biocompatible solution. The peptide has a folding group having a plurality of charged amino acid residues and hydrophobic amino acid residues arranged in a substantially alternative pattern and a turn sequence. The peptide is configured to self-assemble into a hydrogel. The methods include combining the preparation with a buffer having an effective amount of an ionic salt and a biological buffering agent to form the hydrogel. The methods include administering the hydrogel and the biological material to a subject. Methods of biofabricating a hydrogel having biological material are also provided. Methods of facilitating cell therapy are also provided.

Specification includes a Sequence Listing.



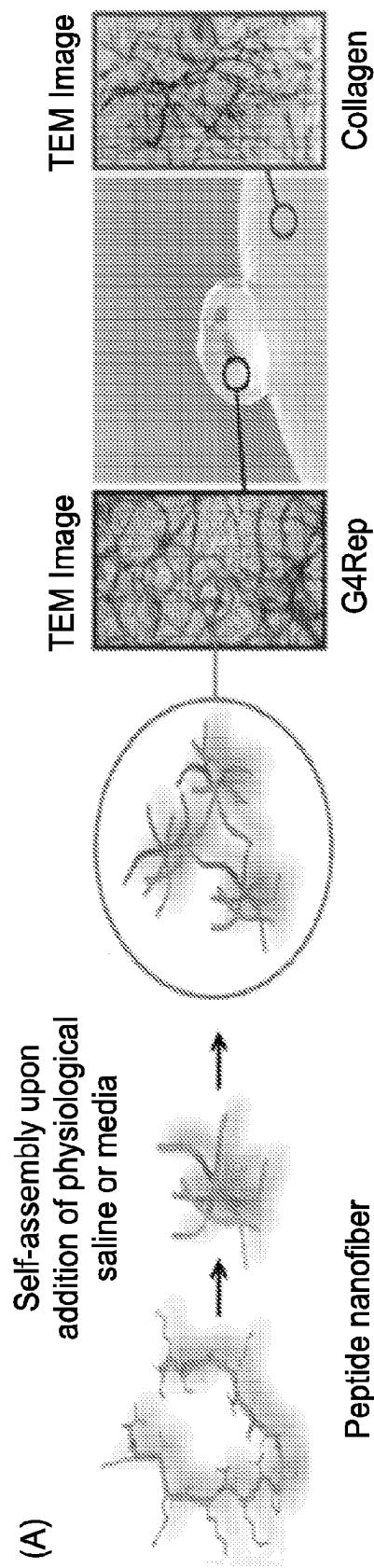


FIG. 1A

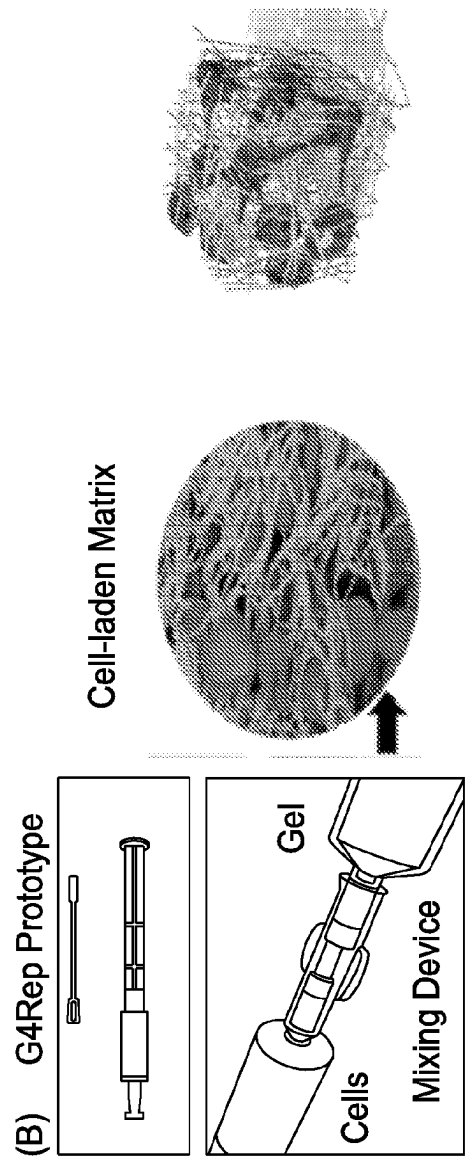


FIG. 1B

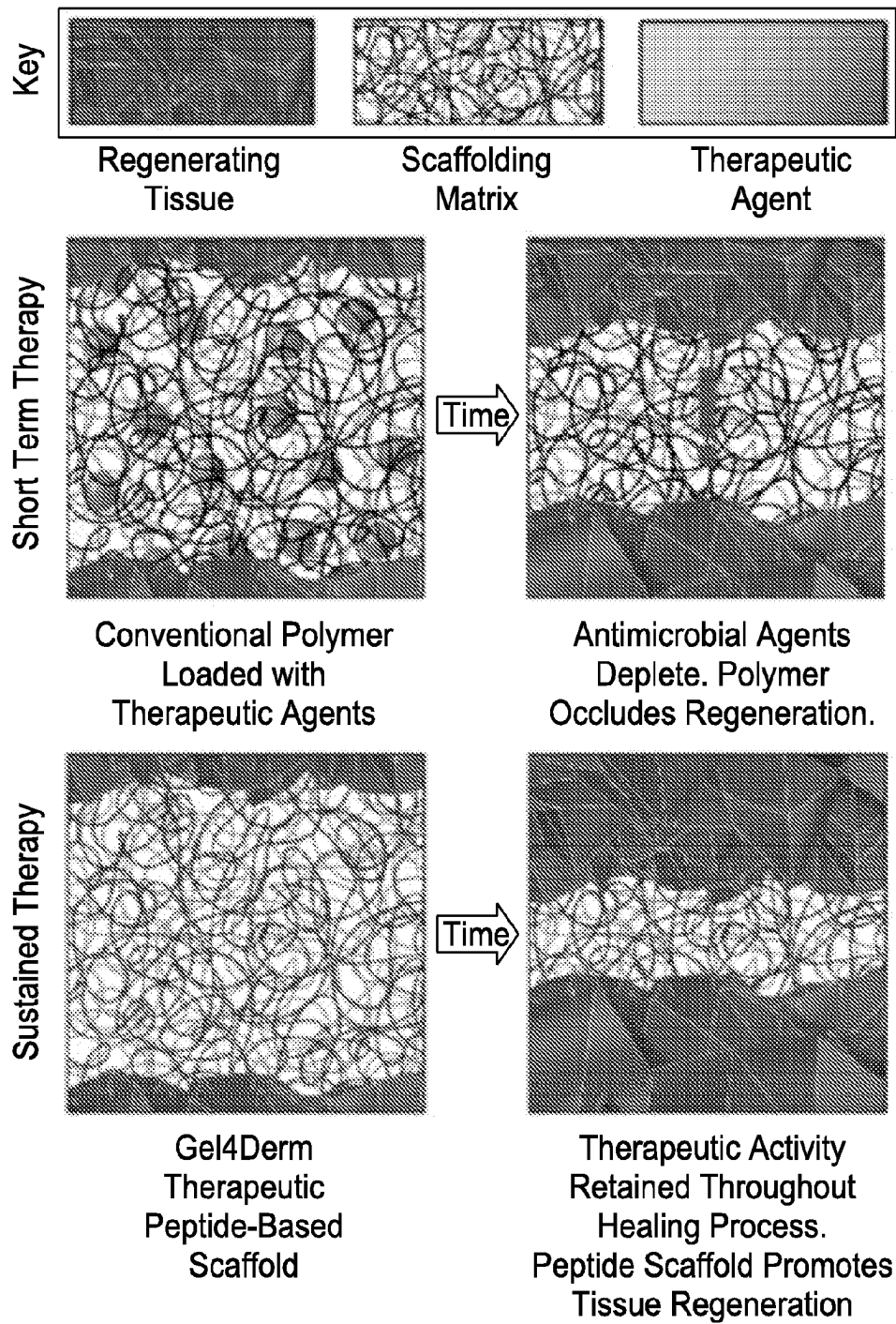


FIG. 2

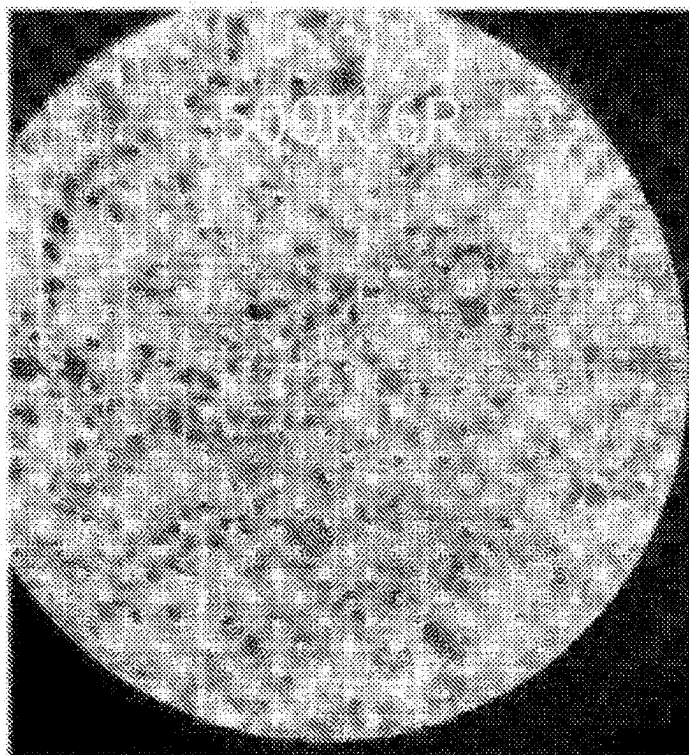


FIG. 3

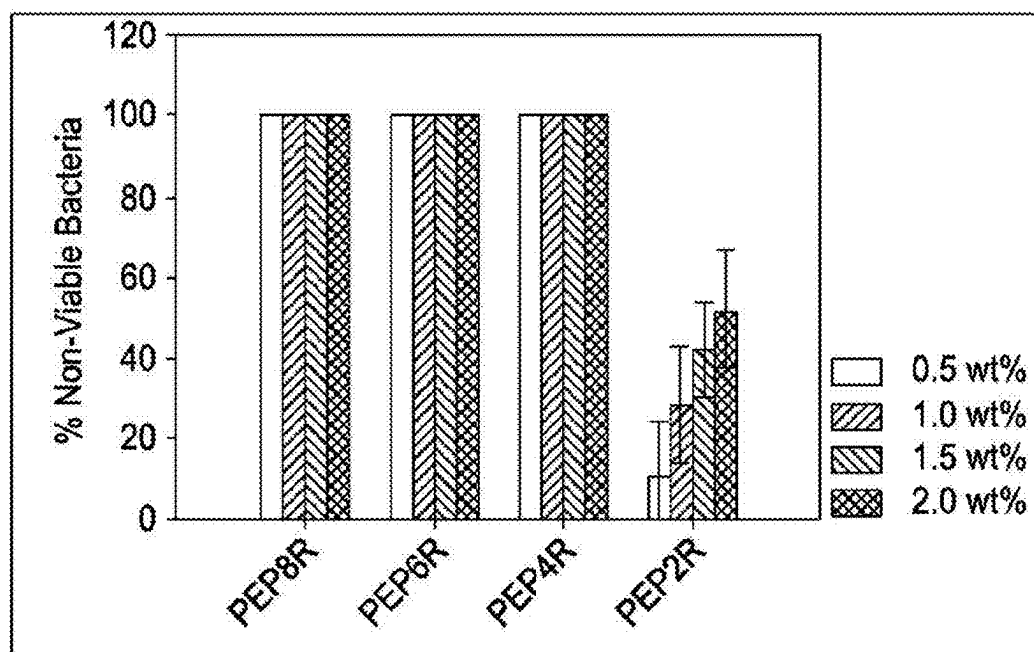
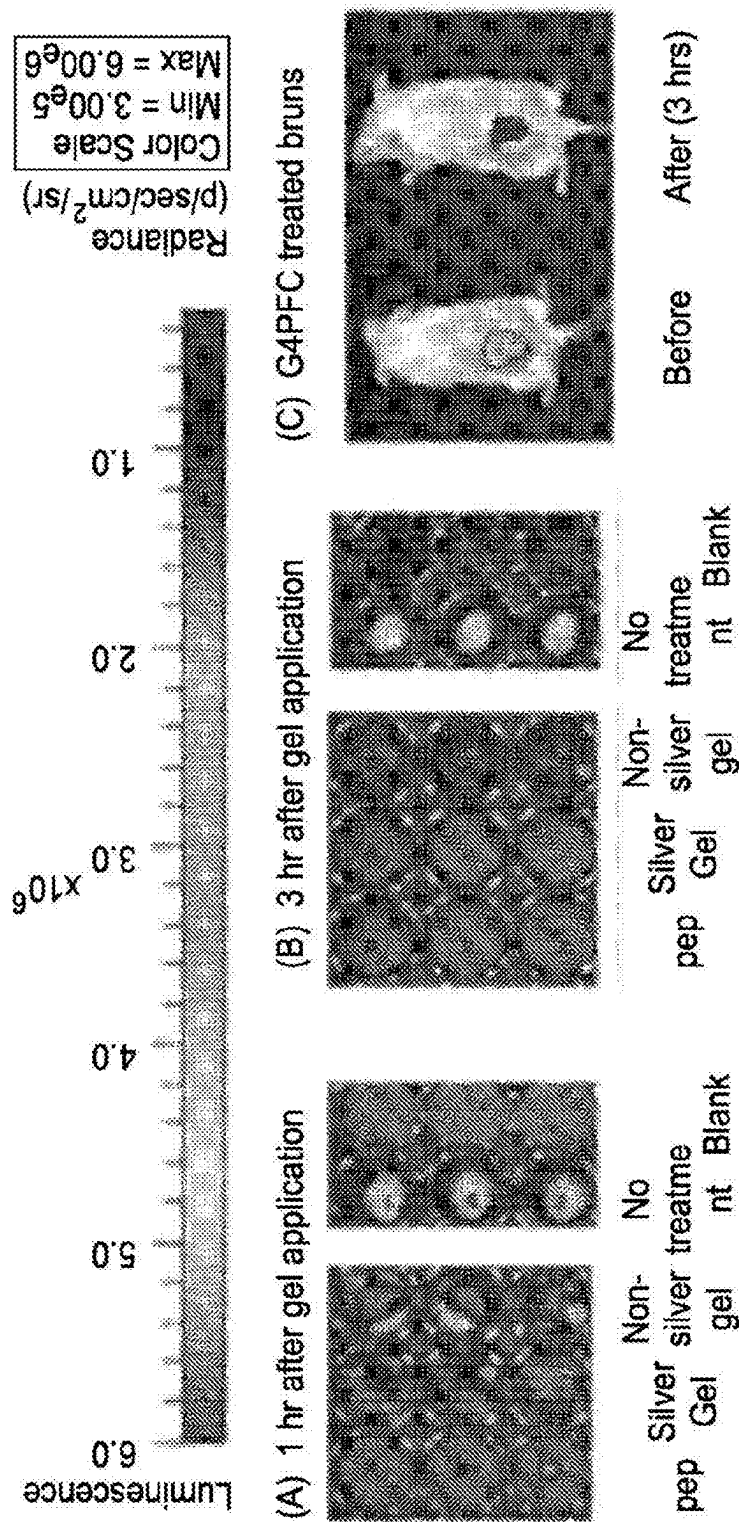


FIG. 4



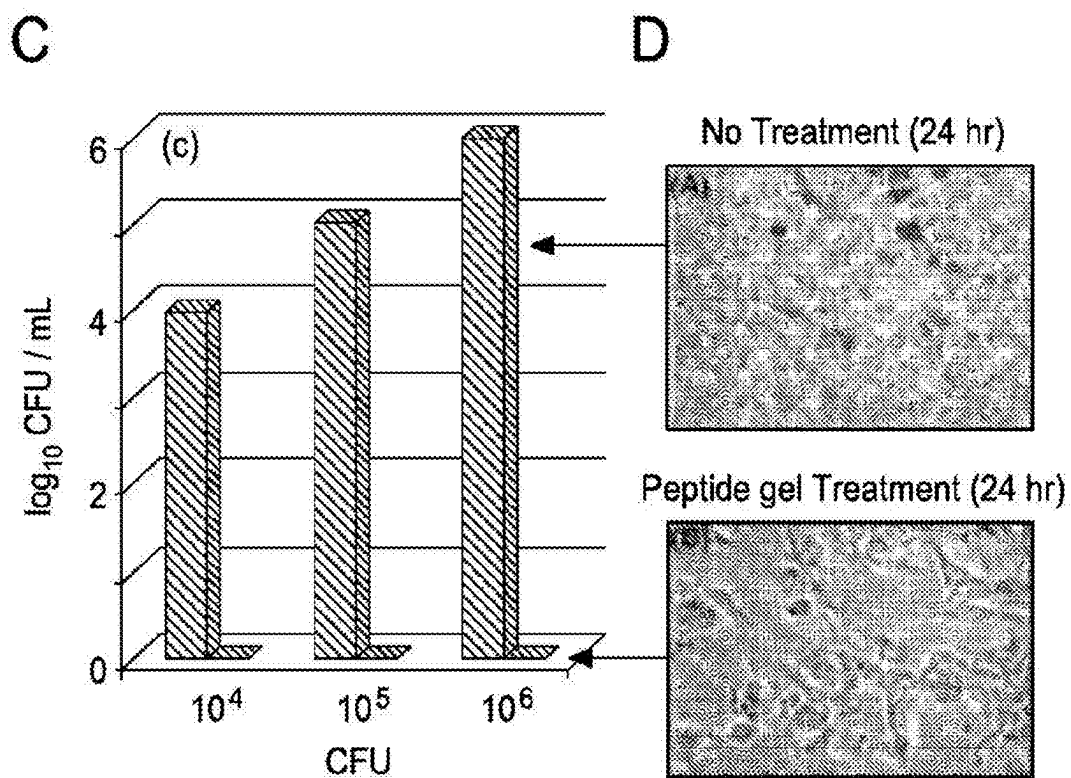
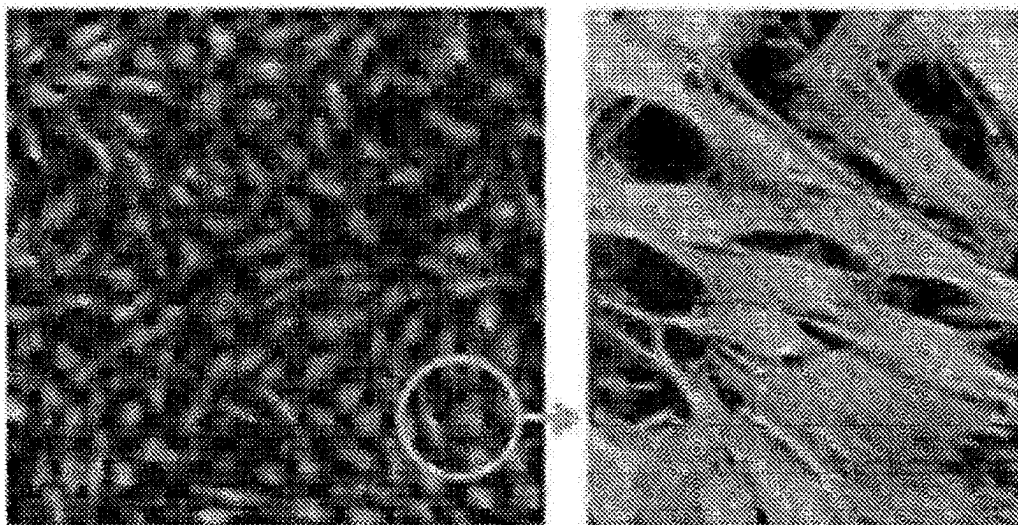


FIG. 6

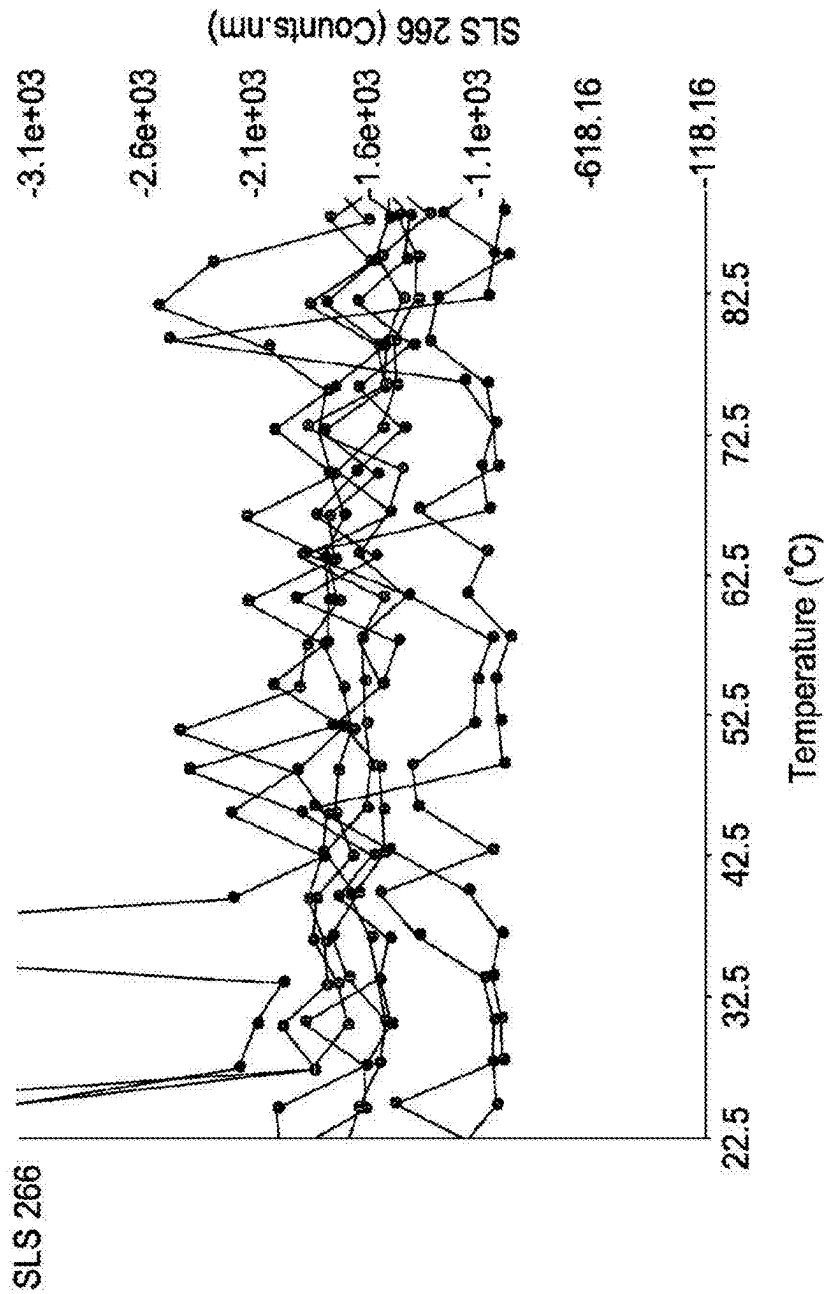


FIG. 7A

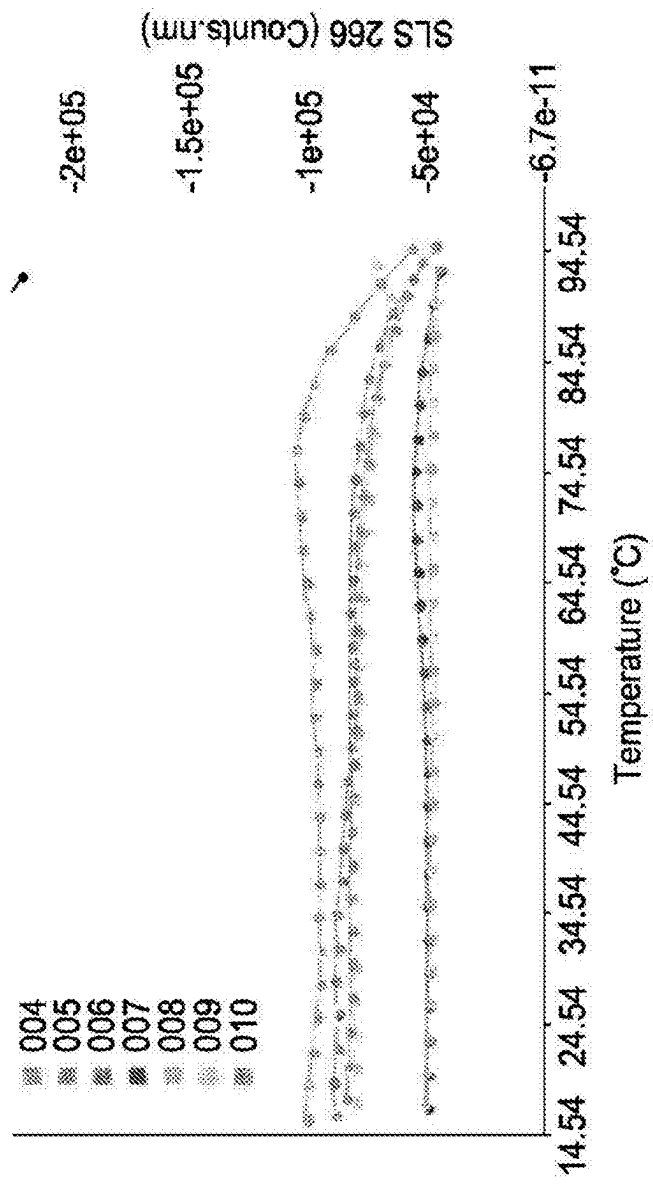


FIG. 7B

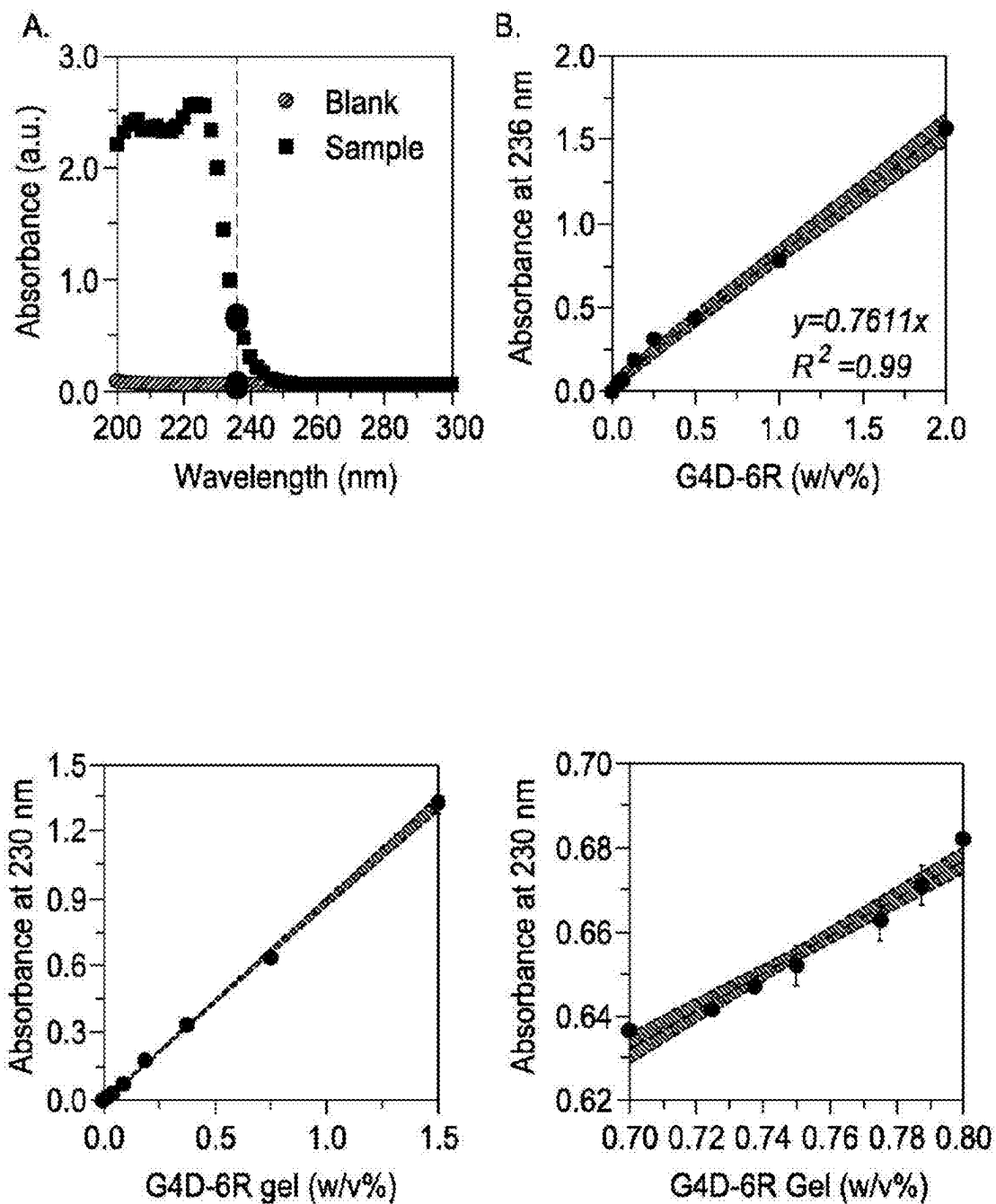


FIG. 8

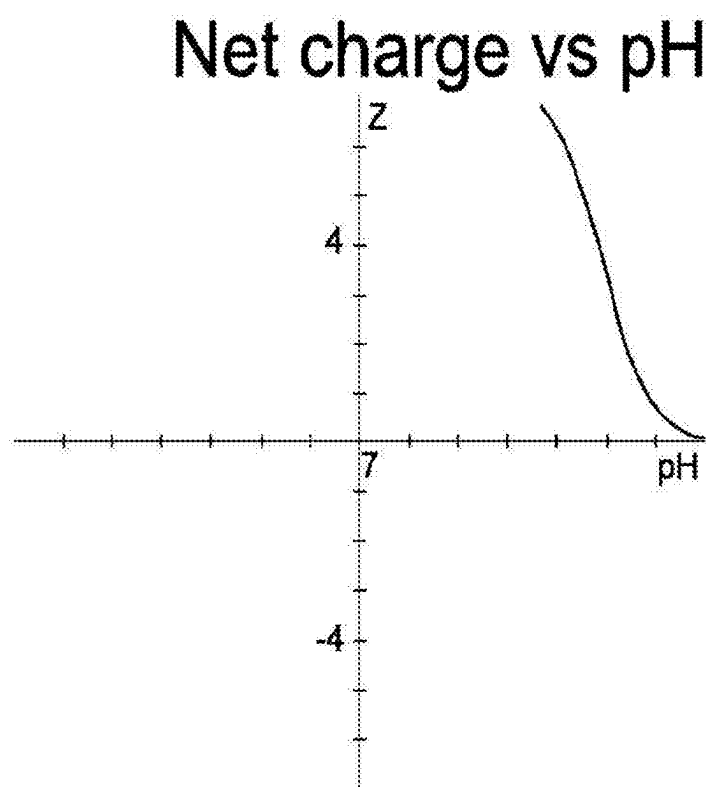


FIG. 9

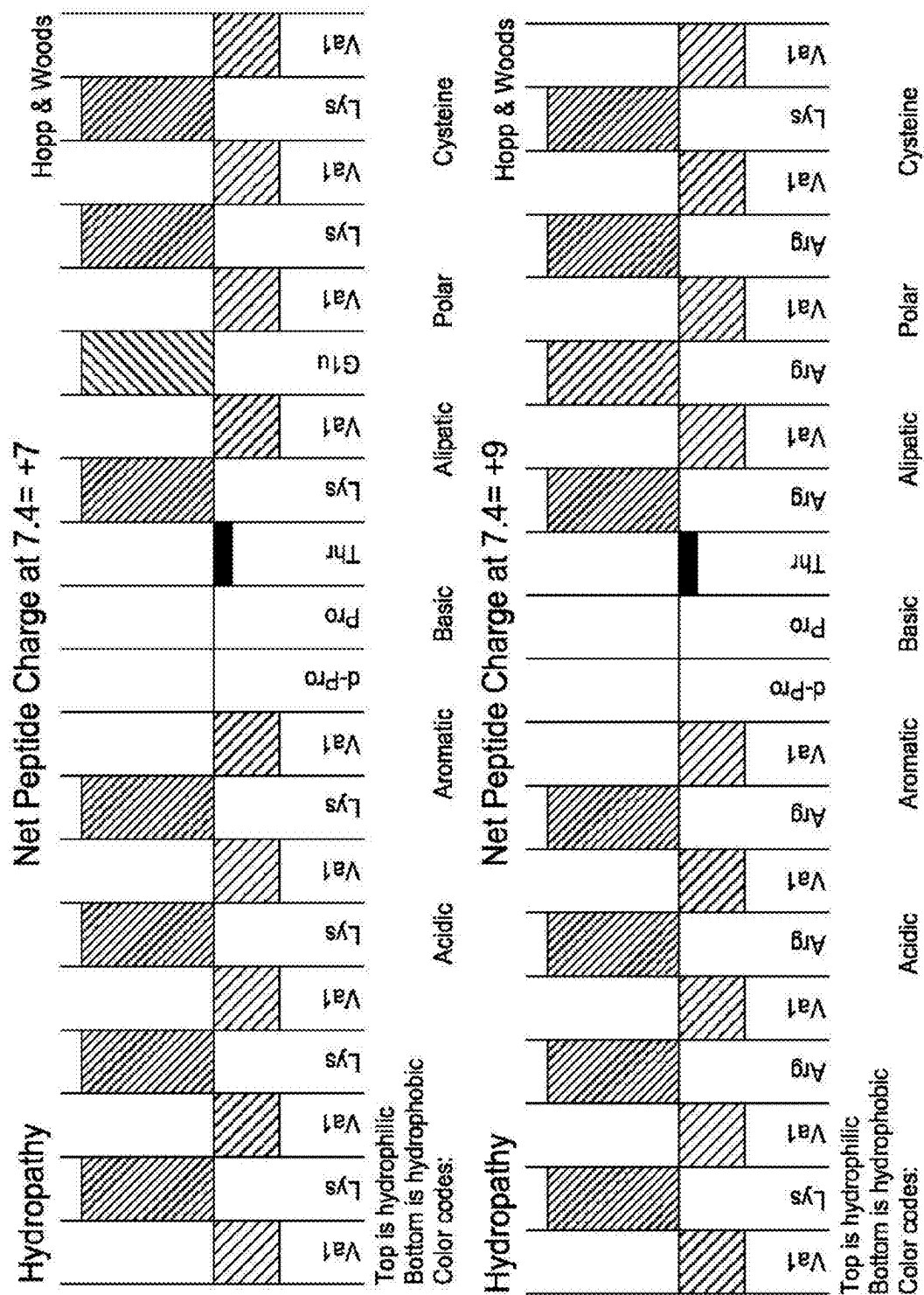
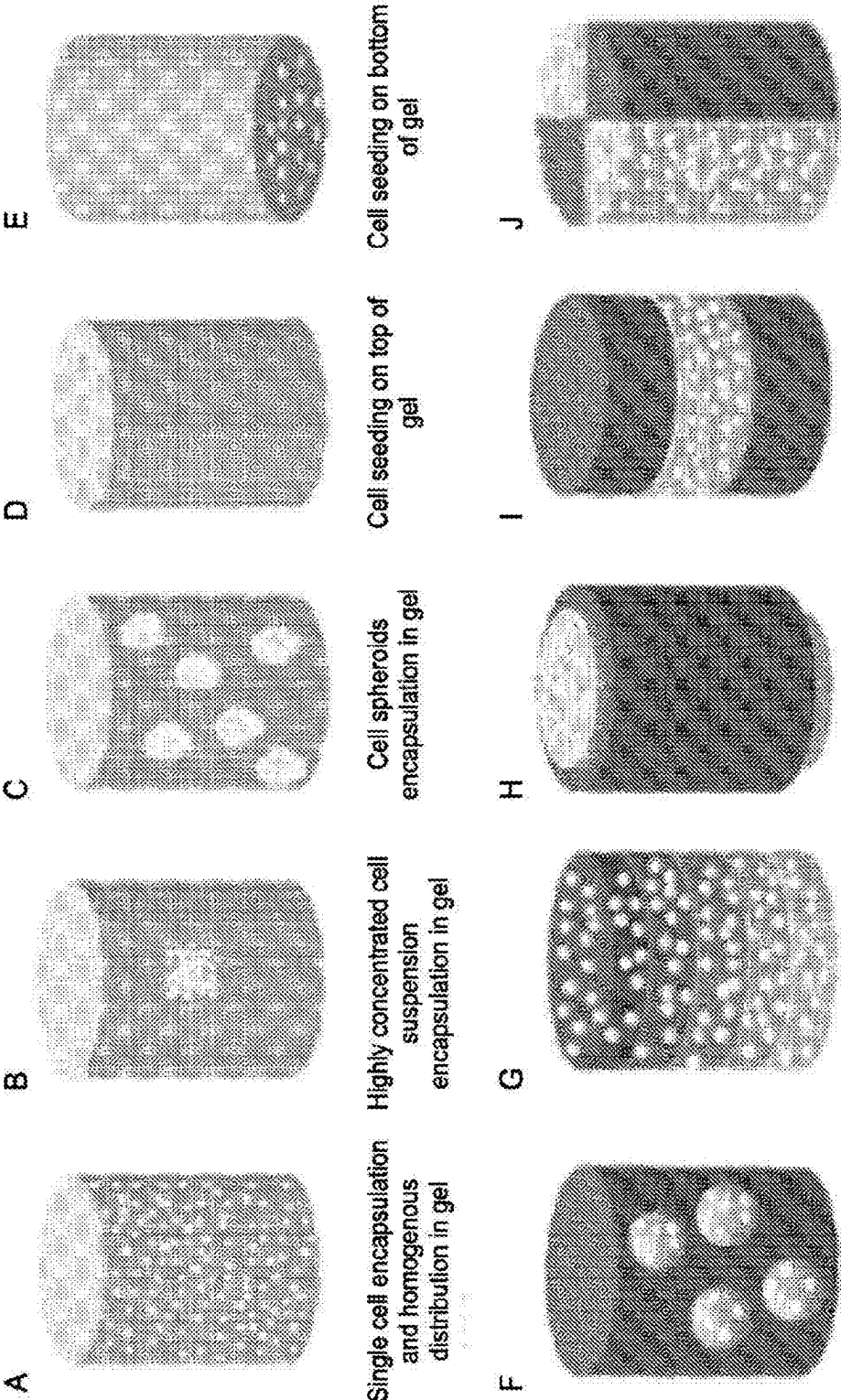


FIG. 10



Single or multiple cell-SAP gel layers

FIG. 11

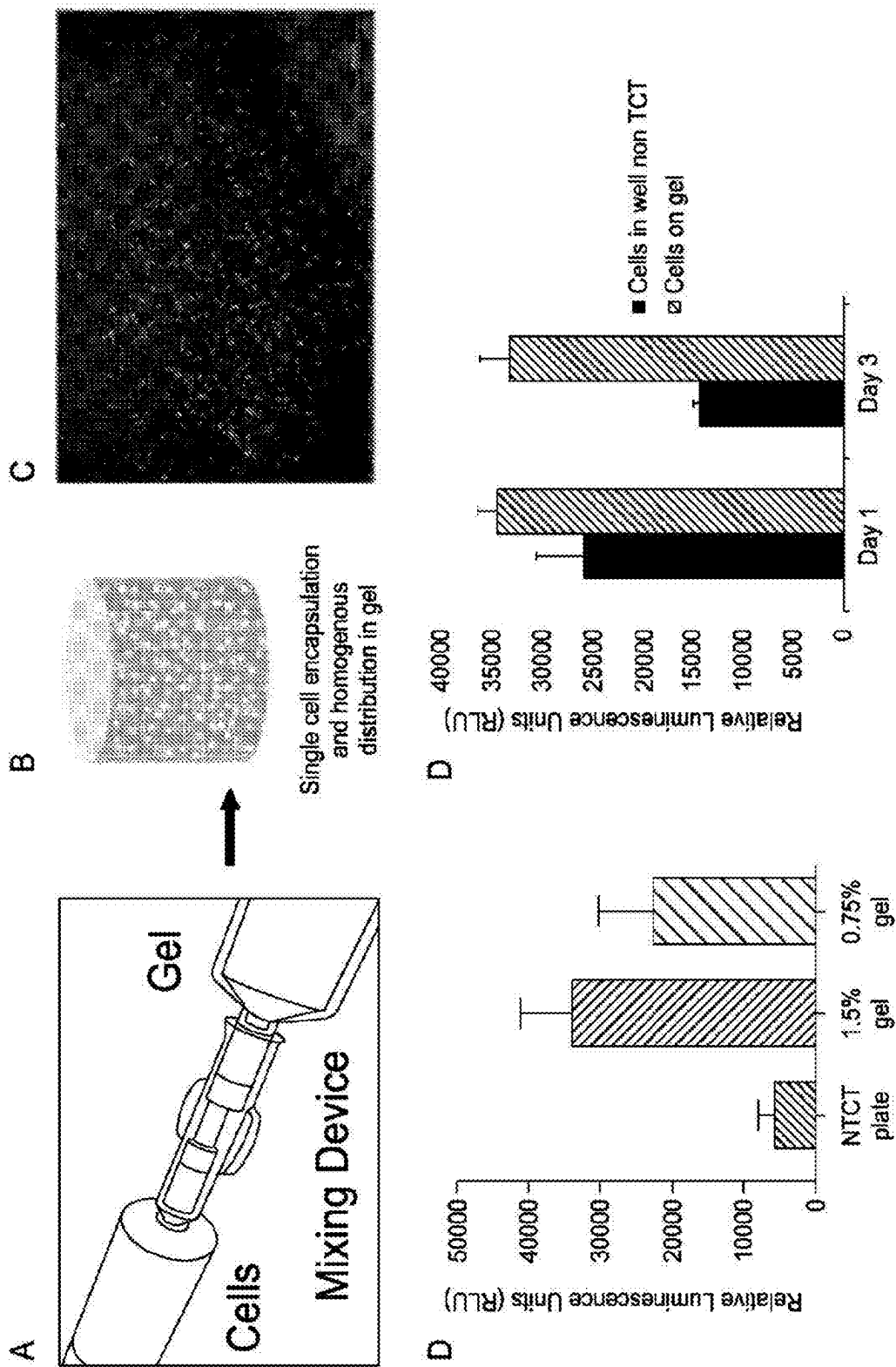


FIG. 12

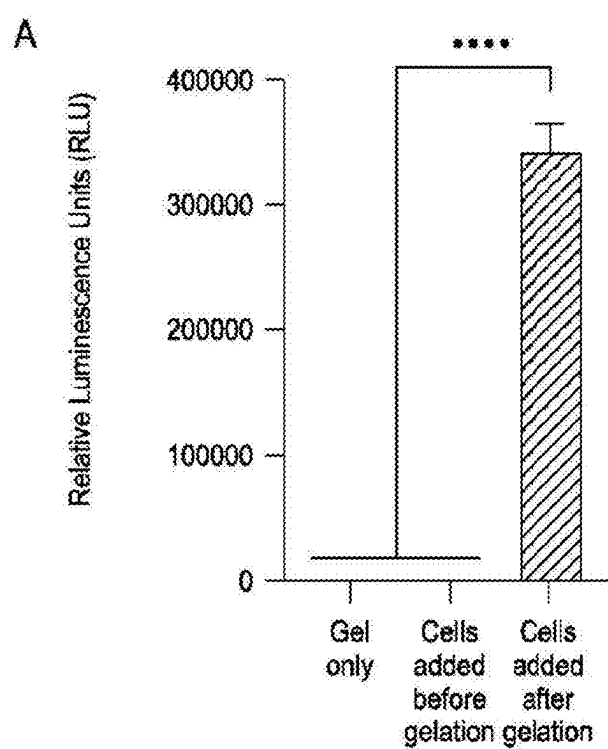


FIG. 13

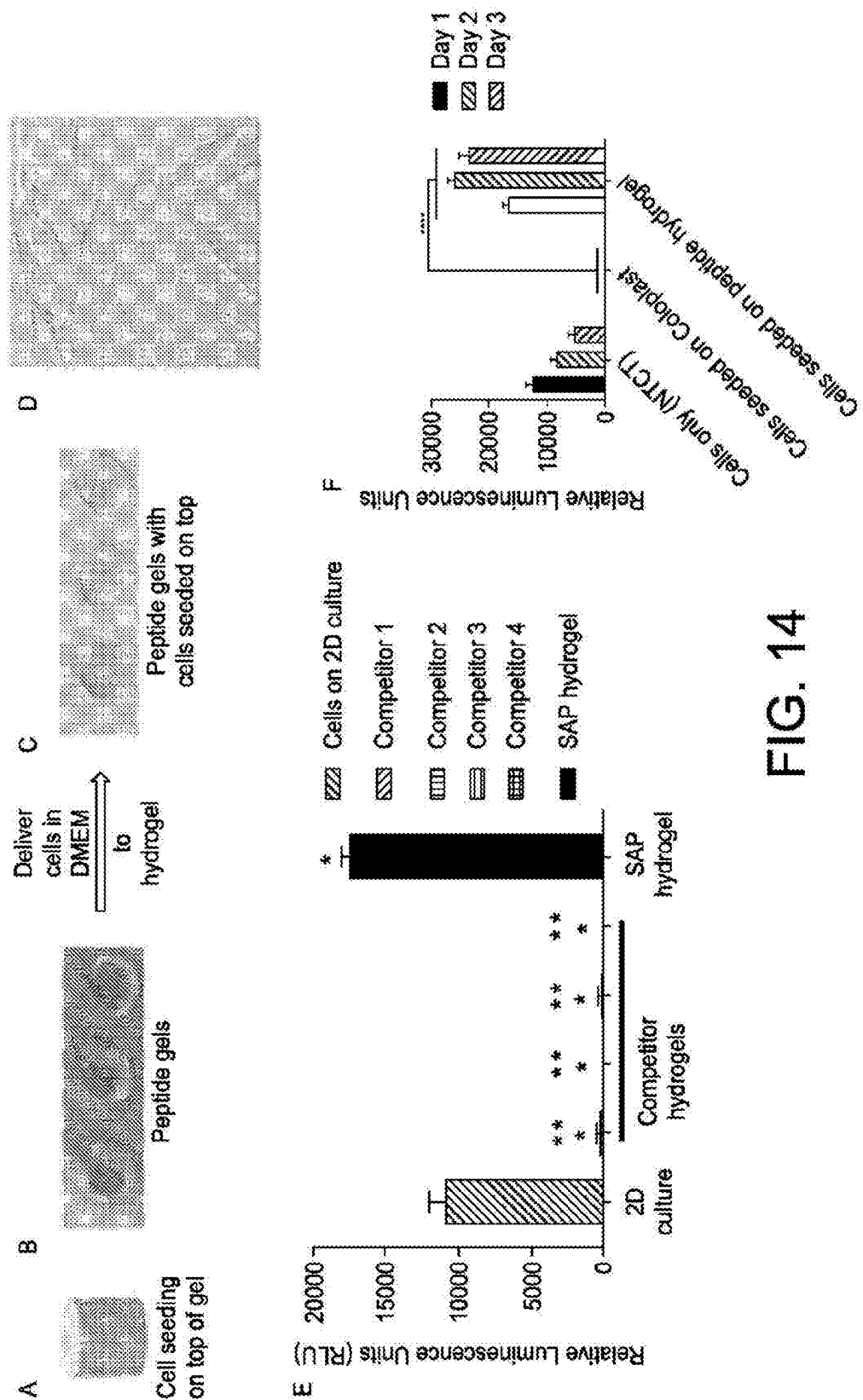


FIG. 14

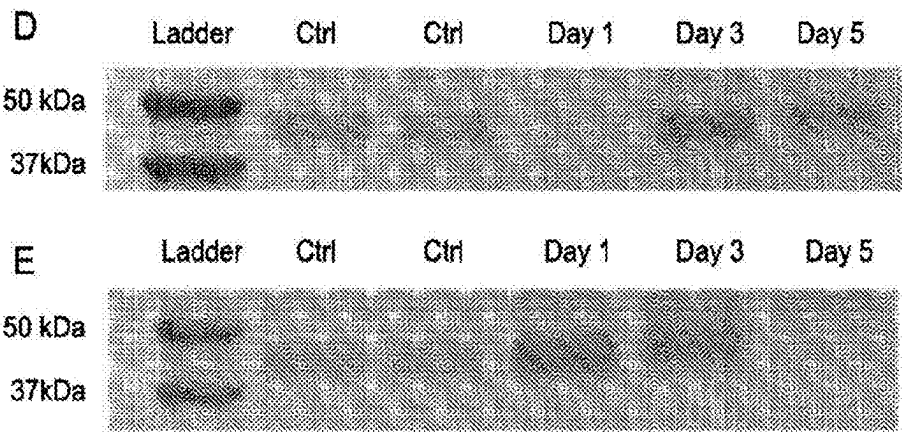


FIG. 15

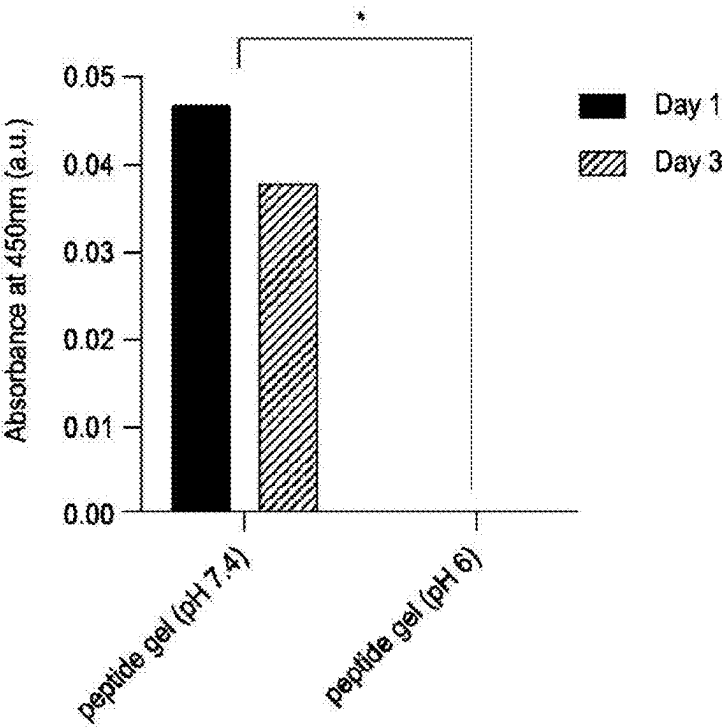


FIG. 16

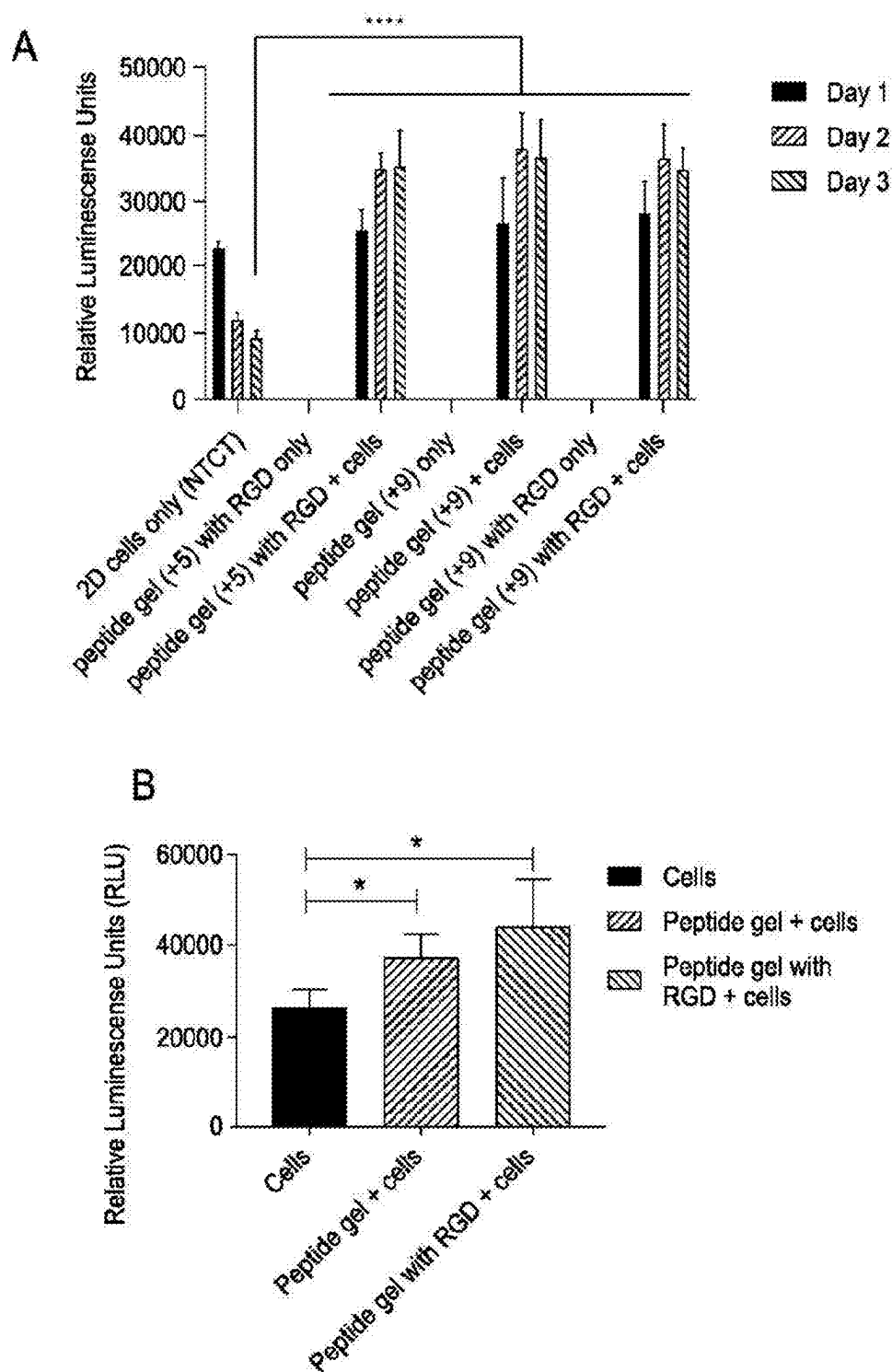


FIG. 17

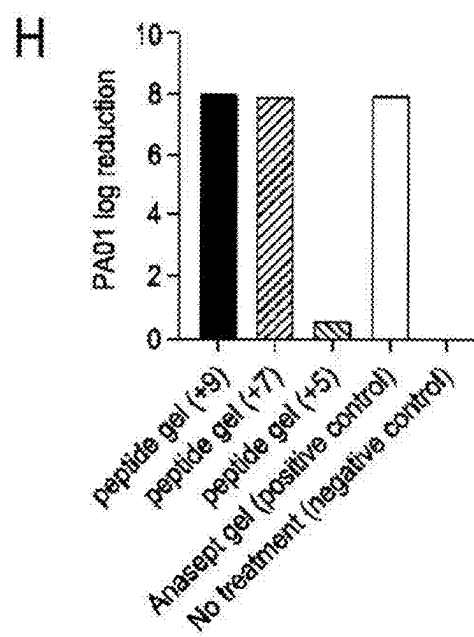
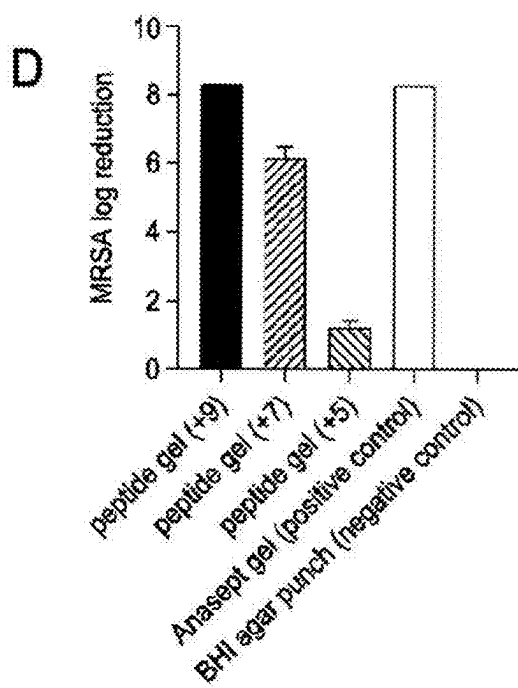
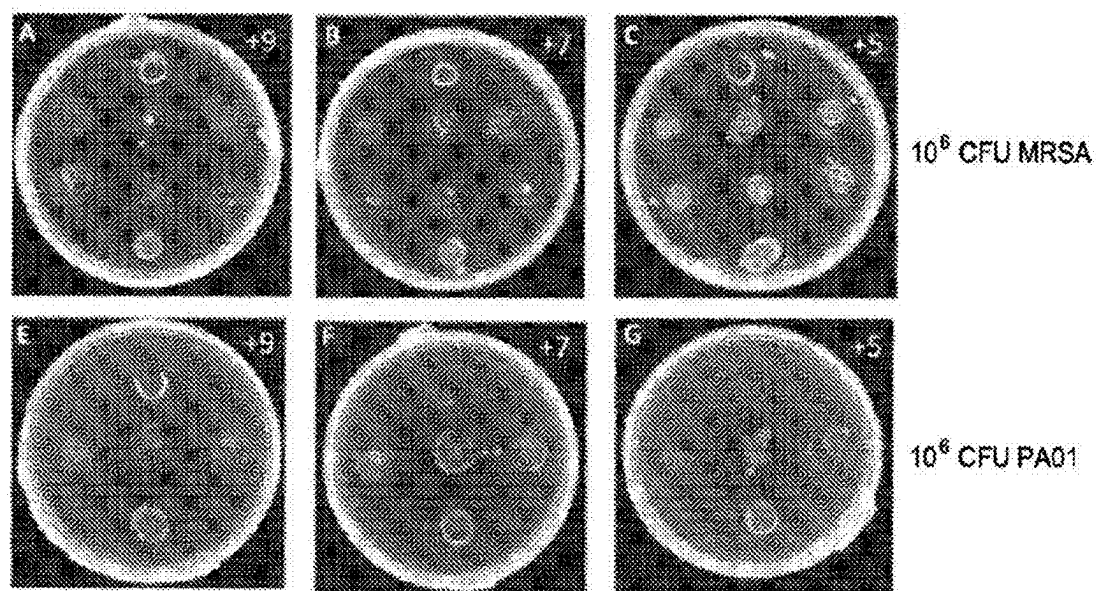


FIG. 18

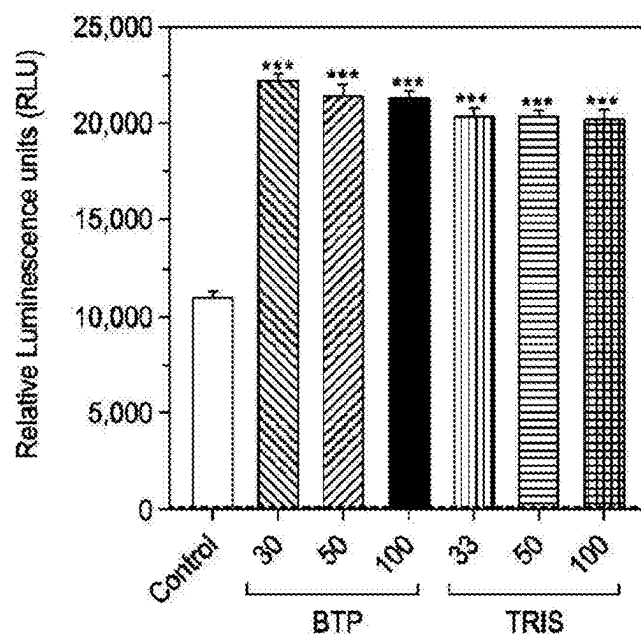


FIG. 19

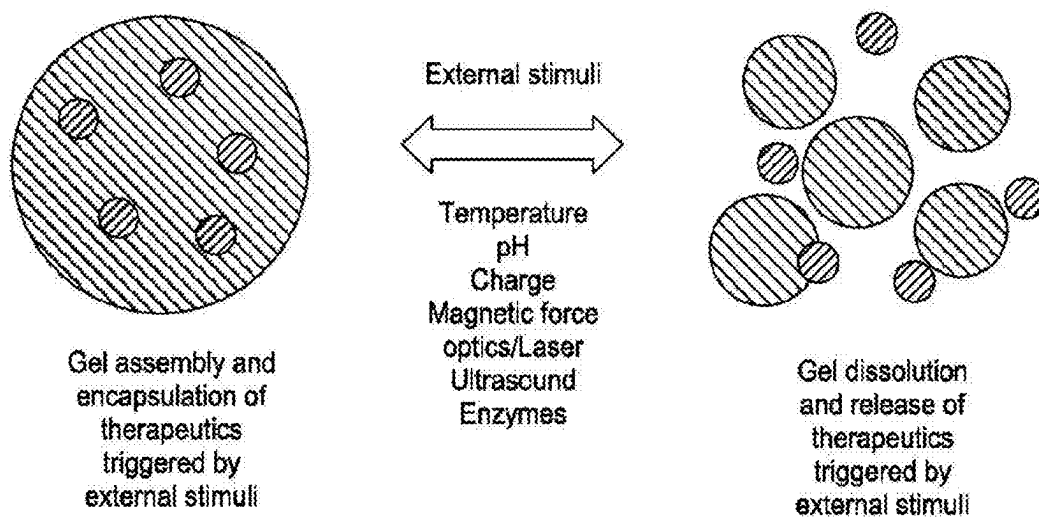
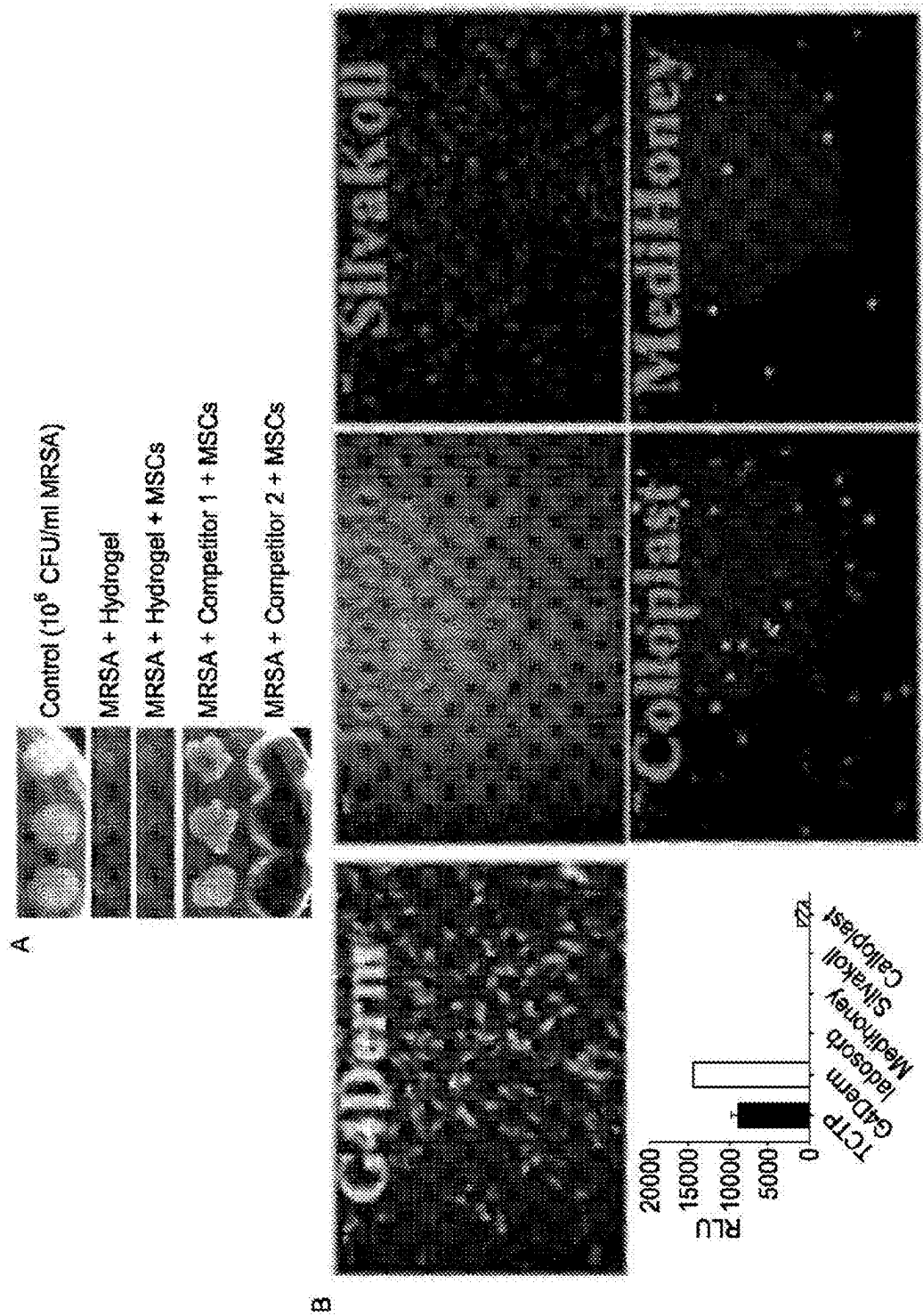


FIG. 20



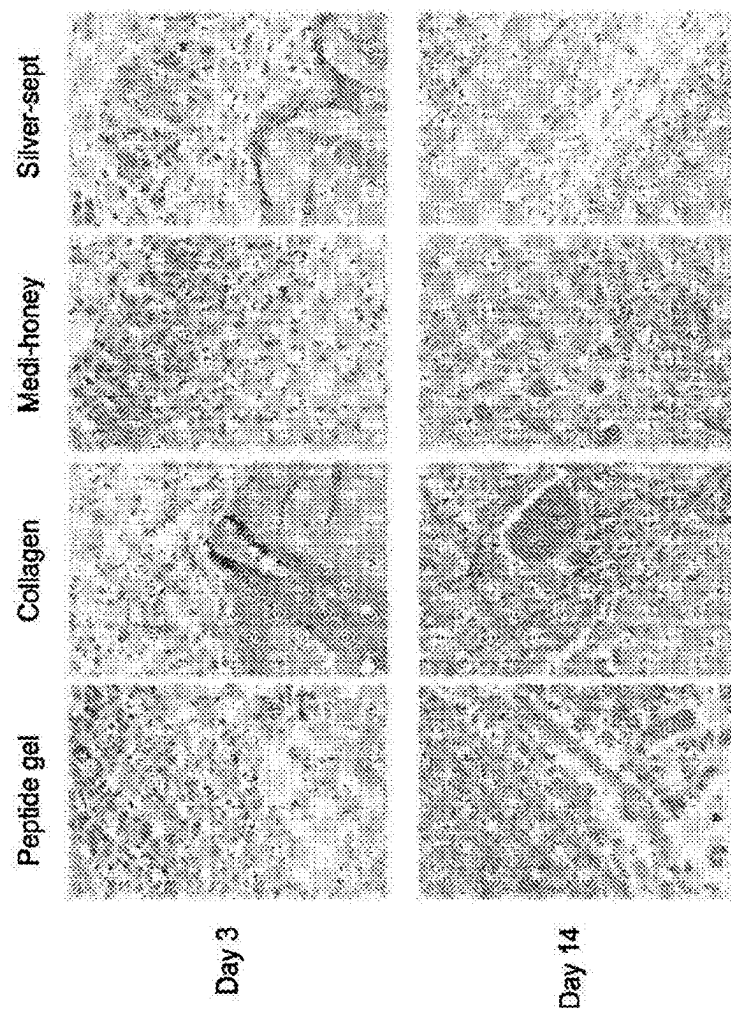


FIG. 22

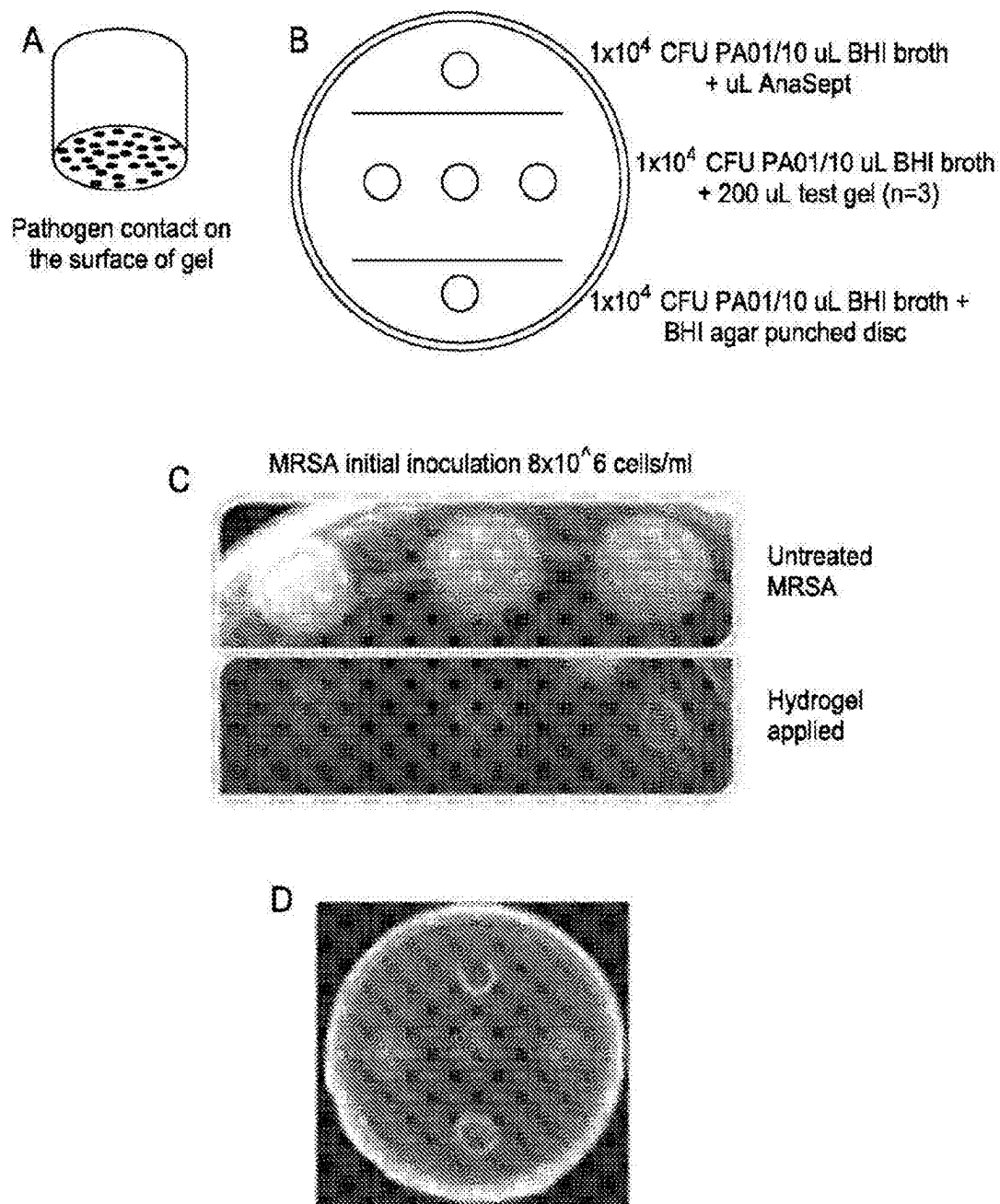


FIG. 23

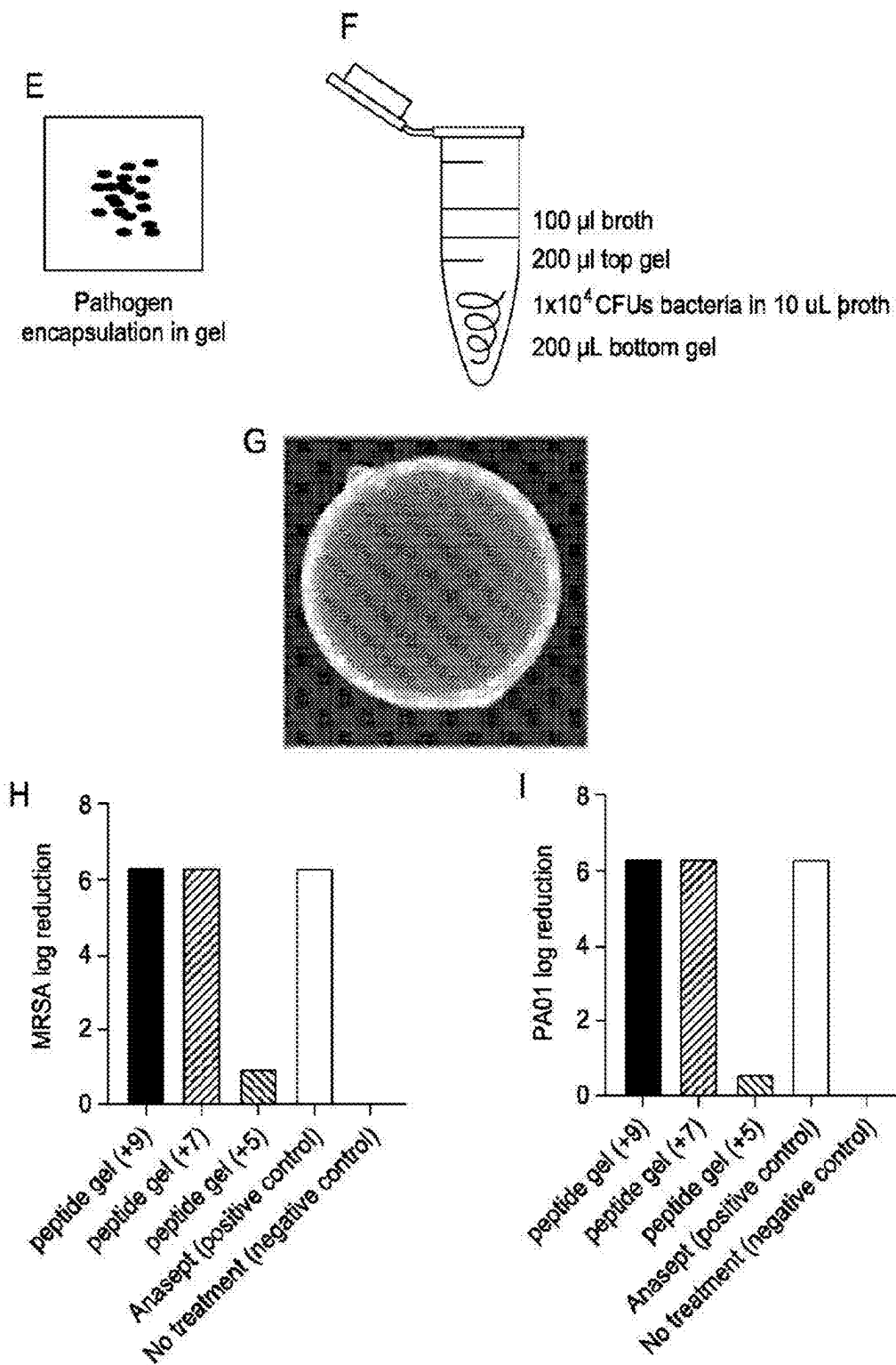


FIG. 23
(CONTINUED)

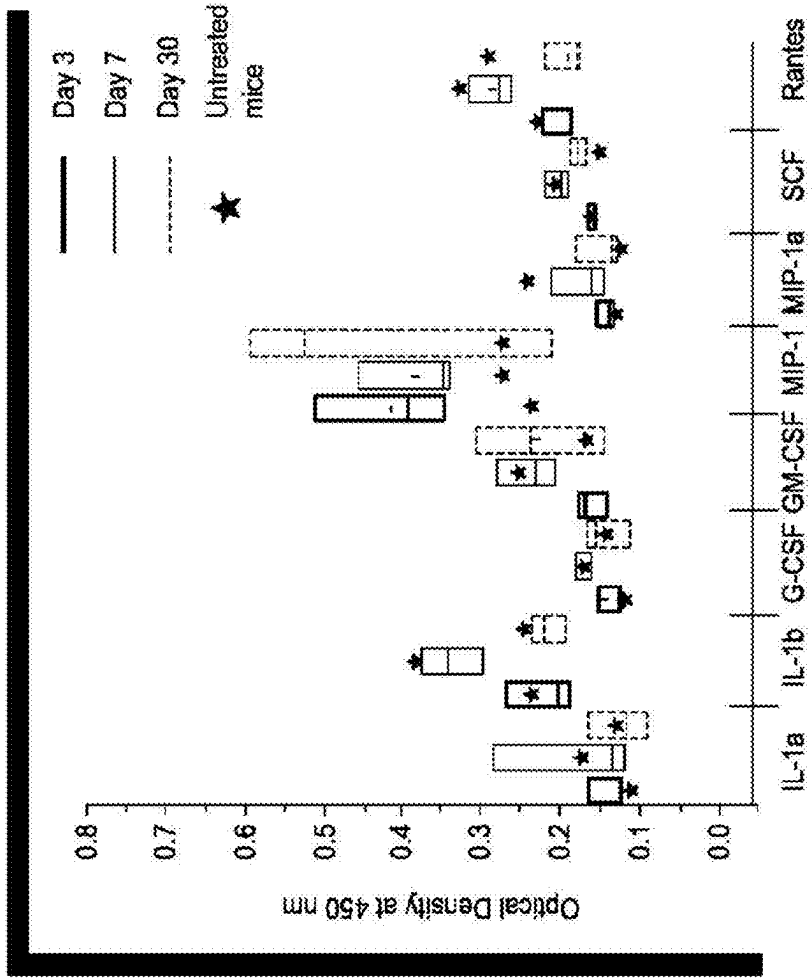


FIG. 24

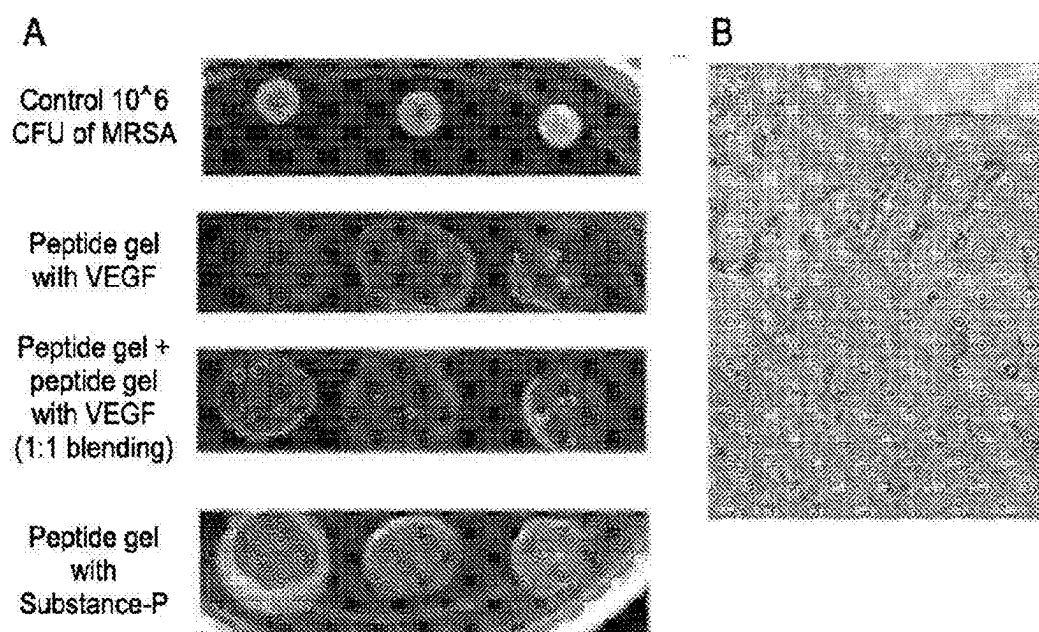


FIG. 25

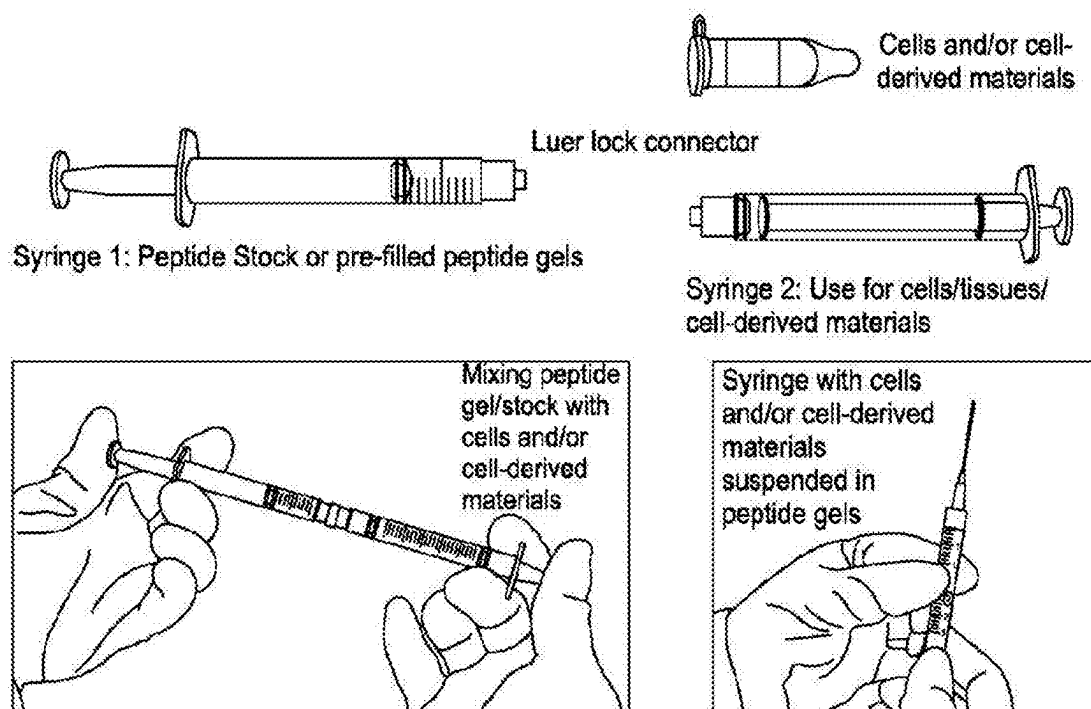


FIG. 26

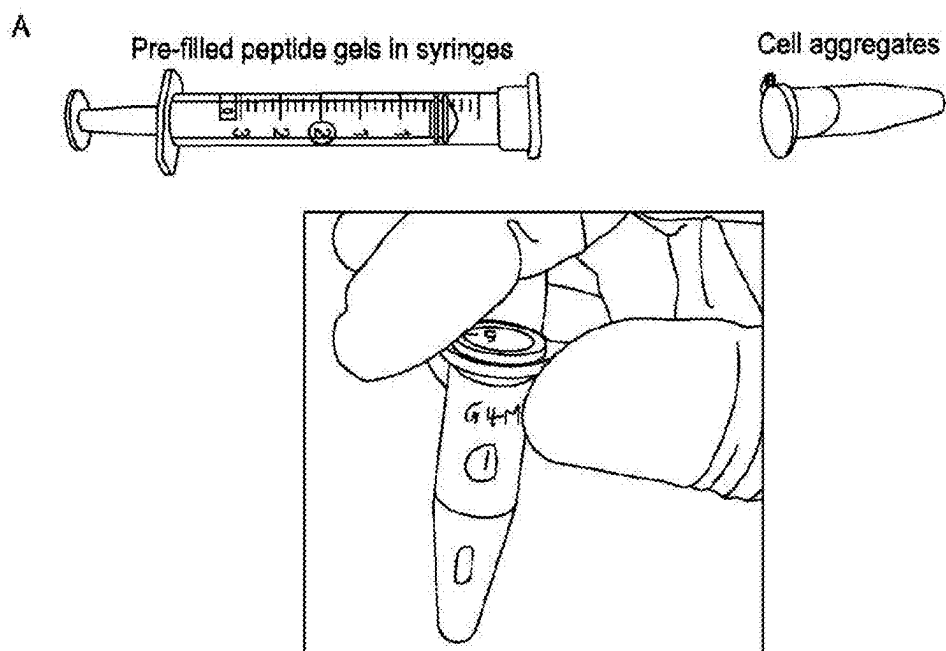


FIG. 27

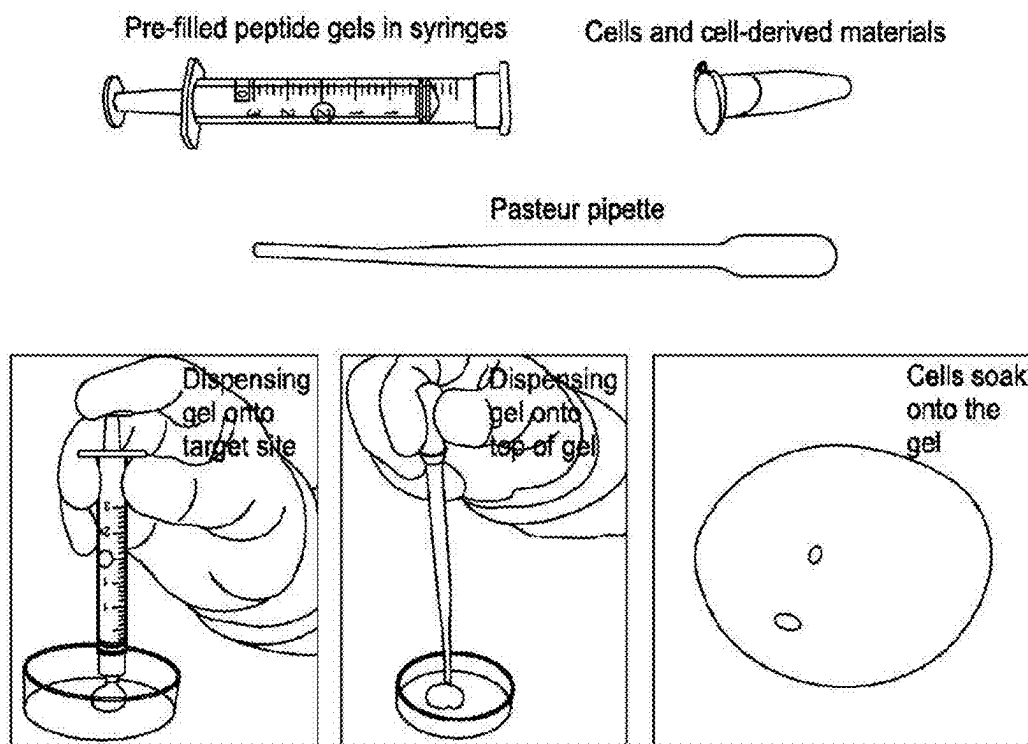


FIG. 28

Standard curve for quantitation of endotoxin-level (EU/mL), from the chromogenic assay

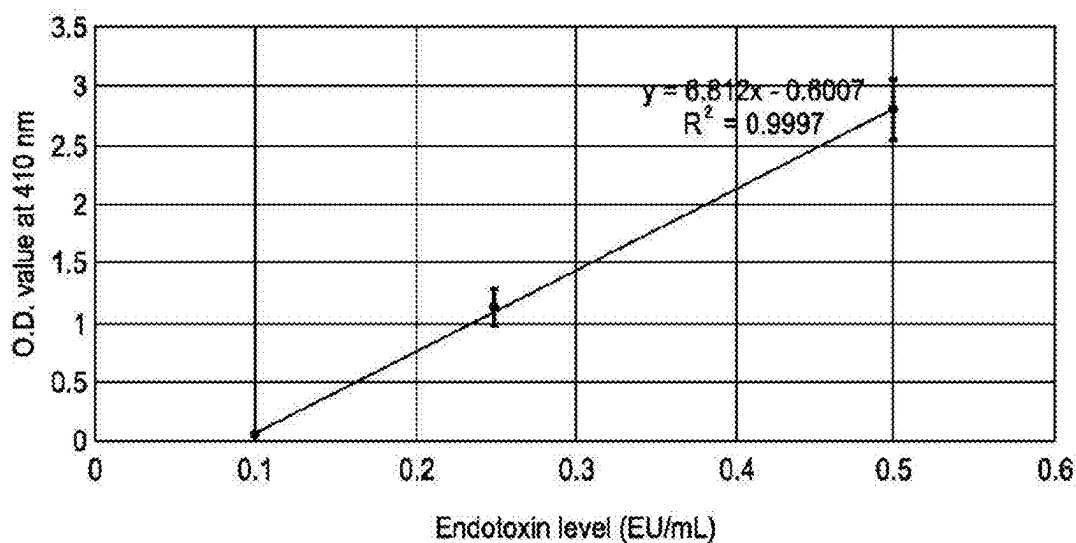


FIG. 29

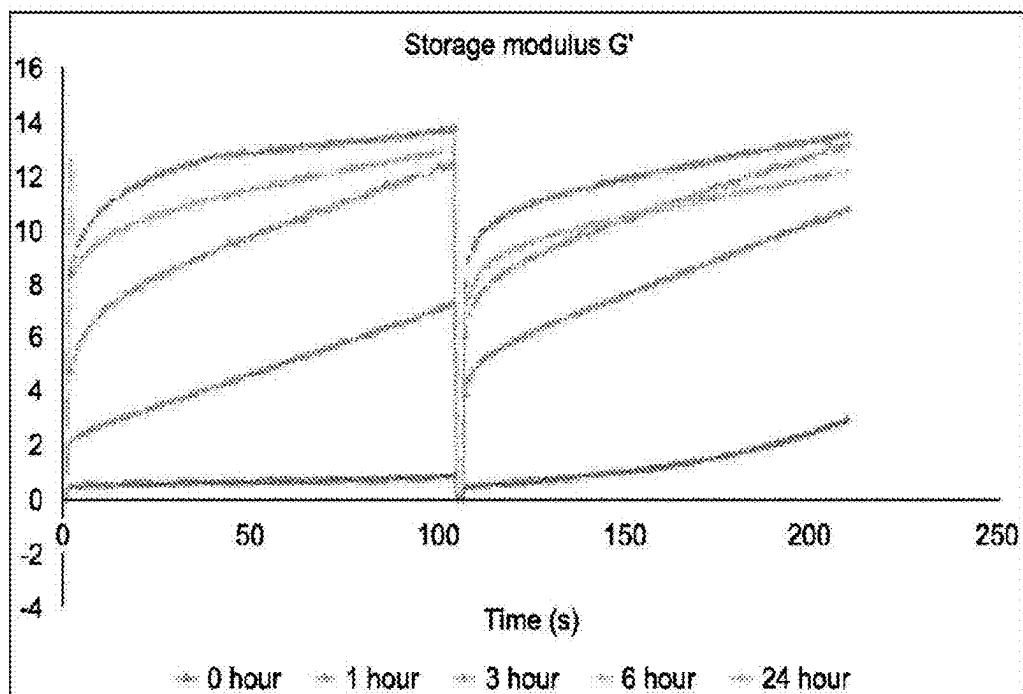


FIG. 30A

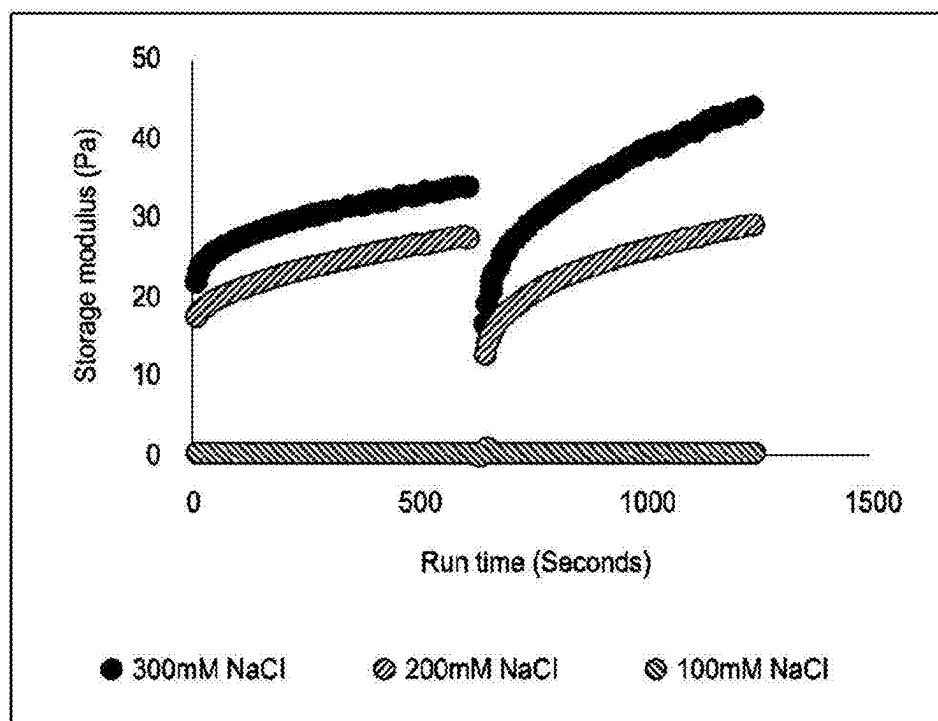


FIG. 30B

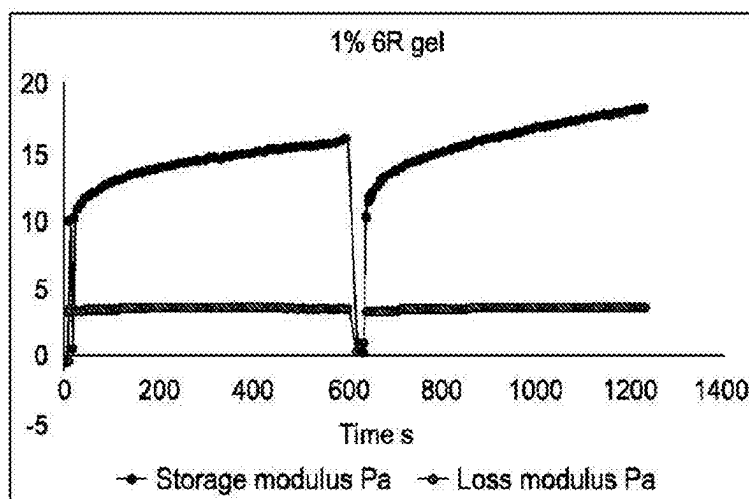
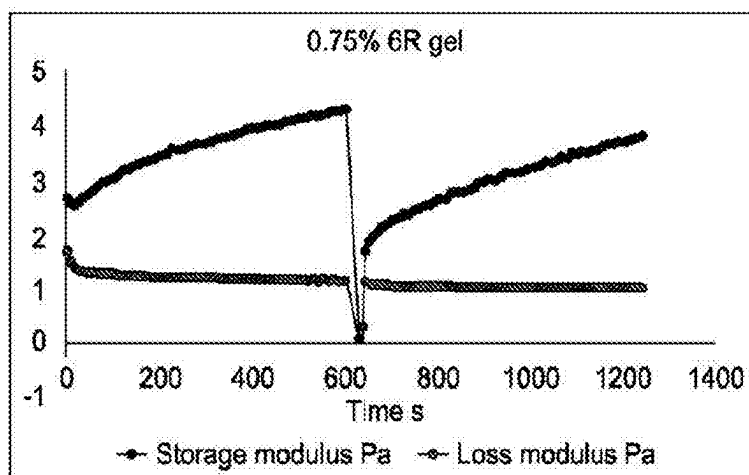
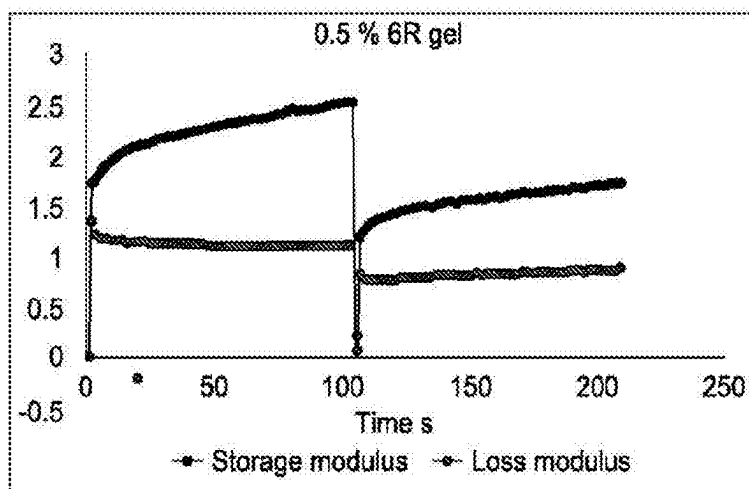


FIG. 30C

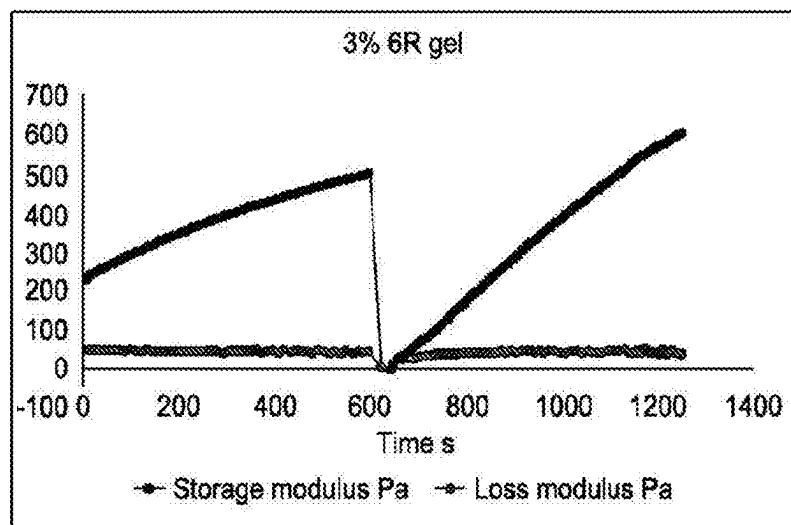
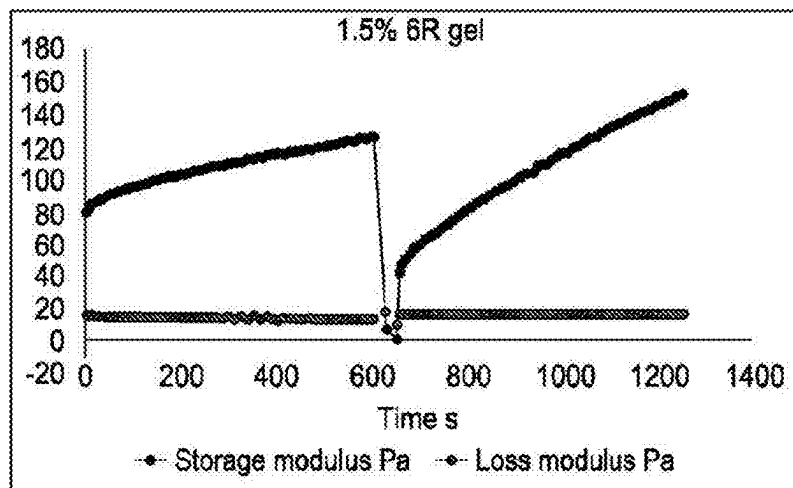


FIG. 30C
(CONTINUED)

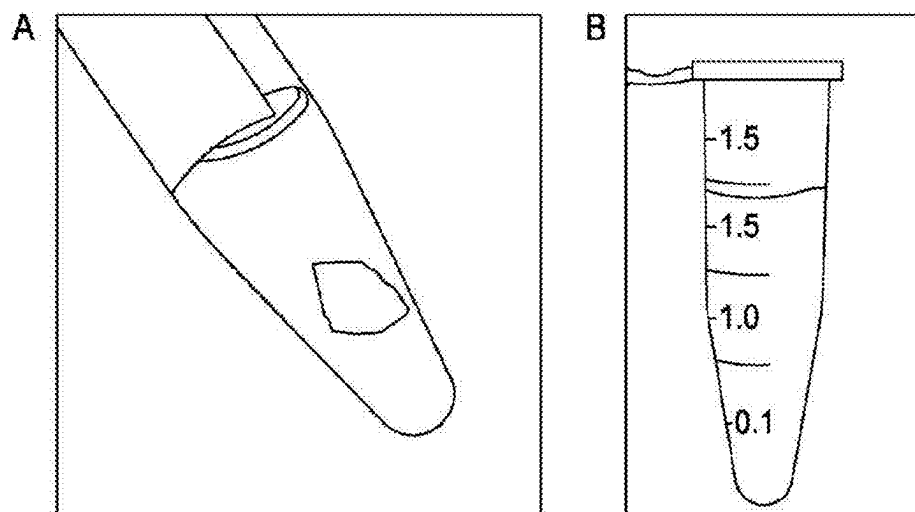


FIG. 31

All steps of biofabrication and delivery of autologous derived therapy done at point of care

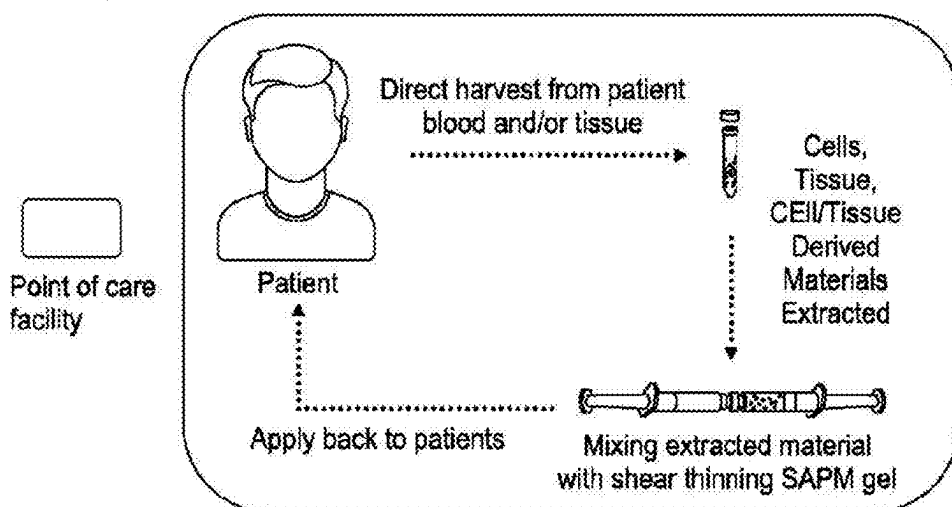


FIG. 32A

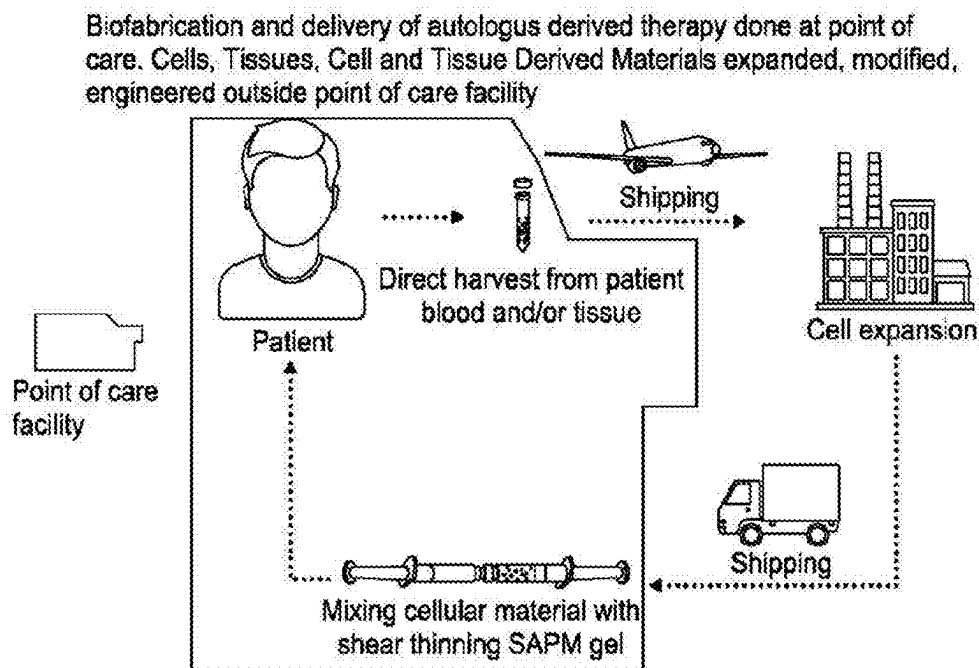


FIG. 32B

Biofabrication and delivery of allogeneic/xenogeneic derived therapy done at point of care. Cells, Tissues, Cell and Tissue Derived Materials expanded, modified, engineered outside point of care facility

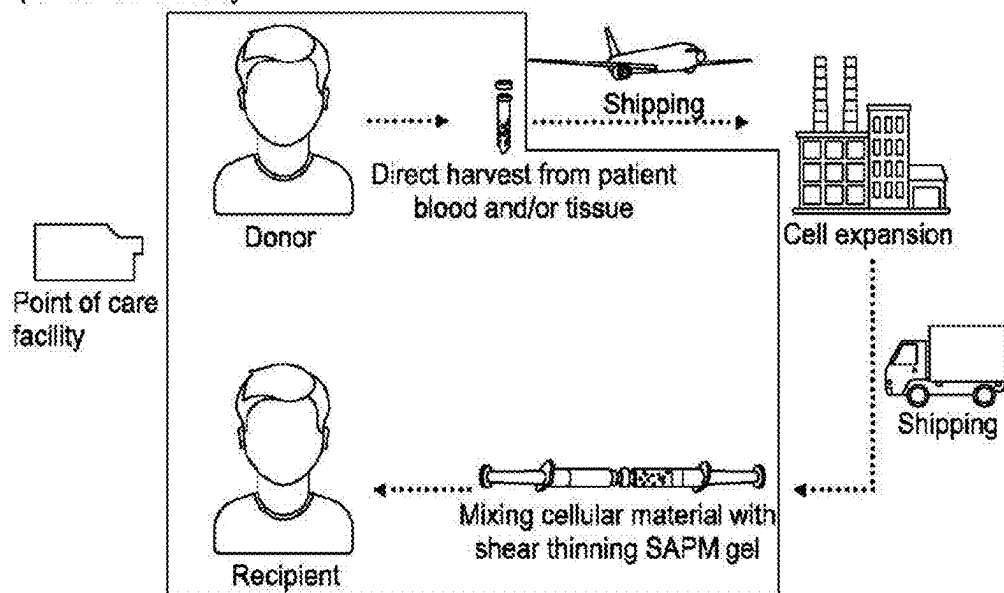
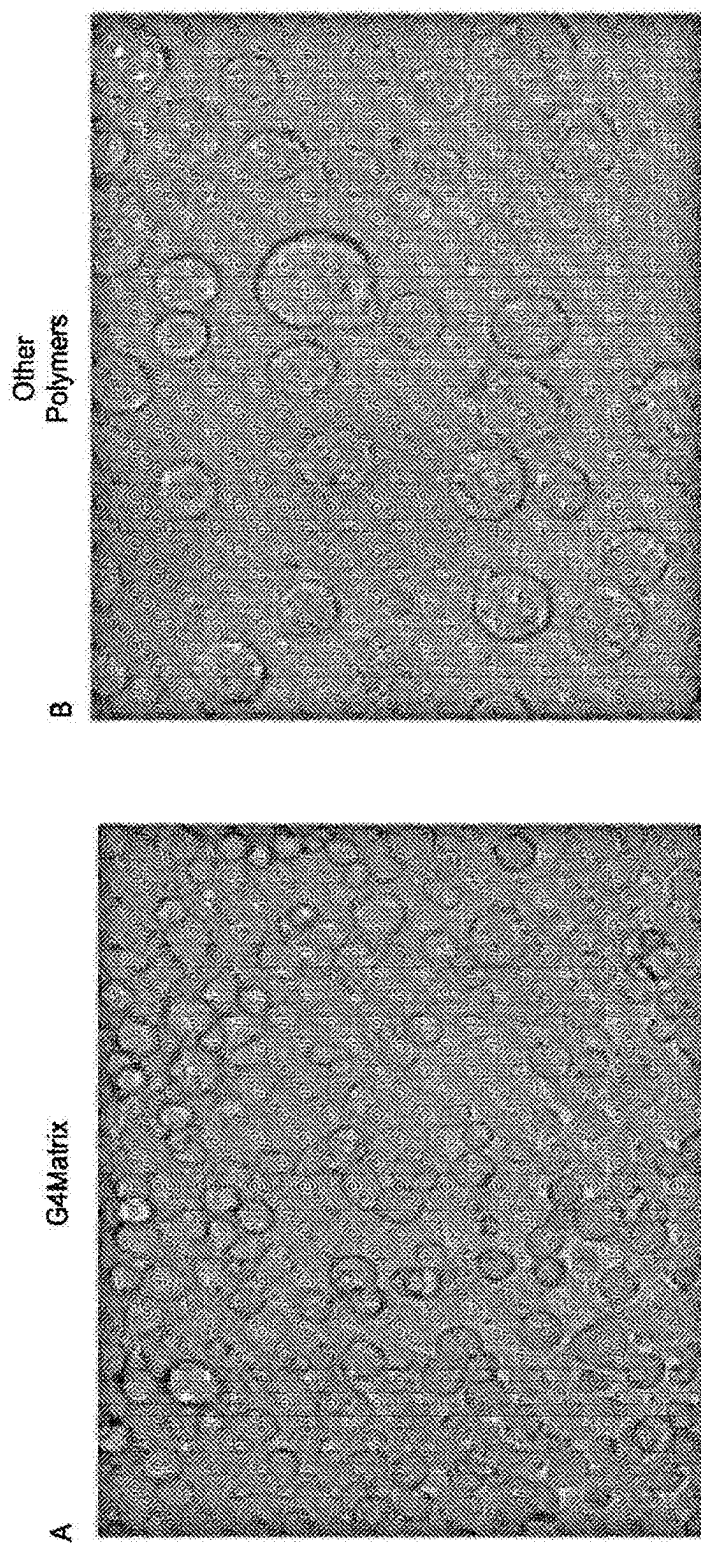


FIG. 32C



Time Lapse imaging of cells over 6 hrs

FIG. 33

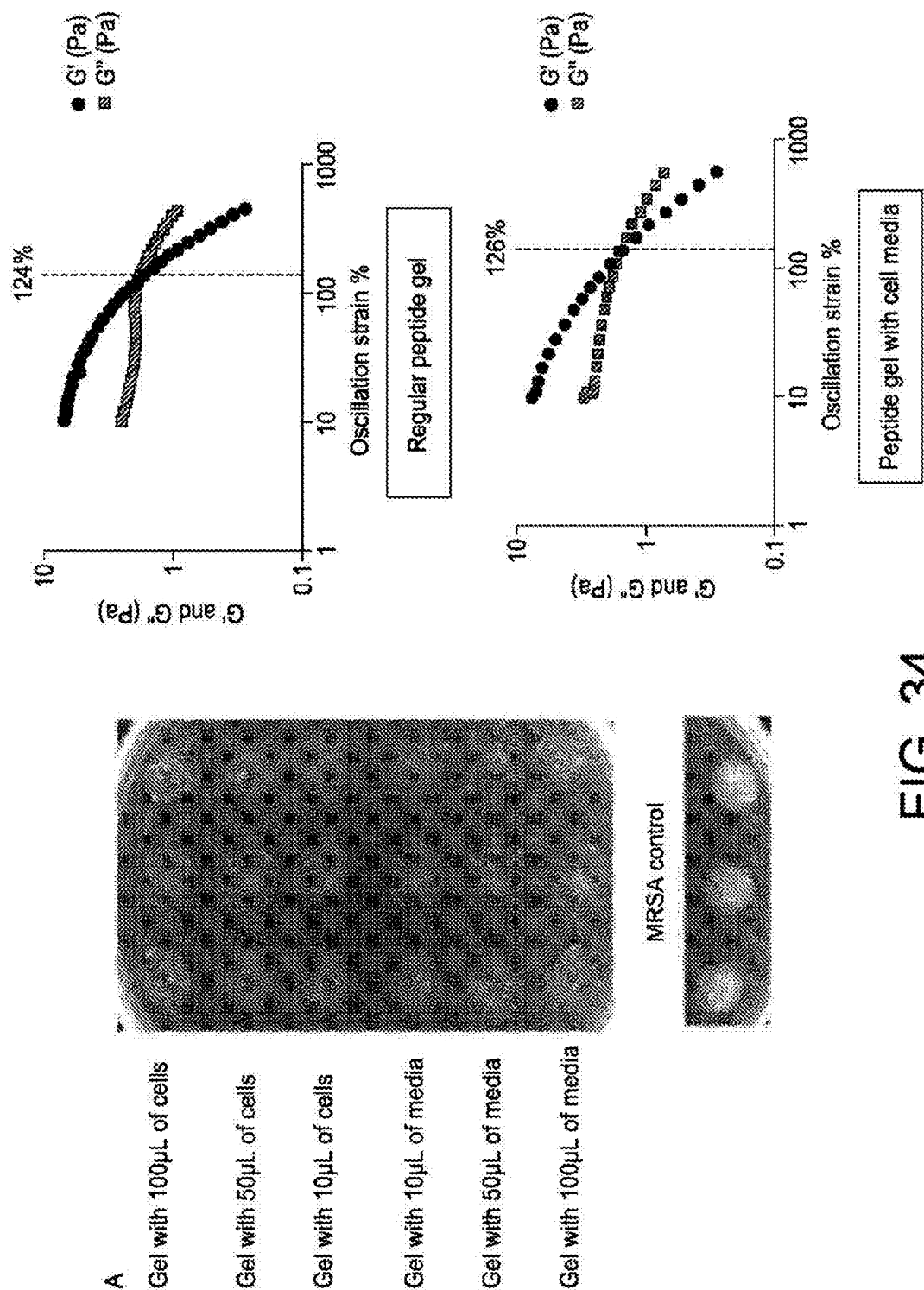


FIG. 34

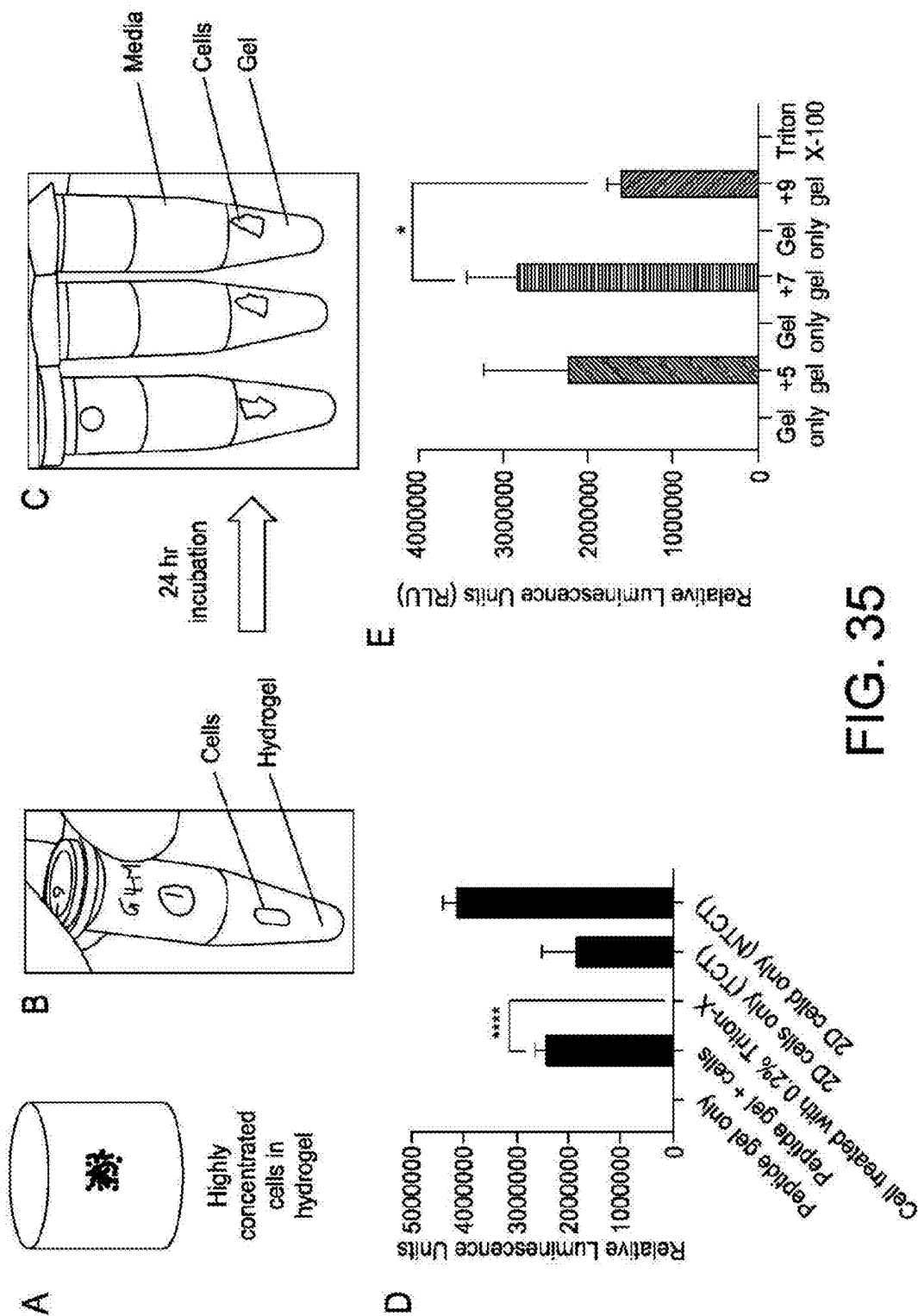


FIG. 35

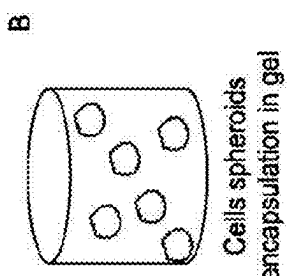
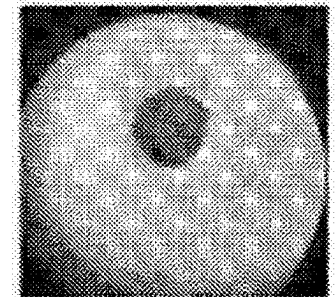
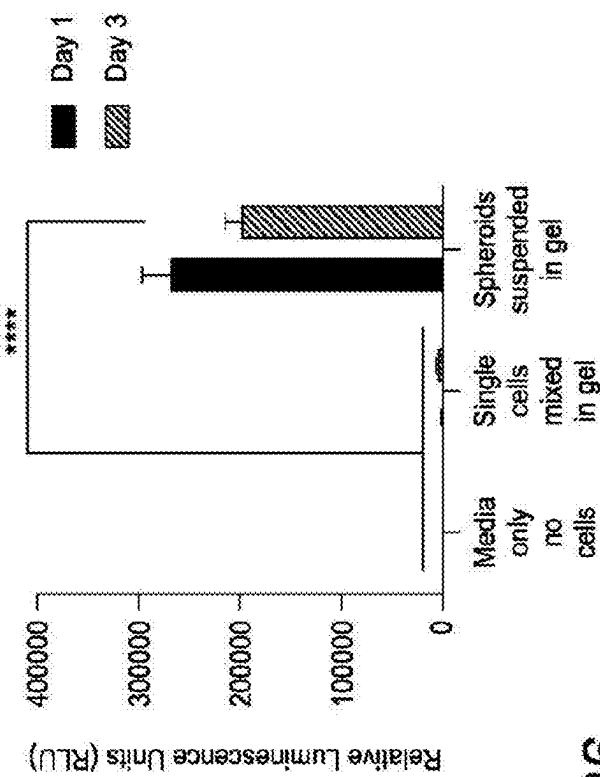


FIG. 36

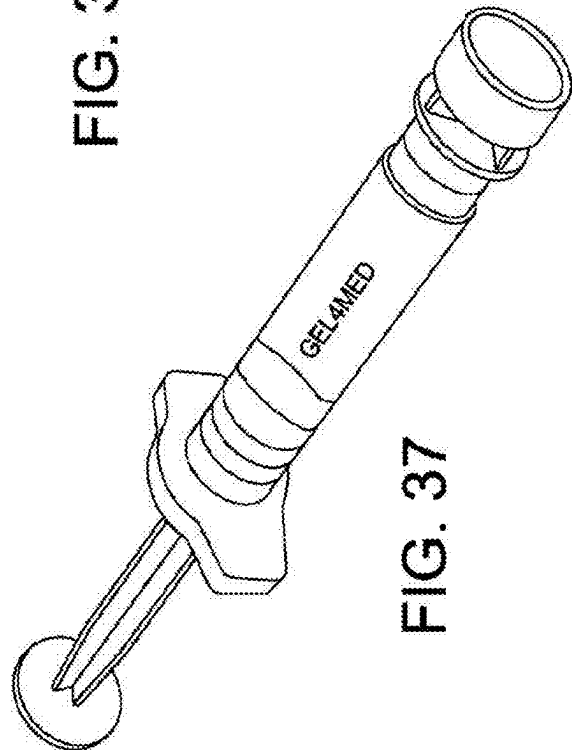


FIG. 37

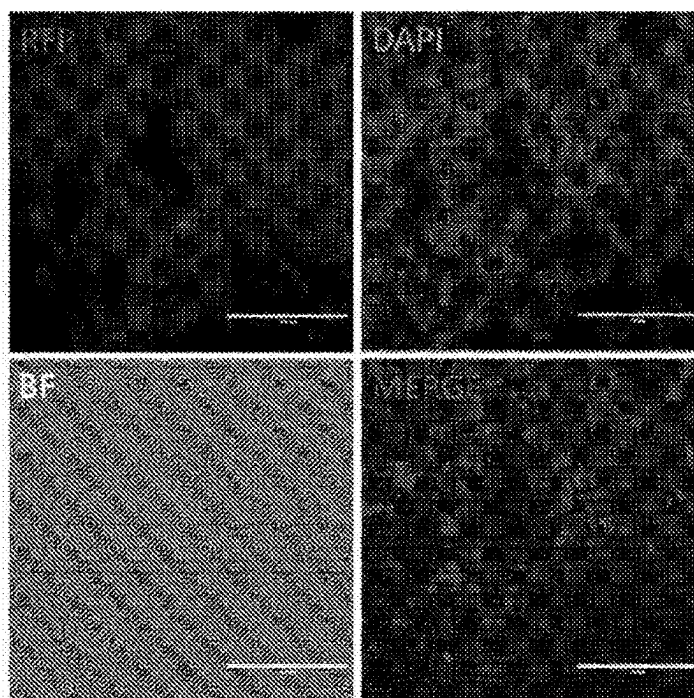


FIG. 38A

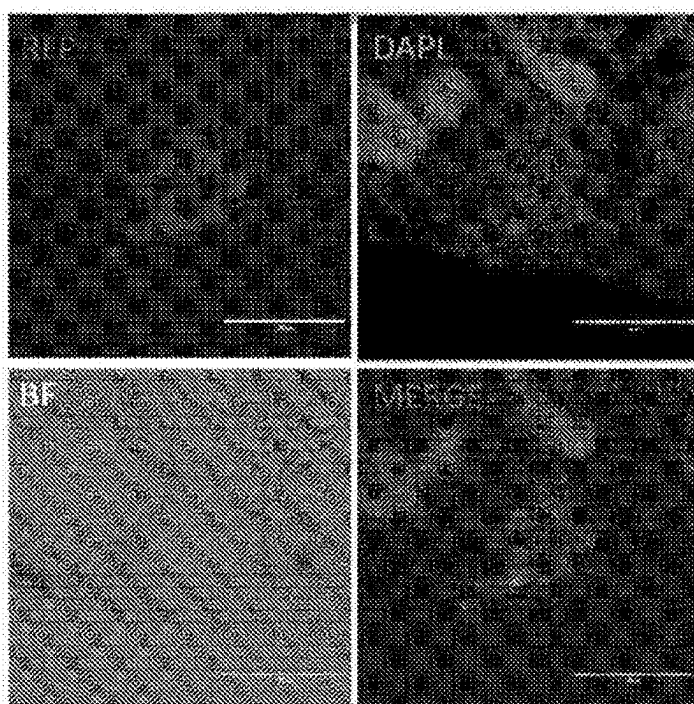
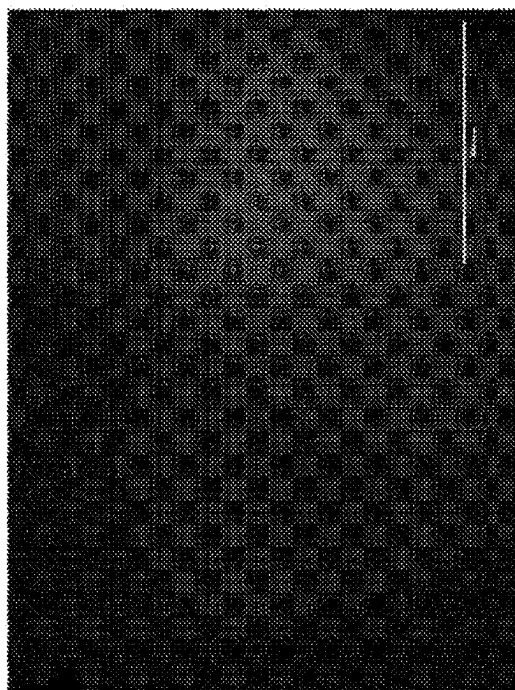
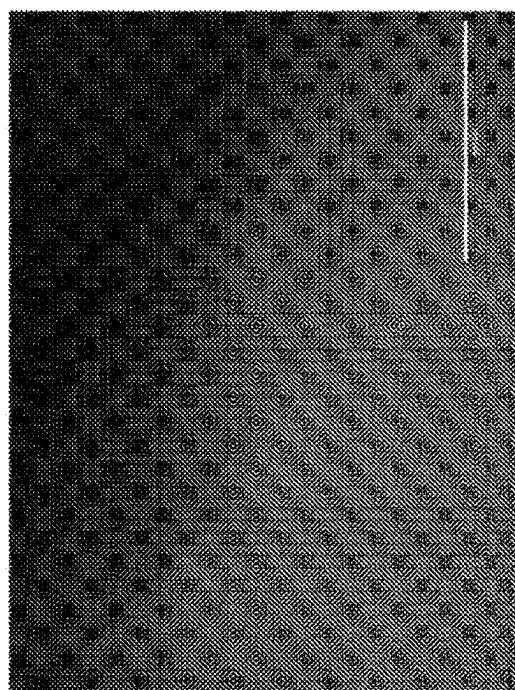


FIG. 38B



MSC's in antimicrobial peptide
gel (20X)

FIG. 39B



MSC's in
non-antimicrobial RGD modified
peptide gels (20X)

FIG. 39A

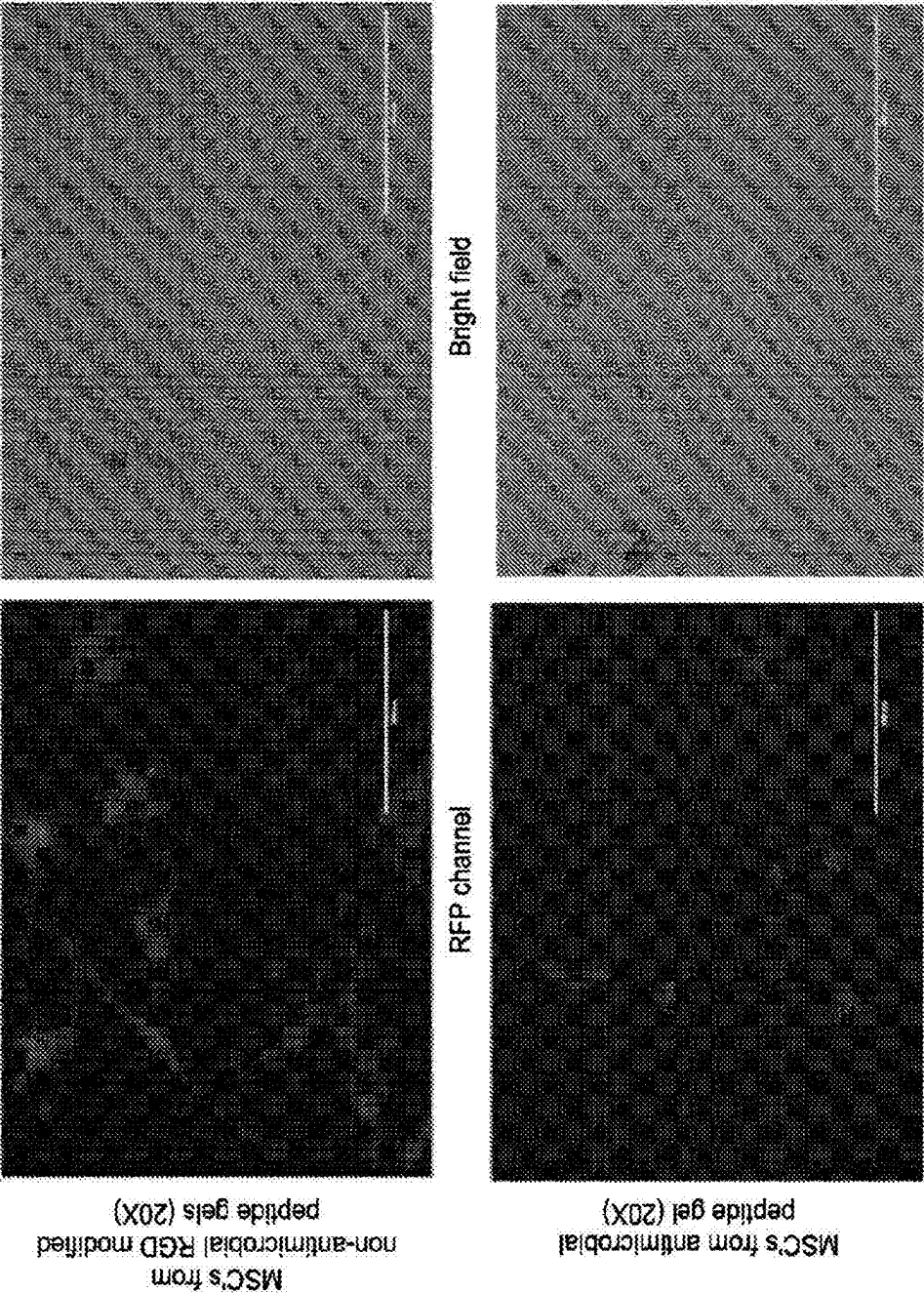


FIG. 40

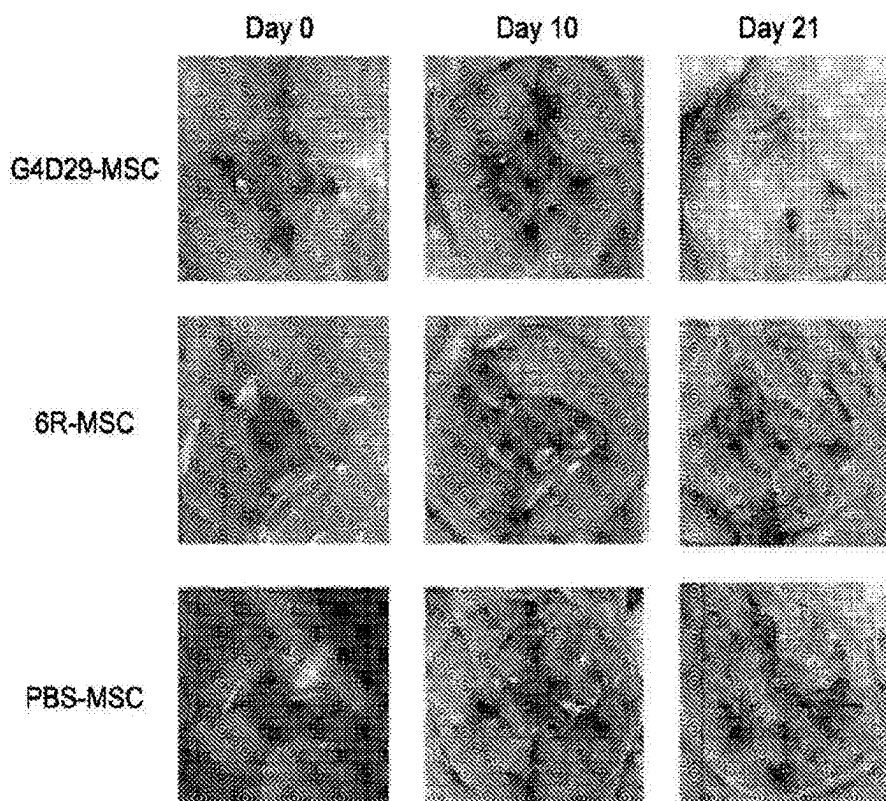


FIG. 41

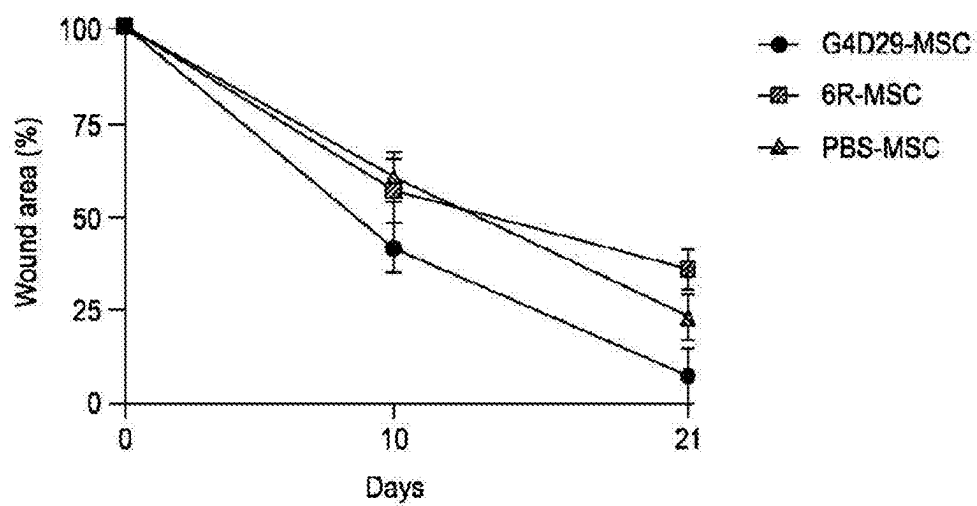


FIG. 42

DELIVERY OF CELLS AND TISSUES WITH SELF-ASSEMBLING PEPTIDE HYDROGEL MATERIALS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority under 35 U.S.C. § 119 (e) to U.S. Provisional Patent Application No. 63/063,747 titled “Delivery of Cells and Tissues with Self-Assembling Peptide Hydrogel Materials” filed Aug. 10, 2020, the entire disclosure of which is herein incorporated by reference in its entirety for all purposes.

SEQUENCE LISTING

[0002] The instant application contains a Sequence Listing which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on Aug. 6, 2021, is named G2093-7000WO_SL.txt and is 9,711 bytes in size.

FIELD OF TECHNOLOGY

[0003] Aspects and embodiments disclosed herein are directed toward systems and methods for the delivery of cells, tissues, and/or cell-derived or tissue-derived materials with hydrogel matrices formed from self-assembling peptide hydrogel materials (SAPM).

BACKGROUND

[0004] Cell therapy is the administration of cell-derived and tissue-derived material into a subject. Often, the cell and tissue-derived materials include biological fluids and/or living cells and tissues. Cell therapy is recognized as an important field in the treatment of disease. There is a need for materials capable of delivering living cells and cell and tissue-derived material to a target tissue site in a safe and effective manner.

SUMMARY

[0005] In accordance with one aspect, there is provided a method of administering biological material to a subject. The method may comprise combining the biological material with a thermally stable preparation comprising a purified amphiphilic peptide in an aqueous biocompatible solution and a buffer comprising an effective amount of an ionic salt and a biological buffering agent to form a hydrogel. The peptide may comprise a folding group having a plurality of charged amino acid residues and hydrophobic amino acid residues arranged in a substantially alternating pattern and a turn sequence. The peptide may be configured to self-assemble into a hydrogel. The method may comprise administering the hydrogel comprising the biological material to a target tissue of the subject.

[0006] In accordance with another aspect, there is provided an in vitro method of preparing a culture of biological material. The method may comprise combining the biological material with a thermally stable preparation comprising a purified amphiphilic peptide in an aqueous biocompatible solution and a buffer comprising an effective amount of an ionic salt and a biological buffering agent to form a hydrogel and prepare the culture of biological material. The peptide may comprise a folding group having a plurality of charged amino acid residues and hydrophobic amino acid residues

arranged in a substantially alternating pattern and a turn sequence. The peptide may be configured to self-assemble into a hydrogel.

[0007] In accordance with another aspect, there is provided a method of grafting biological material in a subject. The method may comprise combining a thermally stable preparation comprising a purified amphiphilic peptide in an aqueous biocompatible solution and a buffer comprising an effective amount of an ionic salt and a biological buffering agent to form a hydrogel. The peptide may comprise a folding group having a plurality of charged amino acid residues and hydrophobic amino acid residues arranged in a substantially alternating pattern and a turn sequence. The peptide may be configured to self-assemble into a hydrogel. The method may comprise administering to a target site of the subject the hydrogel. The method may comprise administering to the target site an effective amount of the biological material to graft the biological material.

[0008] The method may comprise combining the biological material with the preparation to produce a biological material suspension and combining the biological material suspension with the buffer.

[0009] The method may comprise combining the biological material with the buffer to form a biological material buffer suspension and combining the biological material buffer suspension with the preparation.

[0010] The method may comprise combining the preparation with the buffer to form the hydrogel and combining the biological material with the hydrogel.

[0011] The method may comprise combining the preparation with the buffer to form the hydrogel in vitro.

[0012] The method may comprise combining at least two of the biological material, the preparation, and the buffer in vitro.

[0013] The method may comprise combining at least two of the biological material, the preparation, and the buffer in vivo.

[0014] The method may comprise combining at least two of the biological material, the preparation, and the buffer in situ.

[0015] The method may comprise combining the preparation with the buffer to form the hydrogel in vitro, and combining the biological material with the hydrogel in vivo.

[0016] The method may comprise administering an effective amount of the biological material to treat a wound, biofilm, tissue injury, tissue regeneration, microbial contamination, fungal contamination, viral contamination, or a tumor.

[0017] The biological material may comprise at least one of cells, tissue material, cell-derived material, tissue-derived material, and biological fluids.

[0018] The biological material may comprise active or inactive forms of at least one of eukaryote cells, virus, prokaryote cells, adjuvants, cytokines, and growth factors.

[0019] The cells may comprise progenitor cells, multipotent cells, induced pluripotent cells, immune cells, specialized cells, terminally-differentiated cells, bone marrow mononuclear cells, islet cells, or combinations thereof.

[0020] The tissue material may comprise bone tissue, connective tissue, neural tissue, adipose tissue, cartilage, epithelial tissue, muscle tissue, bone marrow, or combinations thereof.

[0021] The biological material may be autologous.

[0022] The biological material may be allogeneic.

[0023] The biological material may be xenogeneic.

[0024] The biological material may be synthetic.

[0025] The method may comprise collecting the biological material from the subject or from a donor subject.

[0026] The method may comprise culturing the biological material in the hydrogel for a predetermined period of time prior to administration.

[0027] In some embodiments, the hydrogel may comprise a non-homogeneous suspension of the biological material. For example, the suspension may comprise or produce clusters or spheroids.

[0028] The method may comprise combining the preparation and the buffer at a point of use. The method may comprise combining the preparation and the buffer less than about 1 minute, less than about 2 minutes, less than about 5 minutes, or less than about 10 minutes prior to administration.

[0029] The method may comprise administering the hydrogel or the biological material topically or parenterally to the subject.

[0030] In some embodiments, the target tissue or target site may be internal relative to the subject.

[0031] In some embodiments, the target tissue or target site may be external relative to the subject.

[0032] The peptide may comprise an effective amount of counterions.

[0033] The peptide may comprise an effective amount of acetate, citrate, and/or chloride counterions.

[0034] The peptide may be substantially free of chloride counterions.

[0035] The buffer may comprise between about 10 mM and 150 mM sodium chloride and between about 10 mM and 100 mM Bis-tris propane (BTP).

[0036] The method may further comprise combining the biological material with a cell culture media, cell maintenance agent, cell growth agent, cell culture serum, or combination thereof.

[0037] In some embodiments, administration to a target tissue or target site may comprise administration to a tissue selected from mesenchymal tissue, connective tissue, muscle tissue, nervous tissue, embryonic tissue, dermal tissue, bone tissue, dental tissue, corneal tissue, cutaneous tissue, integumental tissue, soft tissue, and hard tissue.

[0038] In some embodiments, administration to a target tissue or target site may comprise administration to a biological fluid selected from tears, mucus, urine, menses, blood, wound exudates, and mixtures thereof.

[0039] The hydrogel may be administered by spray, dropper, film, squeeze tube, or syringe. The hydrogel may be administered in combination with a surgical procedure.

[0040] The method may comprise administering a first dosage of the preparation and/or the biological material.

[0041] The method may comprise administering at least one booster dosage of the preparation and/or the biological material.

[0042] In some embodiments, the hydrophobic amino acid residues may be independently selected from glycine, alanine, valine, leucine, isoleucine, proline, phenylalanine, methionine, threonine, tryptophan, and combinations thereof.

[0043] In some embodiments, the charged amino acid residues may be independently selected from arginine, lysine, histidine, and combinations thereof.

[0044] In some embodiments, the folding group may have a sequence comprising Y[XY]N[T][YX]MY, where X is 1-3 charged amino acids, Y is 1-3 hydrophobic amino acids, T is 2-8 turn sequence amino acids, and N and M are each independently between 2 and 10.

[0045] In some embodiments, the turn sequence amino acids may be independently selected from a D-proline, an L-proline, aspartic acid, threonine, asparagine, and combinations thereof.

[0046] The peptide may be configured to self-assemble into a substantially biocompatible hydrogel.

[0047] The peptide may be configured to self-assemble into a hydrogel having at least one property selected from a cell-friendly hydrogel, a substantially biodegradable, non-inflammatory, and/or non-toxic hydrogel, a hydrogel having substantially low hemolytic activity, and a hydrogel having substantially low immunogenic activity.

[0048] The method may further comprise administering at least one combination treatment selected from: an antibacterial composition, an antifungal composition, an antiviral composition, an anti-tumor composition, an anti-inflammatory composition, an anti-odor composition, a cell culture media, a cell culture serum, a hemostatic composition, and an analgesic or pain-relief composition.

[0049] The combination treatment may be administered prior to the preparation.

[0050] The combination treatment may be administered after the preparation.

[0051] The combination treatment may be administered concurrently with the preparation.

[0052] In some embodiments, the peptide may be at least 80% purified, for example, at least 85%, at least 90%, at least 92%, at least 95%, at least 98%, at least 99%, or at least 99.9%.

[0053] The purified peptide may have less than 10% residual organic solvent by weight, for example, less than 8%, less than 5%, less than 2%, less than 1%, or less than 0.1%.

[0054] 25 The organic solvent may comprise at least one of trifluoroacetic acid (TFA), acetonitrile, isopropanol, N,N-Dimethylformamide, triethylamine, Ethyl Ether, and acetic acid.

[0055] The preparation may have a residual Trifluoroacetic acid (TFA) concentration of less than about 1% w/v, a residual acetonitrile concentration of less than about 410 ppm, a residual N,N-Dimethylformamide concentration of less than about 880 ppm, a residual triethylamine concentration of less than about 5000 ppm, a residual Ethyl Ether concentration of less than about 1000 ppm, a residual isopropanol concentration of less than about 100 ppm, and/or a residual acetic acid concentration of less than 0.1% w/v.

[0056] The peptide may include a functional group.

[0057] The functional group may have between 3 and 30 amino acid residues.

[0058] The functional group may be engineered to express a bioactive property.

[0059] The functional group may be engineered to control or alter charge of the peptide or preparation.

[0060] The functional group may be engineered to control or alter pH of the peptide or preparation.

[0061] The functional group may be engineered for a target indication.

[0062] The target indication may be selected from cell culture, cell delivery, wound healing, treatment of biofilm, and combinations thereof.

[0063] The functional group may have a sequence selected from RGD, IKVAV, YIGSR, LKKTETQ, SNKPGVL, PKPQQFFGLM, GKLTWQELYQLKYKGI, and GGG.

[0064] The peptide may be configured to self-assemble into an ionically-crosslinked hydrogel.

[0065] The peptide may be configured to self-assemble into a shear-thinning hydrogel. The peptide may be configured to self-assemble into a substantially transparent hydrogel.

[0066] The buffer may comprise from about 5 mM to about 200 mM ionic salts.

[0067] The ionic salt may dissociate into at least one of sodium, potassium, calcium, magnesium, iron, ammonium, pyridium, quaternary ammonium, chloride, and sulfate ions. The ionic salt may comprise sodium chloride, ammonium chloride, magnesium chloride, potassium chloride, calcium chloride, ammonium sulfate, magnesium sulfate, sodium sulfate, potassium sulfate, calcium sulfate, sodium bicarbonate, and combinations thereof.

[0068] The buffer may comprise from about 10 mM to about 150 mM sodium chloride.

[0069] The peptide may have a bacterial endotoxin level of less than about 10 EU/mg.

[0070] The preparation may comprise between 0.1% w/v and 8.0% w/v of the peptide.

[0071] The preparation may comprise between 0.5% w/v and 6.0% w/v of the peptide.

[0072] The preparation may comprise between 0.5% w/v and 3.0% w/v of the peptide.

[0073] The preparation may comprise between 0.5% w/v and 1.5% w/v of the peptide.

[0074] The preparation may comprise between 0.5% w/v and 1.0% w/v of the peptide.

[0075] The preparation may comprise between 0.7% w/v and 2.0% w/v of the peptide.

[0076] The preparation may comprise between 0.7% w/v and 0.8% w/v of the peptide.

[0077] The hydrogel may comprise between 0.25% w/v and 6.0% w/v of the peptide.

[0078] The hydrogel may comprise between 1.5% w/v and 6.0% w/v of the peptide.

[0079] The hydrogel may comprise between 0.25% w/v and 3.0% w/v of the peptide.

[0080] The peptide may be configured to self-assemble into a hydrogel having between 90% w/v and 99.9% w/v aqueous solution.

[0081] The peptide may have a net charge of from -7 to +11.

[0082] The peptide may have a net charge of from +2 to +9.

[0083] The peptide may have a net charge of from +5 to +9.

[0084] The peptide may be lyophilized.

[0085] The preparation may be sterile.

[0086] The preparation may be substantially free of a preservative.

[0087] The preparation may be thermally stable between -20° C. and 150° C.

[0088] The preparation may be sterilized by autoclave sterilization.

[0089] The preparation may be administered as a hemostat, antimicrobial barrier dressing, antifungal barrier dressing, antiviral barrier dressing, and/or autolytic debridement agent.

[0090] The method may comprise debridement of the target tissue prior to administration of the preparation.

[0091] The method may comprise providing at least one of the biological material, the preparation, and the buffer.

[0092] The method may comprise providing at least one of the biological material, the peptide, the biocompatible solution, and the buffer separately.

[0093] In accordance with another aspect, there is provided a method of treating a wound or biofilm in a subject, comprising administering an effective amount of the hydrogel and the biological material to a target site of the wound or biofilm of the subject.

[0094] In accordance with another aspect, there is provided a method of treating tissue injury or providing tissue regeneration in a subject, comprising administering an effective amount of the hydrogel and the biological material to a target site of the tissue injury or a tissue to be regenerated of the subject.

[0095] In accordance with another aspect, there is provided a method of treating a tumor in a subject, comprising administering an effective amount of the hydrogel and the biological material to the tumor of the subject.

[0096] In accordance with another aspect, there is provided a method of biofabricating a hydrogel having suspended biological material. The method may comprise obtaining the biological material from a donor subject. The method may comprise combining, at a point of care, the biological material with a thermally stable preparation comprising a purified amphiphilic peptide in an aqueous biocompatible solution and a buffer comprising an effective amount of an ionic salt and a biological buffering agent to form the hydrogel and biofabricate the hydrogel having suspended biological material. The peptide may comprise a folding group having a plurality of charged amino acid residues and hydrophobic amino acid residues arranged in a substantially alternating pattern and a turn sequence. The peptide may be configured to self-assemble into a hydrogel.

[0097] The method may further comprise administering the hydrogel to a target tissue of a recipient subject at the point of care.

[0098] In some embodiments, the donor subject may be the recipient subject.

[0099] In some embodiments, the biological material may be obtained at the point of care.

[0100] The method may further comprise treating the biological material at the point of care prior to suspending the biological material in the preparation.

[0101] In accordance with yet another aspect, there is provided a method of facilitating cell therapy in a subject. The method may comprise providing a thermally stable preparation comprising a purified amphiphilic peptide in an aqueous biocompatible solution, the peptide comprising a folding group having a plurality of charged amino acid residues and hydrophobic amino acid residues arranged in a substantially alternating pattern and a turn sequence, the peptide being configured to self-assemble into a hydrogel. The method may comprise providing instructions to combine cells with the preparation. The method may comprise providing instructions to combine cells with the preparation and a buffer comprising an effective amount of an ionic salt

and a biological buffering agent to form the hydrogel. The method may comprise providing instructions to agitate the hydrogel comprising the cells to produce a non-homogeneous cell suspension hydrogel. The method may comprise providing instructions to administer an effective amount of the non-homogeneous cell suspension hydrogel to a target site of the subject to provide cell therapy to the subject.

[0102] In some embodiments, the method may further comprise providing the buffer.

[0103] The method may further comprise providing the cells.

[0104] The method may further comprise providing at least one of a mixing device configured to agitate the hydrogel and a delivery device configured to administer the non-homogeneous cell suspension hydrogel.

[0105] The disclosure contemplates all combinations of any one or more of the foregoing aspects and/or embodiments, as well as combinations with any one or more of the embodiments set forth in the detailed description and any examples.

BRIEF DESCRIPTION OF DRAWINGS

[0106] The accompanying drawings are not intended to be drawn to scale. In the drawings, each identical or nearly identical component that is illustrated in various figures is represented by a like numeral. For purposes of clarity, not every component may be labeled in every drawing. In the drawings:

[0107] FIG. 1A includes schematic and microscopic images of an assembled peptide hydrogel matrix with encapsulated cells as compared to collagen, according to one embodiment;

[0108] FIG. 1B includes a schematic drawing of a mixing device and a schematic representation of cells in a hydrogel matrix, according to one embodiment;

[0109] FIG. 2 includes images sustained therapeutic activity of the administered peptide hydrogel compared to conventional polymer, according to one embodiment;

[0110] FIG. 3 is a microscopy image of positively-charged peptide hydrogels, according to one embodiment;

[0111] FIG. 4 is a graph showing the antimicrobial activity of the peptide hydrogel, in accordance with one embodiment;

[0112] FIG. 5 includes images of a mouse model post-burn injury with bacterial infection showing antimicrobial activity of the peptide hydrogel, in accordance with one embodiment;

[0113] FIG. 6 includes microscopy images of cells engrafted on the peptide hydrogels and a graph showing antimicrobial efficacy of the peptide hydrogel, according to one embodiment;

[0114] FIG. 7A is a graph of static light scattering (SLS) at 266 nm of exemplary peptides as a function of temperature, according to some embodiments;

[0115] FIG. 7B is a graph of static light scattering (SLS) at 266 nm of exemplary peptides as a function of temperature, according to some embodiments FIG. 8 includes graphs showing absorbance of a peptide hydrogel as a function of peptide concentration, according to one embodiment;

[0116] FIG. 9 is a graph showing net charge of a peptide preparation as a function of pH value, according to one embodiment;

[0117] FIG. 10 is a visual representation of net peptide charge at a pH of 7.4 for several amino acid residues, according to one embodiment;

[0118] FIG. 11 includes schematic drawings showing methods of incorporating biological material in a hydrogel, according to one embodiment;

[0119] FIG. 12 includes a schematic drawing of a mixing device, images showing cell distribution in the hydrogel after use of the mixing device, and graphs showing results from a luciferase assay demonstrating viability, according to one embodiment;

[0120] FIG. 13 is a graph showing cell viability in the hydrogel, according to one embodiment;

[0121] FIG. 14 includes a schematic drawing and bright-field microscopy images showing cells seeded on top of a hydrogel and graphs showing results from a luciferase viability assay, according to one embodiment;

[0122] FIG. 15 includes images of polyacrylamide gel electrophoresis (PAGE) showing release of encapsulated enzyme from hydrogels with varying peptide concentrations, according to one embodiment;

[0123] FIG. 16 is a graph showing cell viability in hydrogels of varying pH value, according to one embodiment;

[0124] FIG. 17 includes graphs showing cytocompatibility of hydrogels formed of peptides having functional groups, according to one embodiment;

[0125] FIG. 18 includes graphs from antimicrobial assays and images of the cultured plates, according to one embodiment;

[0126] FIG. 19 includes a graph from a luciferase cell viability assay showing cell viability in hydrogels prepared with different buffers, according to one embodiment;

[0127] FIG. 20 is a schematic drawing showing assembly and disassembly of the hydrogel responsive to external stimuli, according to one embodiment;

[0128] FIG. 21 includes images of cells and peptide hydrogel, according to one embodiment;

[0129] FIG. 22 includes images of hematoxylin and eosin (H&E) stained tissue samples, treated according to one embodiment;

[0130] FIG. 23 includes images of antimicrobial efficacy testing of the self-assembling peptide hydrogel, according to one embodiment;

[0131] FIG. 24 includes an image of a hematoxylin and eosin (H&E) stained tissue sample and a graph showing data from a tissue cytokine array, according to one embodiment

[0132] FIG. 25 includes images of peptide hydrogel cultured with MRSA and an image of alpha-SMA stained tissue sample after treatment with the peptide hydrogel, according to one embodiment;

[0133] FIG. 26 includes photographs of a mixing device, according to one embodiment;

[0134] FIG. 27 includes photographs of a peptide hydrogel encapsulating cell aggregates, according to one embodiment;

[0135] FIG. 28 includes photographs of a peptide hydrogel having cells seeded thereon, according to one embodiment;

[0136] FIG. 29 is a graph showing the standard curve for quantification of endotoxin level of the peptide hydrogel, according to one embodiment;

[0137] FIG. 30A is a graph of storage modulus of the hydrogel as a function of post-formulation time, according to one embodiment;

[0138] FIG. 30B is a graph of storage modulus of the hydrogel as a function of salt concentration, according to one embodiment;

[0139] FIG. 30C includes graphs of storage modulus of hydrogel formulations having different peptide concentrations, according to one embodiment;

[0140] FIG. 31 includes photographs of multi-layered self-assembling peptide hydrogel matrices, according to one embodiment;

[0141] FIG. 32A is a schematic diagram of a method for biofabricating hydrogels, according to one embodiment;

[0142] FIG. 32B is a schematic diagram of a method for biofabricating hydrogels, according to one embodiment;

[0143] FIG. 32C is a schematic diagram of a method for biofabricating hydrogels, according to one embodiment;

[0144] FIG. 33 includes microscopy images of a cells on a self-assembling peptide hydrogel and a conventional polymer, according to one embodiment;

[0145] FIG. 34 includes images of peptide hydrogel cultured with MRSA and graphs of the storage modulus of the peptide hydrogels, according to one embodiment;

[0146] FIG. 35 includes photographs of highly concentrated cells in self-assembling peptide hydrogels and graphs of luciferase viability assays, according to one embodiment;

[0147] FIG. 36 includes a schematic illustration and photograph of cell spheroids in self-assembling peptide hydrogel and a graph showing results of a luciferase cell viability assay, according to one embodiment;

[0148] FIG. 37 is a photograph of the preparation provided in an end-use container, according to one embodiment;

[0149] FIG. 38A includes microscopy images of peptide hydrogels after implant in vivo, according to one embodiment;

[0150] FIG. 38B includes microscopy images of peptide hydrogels after implant in vivo, according to one embodiment;

[0151] FIG. 39A is a microscopy image of mesenchymal stem cells (MSC) in a non-antimicrobial peptide hydrogel;

[0152] FIG. 39B is a microscopy image of mesenchymal stem cells (MSC) in an antimicrobial peptide hydrogel, according to one embodiment;

[0153] FIG. 40 includes microscopy images of MSCs in a non-antimicrobial peptide hydrogel and images of MSCs in an antimicrobial peptide hydrogel, according to one embodiment;

[0154] FIG. 41 includes images of wound area after administration of the peptide hydrogel, according to one embodiment; and

[0155] FIG. 42 is a graph of wound area after administration of the peptide hydrogel, according to one embodiment.

DETAILED DESCRIPTION

[0156] Cell therapy is the administration of cell-derived and tissue-derived material to a subject. Conventional cell therapy efforts have been limited by challenges in implantation and biofabrication of cell and tissue implants, suspension, encapsulation, retention, survival, delivery, expansion, and protection of biological materials upon administration. Disclosed herein are compositions and methods for performing and facilitating cell therapy with a self-assembling peptide hydrogel materials (SAPM).

[0157] Biofabrication involves the preparation and production of cells, tissues, cell-derived and tissue-derived materials, biological fluids, suspension materials, scaffolds,

biomaterials, polymers, or any combination thereof, for therapeutic administration to a subject. Biofabrication may be performed at the point of patient care, in a laboratory, at manufacturing facilities, or any facility under sterile conditions, as approved by regulatory bodies. Disclosed herein are compositions and methods of biofabrication of cells, cell-derived and tissue-derived materials, biological fluids, or any combination thereof, with self-assembling peptide hydrogel materials (SAPM) for treatment of disease and injury.

[0158] Methods of administering cells, tissues, implants, or biological material to a subject are disclosed herein. The methods may generally include suspending the material in a hydrogel which may serve as a cell or tissue scaffold or matrix when implanted. The methods may comprise combining the biological material with a solution comprising a self-assembling peptide and a buffer configured to induce self-assembly. The methods may comprise administering an effective amount of the suspension to a target site of the subject. The solution may comprise water, media, or buffer. The peptide may include functional groups to provide desired physical or chemical properties upon administration. The preparation composition may be designed to control the assembled hydrogel's physical or chemical properties. In general, the suspension may be designed to have shear-thinning properties and a substantially physiological pH. However, the stiffness, rheology, viscosity, pH, net charge, and other properties may be controlled by peptide design. In certain embodiments, the self-assembled hydrogel may have antimicrobial properties.

[0159] For instance, the peptide net charge may be selected or designed by amino acid selection, N and C Terminus modifications, and amino acid positioning within the peptide sequence, to enhance and promote properties of the hydrogel such as cell cytocompatibility, cell attachment, cell proliferation, cell viability, cell programming, cell deployment. The preparation may be designed to exhibit shear-thinning kinetics that reversibly transition the peptide from hydrogel to solution responsive to an external stimulus. The shear-thinning kinetics of the preparation may generally enable effective cell encapsulation (biofabrication) within the formed hydrogel, and thus effective delivery of biological material to a target site.

[0160] The physical and chemical properties of the hydrogel may be controlled to engineer a release rate of the biological material after administration to the target site. For example, hydrogel systems with tunable degradability may be used for three-dimensional cell encapsulation and drug delivery. Degradable hydrogels may provide a temporary scaffold for the growth of implanted cells and reduce or eliminate the need for additional surgical treatment for removal of the implanted substrate. The methods disclosed herein facilitate the design and engineering of implantable hydrogels by controlling degradation profile. Degradation profile may be controlled to select a desired rate of drug release or tissue regeneration and provide an improved therapeutic effect at the target site and systemically.

[0161] The methods disclosed herein may comprise administering an effective amount of the cells, tissues, biological materials, biological fluids, and combinations thereof, to a target site of the subject. The methods disclosed herein may achieve a therapeutic effect locally or systemically.

[0162] Preparations comprising self-assembling peptide hydrogels are disclosed herein. The self-assembled peptide may be amphiphilic. The peptide may generally have a folding group having a plurality of charged amino acid residues and hydrophobic residues arranged in a substantially alternating pattern. The peptide may include functional groups to provide desired physical or chemical properties upon administration. The purified peptide may include counterions that improve biocompatibility of the preparation. The counterions may control the self-assembly, physical and chemical properties of the peptide. The counterions may enhance the therapeutic functional properties of the peptide. The preparation may include the peptide in an aqueous biocompatible solution. The preparation may include a buffer solution capable of inducing self-assembly of the peptide upon contact. The buffer solution may contain a buffering agent and ionic salts. The buffer solution composition may be designed to control the assembled hydrogel's physical or chemical properties. The preparation may be designed to be thermally stable.

[0163] In general, the preparation may have shear-thinning properties and a substantially physiological pH level. The self-assembled hydrogel may have antimicrobial, antiviral, and/or antifungal properties. The preparation may be administered topically or parenterally. The preparation may be administered for tissue engineering applications. Certain exemplary applications include cell delivery, cell culture, treatment and prevention of fungal infections, treatment and prevention of bacterial infections, wound healing, biofilm treatment, biofilm management, and prevention of biofilm and wound infection, including infection of chronic wounds. Other tissue engineering applications are within the scope of the disclosure.

[0164] Methods of administering the preparation to a subject are disclosed herein. The methods may generally include selecting a target site for administration and administering the preparation to the target site. Methods of administering the preparation may also include mixing the preparation with a buffer configured to induce self-assembly of the peptide to form the hydrogel and administering the hydrogel to the target site. In certain exemplary embodiments, the preparation and/or hydrogel may be administered by spray, aerosol, dropper, tube, ampule, instillation, injection, or syringe.

[0165] In certain embodiments, methods of administering cells to a subject are disclosed herein. The methods may generally include suspending the cells in a solution comprising a self-assembling peptide and administering an effective amount of the suspension to a target site of the subject. The methods may comprise combining the solution with a buffer configured to induce self-assembly of the peptide. The solution may be combined with the buffer prior to administration, concurrently with administration, or after administration. The buffer may generally comprise an effective amount of an ionic salt and a biological buffering agent.

[0166] Unlike other peptides in aqueous solution, the peptides disclosed herein undergo self-assembly. The self-assembly may enable the peptides to be administered in a concentrated or localized manner to a target tissue. For example, self-assembling peptides may be administered at higher concentrations when compared to free floating peptides. The self-assembling peptides may exhibit the clinical benefit of reducing offsite toxicity of the peptides, due to the localizing effect upon administration. Additionally, the

therapeutic dosage of peptides may be increased in the vicinity of the target administration site.

[0167] Unlike other polymers in aqueous solution, the peptides disclosed herein may undergo self-assembly in situ at the target site. The in situ self-assembly may enable the peptides to be administered to a target tissue and allow to physically or ionically crosslink, for example, within seconds of administration. For example, self-assembling peptides may be administered directly to target site. Conventional free-floating peptides or polymers usually need a crosslinking agent or exogenous added covalent crosslinking agent. Thus, the self-assembling peptides disclosed herein may provide the clinical benefit of reducing product application and complexity. Additionally, the ionic crosslinking of peptides upon self-assembly may provide the benefit of selecting between product removal and permanent adherence to a target administration site.

Select Definitions

[0168] Hydrogels are a class of materials that have significant promise for use in soft tissue and bone engineering. The general characteristic of hydrogels that make them important materials for these applications are their well hydrated, porous structure. Hydrogels may be designed to be compatible with the adhesion and proliferation of various cell types, e.g., fibroblasts and osteoblasts, making them potential tissue engineering scaffolds for generating connective tissue, such as cartilage, tendons, and ligaments, and bone.

[0169] The hydrogel material may be cytocompatible. Cytocompatibility, defined herein, means that the hydrogel must not be adverse to desired cells, in vitro and/or in vivo. Adversity to cells may be measured by cytotoxicity, cell adhesion, proliferation, phenotype maintenance, and/or differentiation of progenitor cells.

[0170] The hydrogel material may be biocompatible. "Biocompatible," defined herein, means that a material does not cause a significant immunological and/or inflammatory response if placed in vivo. Biocompatibility may be measured according to International Organization for Standardization (ISO) 10993 standards.

[0171] The hydrogel material may be biodegradable affording non-toxic species. The hydrogel material may be proteolytically biodegradable. "Proteolytic" biodegradation, defined herein, refers to local degradation of the material in response to the presence of cell-derived proteases and/or gradual degradation with the proliferation of cells. The hydrogel material may be hydrolytically biodegradable. "Hydrolytic" biodegradation, defined herein, refers to polymer degradation without assistance from enzyme under biologic conditions.

[0172] The hydrogel material may be bioresorbable. Bioresorbable, defined herein, means that the hydrogel material breaks down into remnants that are natural products readily absorbed into the body, resulting in complete loss of original mass.

[0173] The hydrogel material may be shear-thinning. "Shear-thinning," as described herein, refers to a variable apparent viscosity, in particular, a decreasing viscosity with increasing applied stress. For instance, the shear-thinning hydrogel may exhibit non-Newtonian fluid properties. In particular the hydrogels disclosed herein may be administered through a needle or catheter and rapidly resume gelation after removal of the mechanical force.

[0174] The hydrogel and/or other materials disclosed herein may be referred to as having one or more physiological properties. As disclosed herein, physiological properties or values refer to those which are compatible with the subject. In particular, physiological properties or values may refer to those which are compatible with a particular target tissue. In certain embodiments, physiological properties or values may refer to those which are substantially similar to the properties or values of the target tissue. Physiological properties may include one or more of pH value, temperature, net charge, water content, stiffness, and others. “Self-assembling” peptides include such peptides which, typically, after being exposed to a stimulus, will assume a desired secondary structure. The peptides may self-assemble into a higher order structure, for example a three-dimensional network and, consequently, a hydrogel. The self-assembled hydrogel may contain peptides in a tertiary and/or quaternary structure through charge screening, hydrophobic, and disulfide interactions. Peptides have been observed to self-assemble into helical ribbons, nanofibers, nanotubes and vesicles, surface-assembled structures and others. Self-assembling peptides may assemble responsive to certain environmental conditions, e.g., pH, temperature, net charge, exposure to light, applied sound wave, or presence or absence of environmental factors. The environmental conditions may occur upon administration to a subject or by combination with a buffer. In other embodiments, the peptide may assemble spontaneously in solution under neutral pH level. The peptide may assemble spontaneously in solution under physiological conditions and/or in the presence of a cation and/or anion.

[0175] The self-assembling peptides may assemble into an alpha helix, pi-helix, beta sheet, random coil, turn, beta pleated parallel, antiparallel, twist, bulge, or strand connection secondary structure and combinations of thereof. For example, a 20 amino acid peptide which self-assembles into β -strands may comprise alternating valine and lysine residues flanking a tetrapeptide sequence (-VDPPT-). When dissolved in low ionic strength and buffered aqueous solution, the exemplary peptide resides in an ensemble of random coil conformers due to electrostatic repulsions of the positively charged lysine residues. Upon increasing the ionic strength and/or pH of the solution, the lysine-based positive charge is relieved due to either screening of the charge or deprotonating a sufficient amount of the side chain amines. This exemplary action enables peptide folding into an amphiphilic β -hairpin. In the folded state, the exemplary peptide self-assembles via lateral and facial associations of the hairpins to form a non-covalently crosslinked hydrogel containing β -sheet rich fibrils. Thus, the self-assembling peptides may be designed to undergo hydrogelation under varying conditions through rational design of the peptide sequence.

[0176] The self-assembling peptides disclosed herein may assemble into a nano-porous tertiary structure. As disclosed herein, the nano-porous structure is a three-dimensional matrix containing pores having an average size of 1-1000 nm. The pores or voids may constitute between 10% and 90% of the three-dimensional matrix by volume. For example, the pores or voids may constitute 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, or 90% of the three-dimensional matrix by volume. The pores may be permeable and allow diffusion of liquid and/or gas. The nano-porous structure is constructed by physical crosslinks, allowing ionic

bonds to be broken and reformed upon asserted stress. These nano-porous structure may allow for cells to attach and/or migrate through the matrix. The nano-porous structure may also mimic the endogenous extra-cellular matrix environment of tissues and, optionally, be selected to mimic a specific tissue.

[0177] “Disassembly” of the peptides may refer to the ability of the peptide to assume a lower order structure after being exposed to a stimulus. Disassembly may also refer to the ability of the physically crosslinked peptide to temporarily break hydrophobic and disulfide bonds to assume a lower order structure after being exposed to a stimulus. For example, a tertiary structure protein may disassemble into a secondary structure protein, and further disassemble into a primary structure peptide. In accordance with certain embodiments, self-assembly and disassembly of the peptide may be reversible.

[0178] Preparations and formulations disclosed herein may generally be referred to as peptide preparations. The peptide preparations may include a self-assembling peptide and/or a self-assembled hydrogel as disclosed herein. The peptide preparation may include a cytocompatible and/or biocompatible solution. The preparation may include a buffer. While reference is made to a solution, it should be understood that the preparation may be in the form of a liquid, gel, or solid particle. In certain embodiments, for example, the preparation may be in the form of the assembled hydrogel. In other embodiments, the preparation may be in the form of a lyophilized powder.

[0179] The peptide preparation may further include one or more bioactive components for tissue engineering, such as, functionalized peptides, cells, media, serum, collagen and other structure-imparting components, antibodies and antigens, bioactive small molecules, and other bioactive drugs. “Bioactivity” as described herein refers to the ability of a compound to impart a biological effect.

[0180] Cell containing preparations and formulations disclosed herein may be referred to as cell suspensions. Cell suspensions include a plurality of cells, e.g., living cells, suspended in a solution. The solution may be or comprise water, media, or buffer. The suspension may generally further comprise a self-assembling peptide and/or a self-assembled hydrogel, as disclosed herein. While reference is made to cells, it should be understood that the suspension may contain cell fragments and/or tissue, e.g. tissue grafts, in addition to or instead of the cells. For example, the suspension may contain live or dead cells or cell fragments, spheroids, and/or cell aggregates.

[0181] The cells may be isolated from living tissue and subsequently maintained and/or grown in cell culture. The cell culture conditions may vary, but generally include maintaining the cells in a suitable vessel with a substrate or medium that supplies the essential nutrients, e.g., amino acids, carbohydrates, vitamins, minerals, growth factors, hormones, and gases, e.g., CO₂ and O₂, and regulating the physio-chemical environment, e.g., pH, osmotic pressure, temperature. The cells may be maintained in live cell lines, e.g., a population of HeLa cells descended from a single cell and containing the same genetic makeup.

[0182] The term “isolated,” as used herein, refers to material that is removed from its original or native environment (e.g., the natural environment if it is naturally occurring). For example, a naturally-occurring polynucleotide or polypeptide present in a living animal is not isolated, but the

same polynucleotide or polypeptide, separated by human intervention from some or all of the co-existing materials in the natural system, is isolated. Such polynucleotides could be part of a vector and/or such polynucleotides or polypeptides could be part of a composition, and still be isolated in that such vector or composition is not part of the environment in which it is found in nature.

[0183] As used herein, “treatment” of an injury, condition, or disease refers to reducing the severity or frequency of at least one symptom of that injury, condition, or disease, compared to a similar but untreated subject. Treatment can also refer to halting, slowing, or reversing the progression of an injury, condition, or disease, compared to a similar but untreated subject. Treatment may comprise addressing the root cause of the injury, condition, or disease and/or one or more symptoms. “Management” of an injury, condition, or disease may refer to reducing the severity or frequency of at least one symptom of that injury, condition, or disease, to a tolerable level, as determined by the subject or a health care provider.

[0184] As used herein an effective amount refers to a dose sufficient to achieve a desired result. For example, the effective amount may refer to a concentration sufficient to achieve self-assembly of the hydrogel and/or provide desired properties. An effective amount may refer to a dose sufficient to prevent advancement, or to cause regression of an injury, condition, or disease, or which is capable of relieving a symptom of an injury, condition, or disease, or which is capable of achieving a desired result. An effective amount can be measured, for example, as a concentration of peptide or other component in the preparation, solution, or buffer. An effective amount can be measured, for example, as a concentration of bioactive agent or an effect or byproduct of a bioactive agent. An effective amount can be measured, for example, as a number of cells or number of viable cells, or a mass of cells (e.g., in milligrams, grams, or kilograms), or a volume of cells (e.g., in mm³).

[0185] Throughout this disclosure, formulation may refer to a composition or preparation or product.

[0186] Administered “in combination,” as used herein, means that two (or more) different treatments are delivered to the subject during the course of the subject’s affliction with the injury, e.g., the preparation is delivered with a second agent after the subject has been diagnosed with the condition or injury and before the condition or injury has been cured or eliminated. In certain embodiments, administration in combination means the preparation additionally comprises one or more second agent. In some embodiments, the delivery of one treatment is still occurring when the delivery of the second begins, so that there is overlap. This is sometimes referred to herein as “simultaneous” or “concomitant” or “concurrent delivery.” In other embodiments, the delivery of one treatment ends before the delivery of the other treatment begins. This is sometimes referred to herein as “successive” or “sequential delivery.”

[0187] In embodiments of either case, the treatment is more effective because of combined administration. For example, the second agent is a more effective, e.g., an equivalent effect is seen with less of the second agent, or the second agent reduces symptoms to a greater extent, than would be seen if the second agent were administered in the absence of the preparations disclosed herein, or the analogous situation is seen with the preparation. In some embodiments, delivery is such that the reduction in a symptom, or

other parameter related to the disorder is greater than what would be observed with one treatment delivered in the absence of the other. The effect of the two treatments can be partially additive, wholly additive, or greater than additive (i.e., synergistic). The delivery can be such that an effect of the administration of the preparation is still detectable when the second agent is delivered. In some embodiments, one or more treatment may be delivered prior to diagnosis of the patient with the injury.

[0188] As used herein, a subject may include an animal, a mammal, a human, a non-human animal, a livestock animal, or a companion animal. The term “subject” is intended to include human and non-human animals, for example, vertebrates, large animals, and primates. In certain embodiments, the subject is a mammalian subject, and in particular embodiments, the subject is a human subject. Although applications with humans are clearly foreseen, veterinary applications, for example, with non-human animals, are also envisaged herein. The term “non-human animals” of the disclosure includes all vertebrates, for example, non-mammals (such as birds, for example, chickens; amphibians; reptiles) and mammals, such as non-human primates, domesticated, and agriculturally useful animals, for example, sheep, dog, cat, cow, pig, rat, among others. The term “non-human animals” includes research animals, for example, for example, mouse, rat, rabbit, dog, cat, pig, among others.

Properties of the Peptide Sequence and Secondary Structure

[0189] The peptides disclosed herein may have a sequence configured to fold into a desired secondary structure. The secondary structure may refer to a three-dimensional form of local segments of proteins. The secondary structure may comprise, for example, pleated sheet, helical ribbon, nanotube and vesicle, surface-assembled structure, and others. The peptides disclosed herein may have a sequence configured to self-assemble into a desired tertiary structure. The tertiary structure may refer to a three-dimensional organization of secondary structure protein forms. The tertiary structure may comprise, for example, three-dimensional matrix, porous matrix, nano-porous matrix.

[0190] Self-assembling peptides disclosed may be designed to adopt a secondary, for example, B-hairpin, and/or tertiary structure in response to one or more signals. Typically, after adopting the secondary structure, the peptides will self-assemble into a higher order structure, e.g., a hydrogel. In certain embodiments, the self-assembly does not take place unless side chains on the peptide molecules are uniquely presented in the secondary structure conformation. The self-assembling peptides may assemble responsive to certain environmental conditions, e.g., pH, temperature, net charge, exposure to light, applied sound wave, or presence or absence of environmental factors. The environmental conditions which induce self-assembly may occur upon administration to a subject, e.g., upon contact with a target tissue. In some embodiments, the environmental conditions which induce self-assembly may occur upon combination of the peptide preparation with a buffer configured to induce self-assembly. The buffer may have a pH or composition configured to induce self-assembly. For example, the buffer may have a concentration of ions configured to induce self-assembly.

[0191] Self-assembly of the peptides disclosed herein may produce compact structures that exhibit biophysical struc-

tural relationships with the intended function of the peptide. For example, a compact tertiary structure may have a higher number of active amino acid residues per unit area, compared to unassembled peptides. In the particular example of antimicrobial peptides, the tertiary structure may enable a higher concentration of charged, e.g., positively charged, amino acid residues per area, increasing antimicrobial properties (e.g., bacterial membrane destabilization and disruption).

[0192] In certain embodiments, the self-assembling peptide hydrogels may include those disclosed in and/or prepared by the methods disclosed in any of U.S. Pat. Nos. 8,221,773; 7,884,185; 8,426,559; 7,858,585; and 8,834,926, incorporated herein by reference in their entireties for all purposes. For example, the self-assembling peptide hydrogels may be or comprise any of SEQ ID NOS: 1-20 from U.S. Pat. Nos. 8,221,773, 7,884,185, and 7,858,585; and SEQ ID NOS: 1-33 from U.S. Pat. No. 8,834,926. Other self-assembling peptides are known and may be employed to bring about the methods disclosed herein.

[0193] The desired properties of the self-assembling peptides may be controlled by peptide design. The self-assembling peptides may be small peptides, e.g., from about 6 to about 200 residues or from about 6 to about 50 residues or from about 10 to about 50 residues. Any of the amino acid residues may be a D isoform. Any of the amino acid residues may be an L isoform.

[0194] Self-assembling peptides disclosed herein may be designed to be substantially amphiphilic when assembled into the tertiary structure. "Amphiphilic" molecules, e.g., macromolecules or polymers, as disclosed herein, typically contain hydrophobic and hydrophilic components. Peptide amphiphiles are one exemplary class of amphiphilic molecules. Peptide amphiphiles are peptide-based molecules that typically have the tendency to self-assemble into high-aspect-ratio nanostructures under certain conditions. The exemplary conditions may comprise selected pH, temperature, and ionic strength values. One particular type of peptide amphiphiles comprise alternating charged, neutral, and hydrophobic residues, in a repeated pattern, for example, as disclosed herein. A combination of intermolecular hydrogen bonding and hydrophobic and electrostatic interactions may be designed to form well-defined self-assembled nanostructures by assembly of the disclosed peptide amphiphiles.

[0195] The self-assembling peptides may include additional amino acids, for example, an epitope. For example, the self-assembling peptides may include additional functional groups, optionally selected by peptide design. Exemplary functional groups disclosed herein comprise a biologically derived motif, for example, having an effect on biological processes such as cell signal transduction, cell adhesion in the extra-cellular matrix (ECM), cell growth, and cell mobility. The peptide may include one or more modifications, for example, a linker or spacer. In some embodiments, at least one of the N-terminus and the C-terminus may be modified. For example, at least one of the N-terminus and the C-terminus may be amidated. At least one of the N-terminus and the C-terminus may be acetylated. In certain exemplary embodiments, the C-terminus may be amidated and/or the N-terminus may be acetylated. In some embodiments, at least one of the N-terminus and the C-terminus may be free.

[0196] In general, the self-assembling peptides may have a folding group configured to adopt the secondary and/or higher order structure. Exemplary self-assembling peptides may have a folding group designed to adopt a β -hairpin secondary structure. Exemplary self-assembling peptides may have a folding group designed to adopt a three-dimensional nano-porous matrix tertiary structure. Self-assembling peptides disclosed herein may be designed to adopt a β -hairpin secondary structure and/or nano-porous matrix tertiary structure in response to one or more environmental stimulus at the target site, e.g., at a topical or parenteral site. The self-assembling peptides may also be designed to self-assemble into a range of other self-assembled structures, such as spherical micelles, vesicles, bilayers (lamellar structures), nanofibers, nanotubes, and ribbons.

[0197] The self-assembly folding group may have between about 2 and about 200 residues, for example, between about 2 and about 50 residues, between about 10 and about 30 residues, between about 15 and about 25 residues, for example, about 20 residues.

[0198] In accordance with some embodiments, the self-assembling folding group may include hydrophobic amino acids. "Hydrophobic" amino acid residues are those which tend to repel water. Such hydrophobic amino acids may include glycine, alanine, valine, leucine, isoleucine, proline, phenylalanine, methionine, threonine, and tryptophan. In certain embodiments, the hydrophobic amino acid residues may comprise valine.

[0199] The folding group may be functionalized by addition of other functional residues as described herein, or conserved for self-assembly. Exemplary functional residues include basic, neutral, aliphatic, aromatic, and polar amino acid residues.

[0200] The folding group may have a plurality of basic, neutral, aliphatic, aromatic, polar, charged amino acid residues. The folding group may have a plurality of hydrophobic amino acid residues arranged in a substantially alternating pattern with non-hydrophobic amino acid residues. In certain embodiments, the folding group may have a plurality of hydrophobic amino acid residues arranged in a substantially alternating pattern with a plurality of charged amino acid residues.

[0201] The folding group may comprise a turn sequence. The turn sequence may include one or more internal amino acid residues within the folding group. In certain embodiments, the turn sequence may be substantially centrally located within the folding group.

[0202] The turn sequence may have between about 2 and about 20 residues, for example, between about 2 and about 10 residues, between about 2 and about 8 residues, between about 2 and about 5 residues, for example, about 2 residues, about 3 residues, about 4 residues, or about 5 residues.

[0203] In exemplary embodiments, the turn sequence may include one or more of proline, aspartic acid, threonine, and asparagine. The turn sequence may include D-proline and/or L-proline. In some embodiments, the turn sequence may have 1-4 proline residues, for example, 1 proline residue, 2 proline residues, 3 proline residues, or 4 proline residues.

[0204] Exemplary self-assembling peptides may have a folding group sequence comprising $[AY]_M[T][YA]_M$, where A is 1-3 amino acids selected from one or more of basic, neutral, aliphatic, aromatic, polar, and charged amino acids, Y is 1-3 hydrophobic amino acids, T is 2-8 turn sequence amino acids, and N and M are each independently between

2 and 10. Y amino acids may independently be selected from glycine, alanine, valine, leucine, isoleucine, proline, phenylalanine, methionine, threonine, and tryptophan. In some embodiments, the folding group sequence may be $Y[AY]_N[T][YA]_M Y-NH_2$.

[0205] Certain exemplary self-assembling peptides may have a folding group sequence comprising $[XY]_N[T][YX]_M$, where X is 1-3 charged amino acids, Y is 1-3 hydrophobic amino acids, T is 2-8 turn sequence amino acids, and N and M are each independently between 2 and 10. X amino acids may independently be selected from arginine, lysine, tryptophan, and histidine. Y amino acids may independently be selected from glycine, alanine, valine, leucine, isoleucine, proline, phenylalanine, methionine, threonine, and tryptophan. In some embodiments, the folding group sequence may be $Y[XY]_N[T][YX]_M Y-NH_2$.

[0206] Certain exemplary self-assembling peptides may have a folding group sequence comprising $[ZY]_N[T][YZ]_M$, where Z is 1-3 polar or charged amino acids, Y is 1-3 hydrophobic amino acids, T is 2-8 turn sequence amino acids, and N and M are each independently between 2 and 10. Z amino acids may independently be selected from glutamine, asparagine, histidine, serine, threonine, tyrosine, cysteine, alanine, valine, leucine, isoleucine, proline, phenylalanine, arginine, lysine, aspartic acid, and glutamic acid. Y amino acids may independently be selected from glycine, alanine, valine, leucine, isoleucine, proline, phenylalanine, methionine, threonine, and tryptophan. In some embodiments, the folding group sequence may be $Y[ZY]_N[T][YZ]_M Y-NH_2$.

[0207] One exemplary self-assembling peptide may have a folding group comprising-RYRYRYTYRYRYR- where R is an arginine residue, Y is 1-3 hydrophobic amino acids, and T is 2-6 turn sequence amino acids.

[0208] One exemplary self-assembling peptide may have a folding group comprising-VXVXVXVXTXVXVXV- where V is a valine residue, X may be independently selected from charged and neutral amino acid residues serine, glutamic acid, lysine, tryptophan, and histidine, and T is 2-8 turn sequence amino acids. In some embodiments, the exemplary folding group may comprise a series of hydrophobic valine amino acid residues alternating with independently selected hydrophilic and/or neutral amino acid residues.

[0209] One exemplary self-assembling peptide may have a folding group comprising-KYKYKYTYKYKYK- where R is an arginine residue, Y is 1-3 hydrophobic amino acids, and T is 2-6 turn sequence amino acids.

[0210] One exemplary self-assembling peptide may have a folding group comprising -VZVZVZTVZVZVZ- where V is a valine residue, Z is 1-3 hydrophilic amino acids, and T is 2-6 turn sequence amino acids.

[0211] Exemplary self-assembling peptides may have a turn sequence comprising 2-8 turn sequence amino acids, for example 2-5 turn sequence amino acids. The turn sequence amino acids may be selected from proline, for example D-proline and/or L-proline, aspartic acid, and asparagine. In some embodiments, the turn sequence may be (d)PP, (d)PG, or NG.

[0212] Exemplary self-assembling peptides having a turn sequence include VKVRVRVRV(d)PPTRVRVRVKV-NH₂ and VLTKVKTKV(d)PPTKVEVKVLV-NH₂. In the exemplary peptides, the tetrapeptide turn sequence (-V(d)PPT-) was selected to adopt a type II' turn and positioned within the

middle of the peptide sequence. This four-residue turn sequence occupies the i, i+1, i+2 and i+3 positions of the turn. The heterochiral sequence ((d)P (i+1)-P(i+2)) dipeptide was selected for its tendency to adopt dihedral angles consistent with type II' turns. Incorporation of a bulky β -branched residue (valine) at the i position of the turn sequence enforces the formation of a trans prolyl amide bond at the i+1 position. The placement of valine at this position is selected to design against the formation of a cis prolyl bond, which results in β -strands that adopt an extended conformation rather than the intended hairpin. Threonine exhibits a statistical propensity to reside at the i+3 position. Therefore, threonine was selected to be incorporated at this position within the tetrapeptide sequence, which bears a side-chain hydroxyl group capable of hydrogen bonding to the amide backbone carbonyl at the i position, to further stabilize the turn.

[0213] The exemplary folding peptides may be designed to include high propensity β -sheet forming residues flanking the type II' turn sequence. The selection of alternation of hydrophobic and hydrophilic residues along the strands provides an amphiphilic β -sheet when the peptide folds. For example, lysine may be chosen as a hydrophilic residue to provide a side chain pKa value of about 10.5. Side chain amines are generally protonated when dissolved under slightly acid conditions, forming unfavorable electrostatic interactions between β -strands of the hairpin and inhibiting peptide folding and self-assembly. However, as pH is increased to about pH 9, a sufficient portion of the lysine side chains become deprotonated allowing the peptide to fold into an amphiphilic β -hairpin. The electrostatic interactions may be employed to design pH responsiveness of the disclosed peptides.

[0214] While not wishing to be bound by theory, it is believed the amphiphilic β -hairpin is stabilized in the intramolecular folded state by van der Waals contacts between neighboring amino acid side chains within the same hairpin. The formation of intramolecular hydrogen bonds between cross β -strands of the hairpin and the propensity for the turn sequence to adopt at type II' turn may further stabilize the folded conformation. Once in the folded state, the lateral and facial associations of the β -hairpins may be selected to design self-assembly. For example, lateral association of β -hairpins promotes the formation of intermolecular hydrogen bonds and van der Waals contacts between neighboring amino acids.

[0215] Exemplary self-assembling peptides may have a folding group sequence comprising (X)a(Y)b(Z)c-[(d)PP, (d)PG, or NG]-(Z)c(Y)b(X)a, where the turn sequence is (d)PP, (d)PG, or NG, (d)P is a D-proline, X is a charged amino acid, Y is a hydrophobic amino acid, Z is a hydrophobic amino acid or a polar amino acid, and a, b, and c are each independently an integer from 1-10. In certain embodiments, X is independently selected from valine, leucine, isoleucine, phenylalanine, tryptophan, tyrosine, threonine, and combinations thereof. In certain embodiments, Y is independently selected from glutamic acid, serine, alanine, proline, aspartic acid, and combinations thereof. In some embodiments, Z is independently selected from glutamine, glutamic acid, lysine, arginine, and combinations thereof.

[0216] Exemplary self-assembling peptides may have a folding group sequence comprising (Z)c(Y)b(X)a-[(d)PP, (d)PG, or NG]-(X)a(Y)b(Z)c, where the turn sequence is (d)PP, (d)PG, or NG, (d)P is a D-proline, X is a charged

amino acid, Y is a hydrophobic amino acid, Z is a hydrophobic amino acid or a polar amino acid, and a, b, and c are each independently an integer from 1-10. In certain embodiments, X is independently selected from valine, leucine, isoleucine, phenylalanine, tryptophan, tyrosine, threonine, and combinations thereof. In certain embodiments, Y is independently selected from glutamic acid, serine, alanine, proline, aspartic acid, and combinations thereof. In some embodiments, Z is independently selected from glutamine, glutamic acid, lysine, arginine, and combinations thereof.

[0217] Any of the charged, hydrophobic, polar, or amphipathic amino acids disclosed herein may derive one or more of their properties from the composition of the biocompatible solution.

[0218] Hydrophobic amino acids are those which tend to repel water. Hydrophobic amino acids include alanine, valine, leucine, isoleucine, proline, tyrosine, tryptophan, phenylalanine, methionine, and cysteine. The hydrophobic amino acids may be independently selected from alanine, valine, leucine, isoleucine, proline, phenylalanine, methionine, cysteine, and combinations thereof. In some embodiments, the hydrophobic amino acids comprise valine.

[0219] Charged amino acids are those which tend to have an electric charge under the given conditions. Charged amino acids may have side chains which form salt bridges. Charged amino acids include alanine, valine, leucine, isoleucine, proline, phenylalanine, cysteine, arginine, lysine, histidine, aspartic acid, and glutamic acid. The folding group may comprise 2-10 charged amino acids, for example 2-8 charged amino acids.

[0220] The charged amino acids may be positively charged amino acids. The folding group may comprise 2-10 charged amino acids, for example 2-8 charged amino acids. The positively charged amino acids may be independently selected from arginine, lysine, histidine, and combinations thereof. The folding group may comprise 2-8 arginine residues, lysine residues, or a combination of arginine and lysine residues. In some embodiments, the folding group may comprise 6 positively charged residues selected from arginine, lysine, or a combination of arginine and lysine.

[0221] The charged amino acids may be negatively charged amino acids. The folding group may comprise 2-10 negatively charged amino acids, for example, 2-8 negatively charged amino acids. In some embodiments, the negatively charged amino acids may be independently selected from aspartic acid, glutamic acid, and combinations thereof.

[0222] Polar amino acids are those which have an uneven charge distribution. Polar amino acids may tend to form hydrogen bonds as proton donors or acceptors. Polar amino acids include glutamine, asparagine, histidine, serine, threonine, tyrosine, and cysteine. Amphipathic amino acids are those which have both a polar and non-polar component. Amphipathic amino acids may be found at the surface of proteins or lipid membranes. Amphipathic amino acids include tryptophan, tyrosine, and methionine.

[0223] Exemplary self-assembling peptides may have a folding group sequence of any of SEQ ID NOS: 1-23. The self-assembling peptide may have a folding group sequence of SEQ ID NO: 1. In certain embodiments, the self-assembling peptide may have a folding group sequence of SEQ ID NO: 2. The self-assembling peptide may have a folding group sequence of SEQ ID NO: 3. The self-assembling

peptide may have a folding group sequence of SEQ ID NO: 4. The self-assembling peptide may have a folding group sequence of SEQ ID NO: 5.

[0224] The self-assembling peptide may have a folding group sequence of SEQ ID NO: 6. The self-assembling peptide may have a folding group sequence of SEQ ID NO: 7. The self-assembling peptide may have a folding group sequence of SEQ ID NO: 8. The self-assembling peptide may have a folding group sequence of SEQ ID NO: 9. The self-assembling peptide may have a folding group sequence of SEQ ID NO: 10. The self-assembling peptide may have a folding group sequence of SEQ ID NO: 11. The self-assembling peptide may have a folding group sequence of SEQ ID NO: 12. The self-assembling peptide may have a folding group sequence of SEQ ID NO: 13. The self-assembling peptide may have a folding group sequence of SEQ ID NO: 14. The self-assembling peptide may have a folding group sequence of SEQ ID NO: 15. The self-assembling peptide may have a folding group sequence of SEQ ID NO: 16. The self-assembling peptide may have a folding group sequence of SEQ ID NO: 17. The self-assembling peptide may have a folding group sequence of SEQ ID NO: 18. The self-assembling peptide may have a folding group sequence of SEQ ID NO: 19. The self-assembling peptide may have a folding group sequence of SEQ ID NO: 20. The self-assembling peptide may have a folding group sequence of SEQ ID NO: 21. The self-assembling peptide may have a folding group sequence of SEQ ID NO: 22. The self-assembling peptide may have a folding group sequence of SEQ ID NO: 23.

[0225] Exemplary self-assembling peptides which have shear-thinning properties include VKVRVRVRV(d)PP-TRVRVRVKV-NH₂, and VKVRVRVRV(d)PP-TRVEVRVKV-NH₂ (which has a single substitution of glutamic acid at position 15 on the hydrophilic face). The glutamic acid substitution results in a faster rate of gelation of the self-assembling peptide gel in the presence of ionic salts in the biocompatible solution. The negatively charged glutamic acid lowers the overall positive charge of the peptide and enables faster folding and self-assembly.

[0226] Exemplary self-assembling peptides which have shear-thinning properties that can be tuned for net peptide charge include VKVRVRVRV(d)PPTRVEVRVKV-NH₂, and VKVKVKVKV(d)PPTKVEVKVKV-NH₂, (which has an arginine substituted for lysine on the hydrophilic face). The lysine substitution lowers the peptide net charge at physiological pH that allows for better mammalian cell cytocompatibility when compared to peptides with high arginine content (higher net charge). The exemplary peptides are antimicrobial self-assembling peptides.

[0227] Exemplary self-assembling peptides which have shear-thinning properties that can be tuned for peptide gels with faster rate of gelation and increased stiffness include FKFRFRFRV-(d)PPTRFRFRFKF-NH₂, (which has valine substituted for phenylalanine on the hydrophobic face). The phenylalanine substitution increases the hydrophobic face of the peptide that allows for stiffer and faster gelation of peptide gels. The exemplary peptides are antimicrobial self-assembling peptides.

[0228] Exemplary self-assembling peptides which have shear-thinning properties include enantiomer forms of the exemplary sequences listed above, such as an enantiomer form of VKVRVRVRV (d)PPTRVRVRVKV-NH₂, (d) V (d) K (d) V (d) R (d) V (d) R (d) V (d) R (d) V (L) P (d)P (d)

T (d) R (d) V (d) R (d) V (d) R (d) V (d) K (d) V-NH₂, (which has D isoforms of the sequence and an L isoform of P). The isoform substitution may provide control of peptide degradation and increased stability without compromising peptide net charge at physiological pH. The sequence may provide better compatibility with mammalian cells. The peptide may be a complete enantiomer (as shown above) or a partial enantiomer such that any one or more of the amino acids may be an enantiomer of the sequences listed above. The exemplary peptides are antimicrobial self-assembling peptides.

[0229] Other exemplary self-assembling peptides include Ac-VEVSVSVEV(d)PPTEVSVEV EVGGGGRGDV-NH₂ and VEVSVSVEVdPPTEVSVEVEV-NH₂.

[0230] The self-assembling peptide may comprise at least one guanidine moiety. The guanidine moiety may be associated with an organic molecule which makes up part of the peptide chain. In exemplary embodiments, a guanidine group may be incorporated as part of the side chain of an arginine residue. However, the peptide may comprise guanidine moieties which are not associated with arginine residues.

[0231] A guanidine moiety is generally a highly polar group which, when positioned on a cationic peptide, may allow for pairing with hydrophobic and hydrophilic groups forming salt bridges and hydrogen bonds. Such a peptide may display a high capacity to penetrate cell membranes and provide antimicrobial activity. The guanidine moiety may also promote peptide stability by improving peptide folding, physical characteristics and thermal stability of the peptide and/or hydrogel.

[0232] The peptide may generally have 20-50% guanidium content, as measured by number of guanidine groups by total number of amino acid residues of the peptide. For instance, an exemplary peptide sequence having 20 amino acid residues, of which 6 are arginine residues having a guanidine group, has 30% guanidium content. The exemplary peptides may penetrate and disrupt cell membranes.

Properties of the Peptide Hydrogel Preparation

[0233] The preparation may generally comprise the self-assembling peptide in a biocompatible solution. For example, the peptide may be dissolved or substantially dissolved in the biocompatible solution. The preparation may comprise between about 0.1% w/v and about 8.0% w/v of the peptide. The preparation may be formulated for a target indication. For instance, the concentration of the self-assembling peptide may be selected based on a target indication. For example, an exemplary preparation having antimicrobial properties may comprise less than 1.5% w/v of the peptide, for example, between about 0.5% w/v of the peptide and 1.0% w/v of the peptide.

[0234] Exemplary preparations may comprise between about 0.25% w/v and about 6.0% w/v of the peptide, for example, between about 0.5% w/v and about 6.0% w/v of the peptide. When the peptide is purified, the preparation may comprise up to about 6.0% w/v of the peptide. In certain embodiments, the preparation may comprise less than about 3.0% w/v of the peptide, for example, between about 0.25% w/v and about 3.0% w/v of the peptide, between about 0.25% w/v and about 2.0% w/v of the peptide, between about 0.25% w/v and about 1.25% w/v of the peptide, or between about 0.5% w/v of the peptide and about 1.5% w/v of the peptide. The preparation may comprise between about

0.5% w/v and about 1.0% w/v of the peptide, between about 0.7% w/v and about 2.0% w/v of the peptide, or between about 0.7% w/v and about 0.8% w/v of the peptide. For instance, the preparation may comprise about 0.25% w/v, about 0.5% w/v, about 0.7% w/v, about 0.75% w/v, about 0.8% w/v, about 1.0% w/v, about 1.5% w/v of the peptide, about 2.0% w/v, or about 3.0% w/v. In particular embodiments, the preparation may comprise less than about 1.5% w/v of the peptide. The preparation may comprise less than about 1.25% w/v of the peptide or less than about 1.0% w/v of the peptide. In one exemplary embodiment, the preparation may comprise about 0.75% w/v of the peptide.

[0235] After combination with the buffer, the hydrogel may have between about 0.05% w/v and 6.0% w/v of the peptide. For example, the hydrogel may have between about 0.1% w/v, and 6.0% w/v of the peptide, between about 0.25% w/v and 6.0% w/v of the peptide, between about 1.5% w/v and 6.0% w/v of the peptide, between about 0.25% w/v and 3.0% w/v of the peptide, between about 0.25% w/v and 1.0% w/v of the peptide, between about 0.25% w/v and 0.5% w/v of the peptide, or between about 0.3% w/v and 0.4% w/v of the peptide. The peptide preparation and buffer may be combined to form the hydrogel at a ratio of between about 2:1 to 0.5:1 peptide preparation to buffer. In some embodiments, the peptide preparation and buffer may be combined to form the hydrogel at a ratio of about 1:1.

[0236] The peptides in the preparation may be purified. As disclosed herein, "purified" may refer to compositions treated for removal of contaminants. In particular, the purified peptides may have a composition suitable for clinical application. For example, the peptides may be purified to meet health and/or regulatory standards for clinical administration. The peptide may be at least 80% purified, for example, at least 85%, at least 90%, at least 92%, at least 95%, at least 98%, at least 99%, or at least 99.9%.

[0237] In certain embodiments, the peptides may be purified to remove or reduce residual organic solvent content from solid phase synthesis of the peptides. The peptide may comprise less than 20% residual organic solvent by weight. The peptide may comprise less than 15% residual organic solvent by weight. The peptide may comprise less than 10% residual organic solvent by weight. For example, the peptide may comprise less than 8%, less than 5%, less than 2%, less than 1%, or less than 0.1% residual organic solvent by weight. Exemplary organic solvents which may be removed or reduced from the synthesized peptide include trifluoroacetic acid (TFA), acetonitrile, isopropanol, N,N-Dimethylformamide, triethylamine, Ethyl Ether, and acetic acid.

[0238] The purified peptides may be substantially free of Trifluoroacetic acid (TFA). For example, the purified peptides may have less than 1% w/v residual TFA, or between about 0.005% w/v and 1% w/v residual TFA.

[0239] The purified peptide may be substantially free of acetonitrile. In some embodiments, the purified peptide may have less than about 410 ppm residual acetonitrile. The purified peptide may have between about 0.005 ppm and about 410 ppm residual acetonitrile.

[0240] The purified peptide may be substantially free of isopropanol. In some embodiments, the purified peptide may have less than about 400 ppm residual isopropanol. The purified peptide may have less than about 100 ppm residual isopropanol. The purified peptide may have between about 0.005 ppm and 100 ppm residual isopropanol.

[0241] The purified peptide may be substantially free of N,N-Dimethylformamide. In some embodiments, the purified peptide may have less than about 880 ppm residual N,N-Dimethylformamide. The purified may have between about 0.005 ppm and about 880 ppm residual N,N-Dimethylformamide.

[0242] The purified peptide may be substantially free of triethylamine. In some embodiments, the purified peptide may have less than about 5000 ppm residual triethylamine. The purified peptide may have between about 0.005 ppm and about 5000 ppm residual triethylamine.

[0243] The purified peptide may be substantially free of Ethyl Ether. In some embodiments, the purified peptide may have less than about 1000 ppm residual Ethyl Ether. The purified peptide may have between about 0.005 ppm and about 1000 ppm residual Ethyl Ether.

[0244] The purified peptide may be substantially free of acetic acid. For example, the purified peptides may have less than 2% w/v residual acetic acid, for example, less than 1% w/v residual acetic acid, less than 0.5% w/v residual acetic acid, less than 0.1% w/v residual acetic acid, between about 0.0001% w/v and 2% w/v residual acetic acid, or between about 0.005% w/v and 0.1% w/v residual acetic acid.

[0245] In general, the purified peptide and/or biocompatible solution may have properties consistent with regulatory limits defined by the International Council for Harmonization of Technical Requirements for Pharmaceuticals for Human Use (ICH).

[0246] The biocompatible solution of the preparation may refer to a substantially liquid carrier for the peptide. The biocompatible solution may generally be an aqueous solution. The biocompatible solution may comprise water, for example, deionized water. Deionized water may have a resistivity of greater than about 18 MQ2 and a conductivity of less than about 0.056 μ S at 25° C. Deionized water may have a maximum endotoxin specification of 0.03 endotoxin units (EU)/ml and 1 CFU/mL microbial action or less. Deionized water may have a total organic carbon (TOC) concentration of 10 ppb or less. The biocompatible solution may comprise pharmaceutical grade water. Pharmaceutical grade water may have 500 ppb total organic carbon (TOC) or less and 100 CFU/ml microbial action or less. The biocompatible solution may comprise injection grade water. Injection grade water may have a maximum endotoxin specification of 0.25 endotoxin units (EU)/ml and 10 CFU/10 ml microbial action or less. In certain embodiments, the preparation or biocompatible solution may be substantially free of chloride ions.

[0247] The preparation or peptide may comprise counterions. As disclosed herein, a counterion may refer to a charge balancing ion. The preparation or peptide may have an effective amount of counterions to render the preparation substantially electrically neutral. The preparation or peptide may have an effective amount of counterions to render the preparation substantially biocompatible and/or stable. The preparation or peptide may have an effective amount of counterions to control repulsion of anionic or cationic residues of the peptide. The concentration of counterions may be dependent on the peptide sequence and concentration of the peptide and any additives. In exemplary embodiments, the peptide may comprise between 0.1-20% counterions. Additionally, the charge of the counterions may be dependent on the charge of the peptide and any additives. Thus, the counterions may be anions or cations. In general, the coun-

terions may be cytocompatible. In certain embodiments, the counterions may be biocompatible. For instance, the counterions may comprise acetate, citrate, ammonium, fluoride, or chloride. In other embodiments, the preparation or peptide may be substantially free of chloride counterions.

[0248] In exemplary embodiments, the preparation or peptide may comprise an effective amount of acetate counterions. In particular, preparations having a peptide concentration which comprises residual TFA may have an amount of acetate counterions sufficient to balance the residual TFA. Briefly, TFA is commonly used to release synthesized peptides from solid-phase resins. TFA is also commonly used during reversed-phase HPLC purification of peptides. However, residual TFA or fluoride may be toxic and undesirable in peptides intended for clinical use. Furthermore, TFA may interact with the free amine group at the N-terminus and side chains of positively charged residues (for example, lysine, histidine, and arginine). The presence of TFA-salt counterions in the peptide preparation may be detrimental for biological material and may negatively affect the accuracy and reproducibility of the intended peptide activity.

[0249] TFA-acetate salt exchange by acetate or hydrochloride may be employed to counteract some or all of the negative effects of TFA described above. The inventors have determined the acetate counterion is surprisingly well suited for maintaining biological activity of the peptide preparation and for controlling solubility of the peptide and charge for self-assembly of the peptide. Furthermore, acetic acid ($pK_a=4.5$) is weaker than both trifluoroacetic acid (pK_a about 0) and hydrochloric acid ($pK_a=-7$). Acetate counterions may additionally control pH of the peptide preparations to be physiologically neutral.

[0250] The preparation may have variable hydrogelation kinetics. In accordance with certain embodiments, the hydrogelation kinetics of the preparation may be designed for a particular mode of administration. The preparation may be administered as a liquid. The preparation may be administered as a solid or semi-solid. The preparation may be administered as a gel. The preparation may be administered as a combination of hydrogel suspended in a liquid. The preparation may have a variable apparent viscosity. For instance, the preparation may have an apparent viscosity effective to allow injection under the conditions of administration. In certain embodiments, the preparation may have an apparent viscosity which decreases with increasing shear stress.

[0251] The preparation may be configured to reversibly self-assemble and disassemble in response to applied stress, for example, applied mechanical force. The solid or gel preparation may become disrupted with increasing applied stress, to be later restored once the applied stress is reduced. The solid or gel may become fluid in response to applied stress, for example, during delivery through a delivery device. The peptide may be capable of undergoing sequential phase transitions in response to applied stress. The peptide may be capable of recovering after each one or more sequential phase transitions.

[0252] The preparation may be configured to reversibly self-assemble and disassemble responsive to at least one of change in temperature, change in pH, contact with an ion chelator, dilution with a solvent, applied sound wave, lyophilization, vacuum drying, and air drying. The administered fluid may conform to tissue voids before reforming as a solid or gel. Thus, the solid or gel preparation may be

injectable, flowable, or sprayable under the appropriate shear stress. Once administered, the preparation may be restored to a solid or gel form, substantially conforming to the target site. The formation may occur within less than a minute, about one minute, less than about 2 minutes, less than about 3 minutes, less than about 5 minutes, or less than about 10 minutes. The formation may occur within about one minute, less than about 30 seconds, less about 10 seconds, or about 3 to 5 seconds.

[0253] The peptide may be purified. For example, the peptide may be lyophilized. As shown in FIG. 9, net charge may be quantified as a function of pH value. The exemplary peptide measured in FIG. 9 is an arginine-rich peptide having two lysine residues. The exemplary peptide of FIG. 9 has a net charge of +9 at a pH of 7. Other peptides are within the scope of the disclosure. For example, the purified peptide may have a net charge between -9 to +11 at pH 7, for example, -7 to +9 at pH 7. As disclosed herein, "net charge" may refer to a total electric charge of the peptide as a biophysical and biochemical property, typically as measured at a pH of 7.

[0254] The purified peptide may have a net charge of from -7 to +11 at pH 7. In some embodiments, the peptide may have a net charge of from +2 to +9, for example, +5 to +9 or +7 to +9. The purified peptide may have a charge of about +5, +6, +7, +8, +9, +10, or +11 at pH 7. Exemplary peptides having a charge of +5 to +9 include VLTKVTKV(d)PPTKVEVKVLV, VKVRVRVRV (d)PPTRVRVRVKV, and VKVRVRVRV (d)PPTRVEVRVKV. In other embodiments, the purified peptide may be substantially neutral. In other embodiments, the peptide may have a net negative charge. An exemplary peptide having a net negative charge is VEVSVSVEV(d)PPEVSVEVEV. As shown in FIG. 10, a single substitution of glutamic acid in the peptide sequence may alter net peptide charge from +7 (top panel) to +9 (bottom panel) at pH 7, as well as alter isoelectric point from 11.45 to 14. Net charge may be selected by peptide design. Design of electrostatic charge in the peptide hydrogel may allow control of charge interaction with cell membrane and proteins.

[0255] The peptide may be designed to have a charge that adsorbs and/or promotes deactivation of proteins at a target site of administration. For instance, positively charged peptide hydrogels may promote adsorption of negatively and neutrally charged molecules such as small molecules, proteins, and extravesicular membranes. Negatively charged peptide hydrogels may promote adsorption of positively and neutrally charged molecules such as small molecules, proteins, and extravesicular membranes. Furthermore, the peptide may be designed to have regions of positive, neutral, or negative charge, to varying degrees. In certain embodiments, the peptide charge may be designed such that when placed into a rich solution of charged molecules, the peptide may soak out or absorb the molecules into the hydrogels attaching the molecules to the peptides by adsorption. FIG. 3 is a microscopy image showing negatively charged Trypan blue adsorbed on a positively charged hydrogel.

[0256] The purified peptide may have greater than 70% w/v, greater than 80% w/v, or greater than 90% w/v nitrogen, for example, between 70% w/v and 99.9% w/v nitrogen.

[0257] The purified peptide may have a bacterial endotoxin level of less than about 10 EU/mg, for example, less than about 5 EU/mg, less than about 2 EU/mg, or less than about 1 EU/mg. In other embodiments, the purified peptide

may have an endotoxin level of between about -0.010 to -0.015 EU/ml. For instance, the purified peptide may have an OD at 410 nm of between 0.004 to 0.008, for example, about 0.006. The peptide hydrogel may have an OD at 410 nm of between 0.010 to 0.020, for example, about 0.015. In some embodiments, the purified peptide and/or preparation may be substantially free of endotoxins.

[0258] The purified peptide in the biocompatible solution may have a water content of between about 1% w/v and about 20% w/v, for example, at least about 10% w/v, or less than about 15% w/v.

[0259] The purified peptide may have an isoelectric point of between about 7-14. For example, the purified peptide may have an isoelectric point of about 7, 8, 9, 10, 11, 12, 13, or 14.

[0260] The purified peptide may be configured to self-assemble into a hydrogel having a shear modulus of between about 2 Pa to 3500 Pa as determined by rheology testing. For example, the purified peptide may self-assemble into a hydrogel having a shear modulus of greater than 100 Pa, between 100 Pa and 3500 Pa, between 100 Pa and 3000 Pa, between 2 Pa and 1000 Pa, or between 2 Pa and 500 Pa. For example, a formulation having 0.75% w/v peptide may have a shear modulus of between about 2 Pa and 500 Pa. A formulation having 1.5% w/v peptide may have a shear modulus of between about 100 Pa and 3000 Pa. A formulation having 3.0% w/v peptide may have a shear modulus of between about 1000 Pa and 10000 Pa. Thus, shear modulus of the hydrogel may be controlled by selection of peptide concentration in the formulation.

[0261] The peptide may be designed to adopt a predetermined secondary structure. For example, the peptide may be designed to adopt a β -hairpin secondary structure, as previously described. The predetermined secondary structure may comprise a structure preselected from at least one of a B-sheet, an α -helix, and a random coil. In exemplary embodiments, the hydrophobic amino acid residues (for example, quantity, placement, and/or structure of the hydrophobic amino acid residues) may be selected to self-assemble the peptide into a polymer having a majority of β -sheet structures. In particular embodiments, the hydrophobic amino acid residues may be selected to control stiffness of the hydrogel. For example, an amount and type of hydrophobic amino acid residues may be selected to control stiffness of the hydrogel.

[0262] In some embodiments, an external stimulus such as temperature, change in pH, light, and applied sound wave may be used to control and promote preferential secondary structure formation of the self-assembling peptide. Control of the secondary structure formation may enhance biological, biophysical, and chemical therapeutic functions of the peptide. For example, higher cell membrane penetration of self-assembling peptides may be achieved by exposing β -hairpin peptides to high pH (for example, at least pH 9) or high temperatures (for example, at least 125° C.) or low temperatures (for example, 4° C. or lower). The result is hydrogels with a peptide secondary structure having a majority β -sheet or α -helix formation.

[0263] The peptide may be designed to give the preparation shear-thinning properties. In particular, the peptide may be designed to be injectable. For instance, the peptide may be designed to be an injectable solid or gel by employing shear-thinning kinetics. The preparation, in the form of a solid or gel prior to application, may be configured to

shear-thin to a flowable state under an effective shear stress applied during administration by the delivery device. In some exemplary embodiments, the solid or gel may shear-thin to a flowable state during injection or topical application with a syringe. Other modes of administration may be employed. The solid or gel may shear-thin to a flowable state during endoscopic administration. The solid or gel may be configured to shear-thin to flow through an anatomical lumen, for example, an artery, vein, gastrointestinal tract, bronchus, renal tube, genital tract, etc. In some embodiments, the shear thinning properties may be employed during transluminal procedures. The peptide may be designed to be sprayable. For example, the peptide may be designed for administration as a spray or other liquid droplet, for example, other propelled liquid droplet, by employing shear-thinning kinetics, as previously described.

[0264] The shear-thinning kinetics of the hydrogel may be engineered by altering the net charge of the peptides. In some embodiments, the net charge may be altered by controlling one or more of the presence or absence of cationic particles or peptides, the presence or absence of anionic particles or peptides, buffers, salts, peptide concentration, peptide purity, and the presence or absence of peptide counterions. In particular, shear-thinning may be controlled by altering the peptide purity to achieve the desired shear-thinning kinetics. The net charge of the peptide may be positive. The net charge of the peptide may be negative.

[0265] Shear-thinning may be induced by mechanical agitation to the hydrogel or environmental stimulus. Mechanical agitation may be induced, for example, through delivery or sonication mixing. Environmental stimulus may be induced by addition of heat, light, ionic agents, chelator agents, buffers, or proteins, or altering pH level.

[0266] Thus, the preparations may be substantially flowable. The methods may include dispensing the preparation through a cannula or needle. The methods may include conformally filling wound beds of any size and shape. The peptide hydrogels may have shear-thinning mechanical properties. The shear-thinning mechanical properties may allow the gel network to be disrupted and become a fluid during administration, for example, during injection from a needle or administration with a spray nozzle. When the applied stress ceases, the gel network may reform and the elastic modulus may be restored within a predetermined period of time, for example, several minutes. The shear-thinning peptide hydrogels may be employed to protect cells from damage during injection, showing an improved viability over direct injection in saline or media. The shear-thinning hydrogel may display non-Newtonian fluid flow, which may allow for effective mixing of excipients, for example, within minutes to a couple hours. In some embodiments, dyes, small molecules, and large molecules may be substantially homogeneously dispersed within the hydrogel in less than 120 minutes, for example, between 30-120 minutes.

[0267] The peptide may self-assemble into a translucent hydrogel. In some embodiments, the peptide may self-assemble into a substantially transparent hydrogel. The transparency of the hydrogel may enable a user or healthcare provider to view surrounding tissues through the hydrogel. In exemplary embodiments, a surgical procedure may be performed without substantial obstruction of view by the hydrogel. The hydrogel may have at least about 70%, at least

about 80%, at least about 90%, at least about 95%, at least about 99%, or about 100% light transmittance. The hydrogel may be colorless. The light transmittance and color of the hydrogel may be engineered by tuning the sequence of the peptide and/or the composition of the preparation or solution. As shown in the graphs of FIG. 8, transparency of the peptide hydrogels may be quantified by absorbance measurements. The exemplary peptide hydrogels measured in FIG. 8 are substantially transparent.

[0268] In some embodiments, the preparation may include a dye. The dye may be a food-grade dye or a pharmaceutical-grade dye. The dye may be cytocompatible. The dye may be biocompatible. In general, the dye may assist the user or healthcare provider to view the hydrogel after application. The preparation may include an effective amount of the dye to provide a desired opacity of the hydrogel. The hydrogel may comprise an effective amount of the dye to have a light transmittance of less than about 60%, less than about 50%, less than about 40%, less than about 30%, less than about 20%, or less than about 10%. The hydrogel may be substantially opaque when including the dye.

[0269] The peptide may self-assemble into a substantially ionically-crosslinked hydrogel. "Ionic crosslinkage" may refer to ionic bonds between peptides to form secondary structure proteins and/or between secondary structure proteins that form the hydrogel tertiary structure. The shear-thinning properties of the hydrogel may be enabled by physical crosslinks, allowing ionic bonds to be broken and reformed. In accordance with certain embodiments, the hydrogel is formed of a majority of ionic crosslinks. For example, more than 50%, more than 60%, more than 70%, more than 80%, more than 90%, more than 95%, more than 99%, or substantially all of the physical crosslinks of the formed hydrogel may be ionic in nature.

[0270] The preparation and/or assembled hydrogel may be designed to have a substantially physiological pH level. The preparation or hydrogel may have a pH level of between about 4.0 and 9.0. In some embodiments, the preparation or hydrogel may have a pH level of between about 7.0 and 8.0. The preparation or hydrogel may have a pH level of between about 7.3 and 7.5. The substantially physiological pH may allow administration of the preparation at the time of gelation. In some embodiments, the hydrogel may be prepared at a point of care. The methods may comprise mixing the preparation with a buffer configured to induce self-assembly, optionally agitating the mixture, and administering the preparation or hydrogel at a point of care. The administration may be topical or parenteral, as described in more detail below.

[0271] The peptide may be designed to self-assemble in response to a stimulus. The stimulus may be an environmental stimulus, e.g., change in temperature (e.g., application of heat), exposure to light, change in pH, applied sound waves, or exposure to ionic agents, chelator agents, or proteins. The stimulus may be mechanical agitation, e.g., induced through delivery, sonication, or mixing. In some embodiments, the methods may comprise administering the preparation as a liquid. The methods may comprise administering the preparation as a gel. The methods may comprise administering the preparation as a solid or semi-solid.

[0272] In some embodiments, the preparation may be designed to self-assemble after a lapsed period of time. For example, the preparation may be designed such that the peptide is configured to begin self-assembly in less than

about 5 minutes, in less than about 3 minutes, in less than about 2 minutes, in less than about 30 seconds, in less than about 10 seconds, or in less than about 3 seconds. In certain embodiments, the preparation may be designed such that the peptide is configured to self-assemble, i.e. be substantially self-assembled, within about 60 minutes, within about 30 minutes, within about 15 minutes, within about 10 minutes, within about 5 minutes, within about 3 minutes, within about 2 minutes, within about 30 seconds, within about 10 seconds, within about 5 seconds, or within about 3 seconds. The preparation may have a composition configured to control timing of self-assembly. For example, the preparation may have a composition configured for timed release of ionic agents or pH-altering agents. In certain embodiments, the sequence or structure of the peptide may be designed to control self-assembly of the peptide.

[0273] In some embodiments, the methods may comprise combining the preparation with a buffer. The “buffer” may refer to an agent configured to induce gelation, prior to, subsequently to, or concurrently with administration of the preparation to the subject. Thus, in some embodiments, the preparation may comprise a buffer. For example, the preparation may comprise or be combined with up to about 1000 mM of the buffer. The buffer may comprise an effective amount of ionic salts and a buffering agent, for example, to induce gelation and/or provide desired properties. For example, the buffer may be formulated to control or maintain pH of the preparation.

[0274] In particular embodiments, the buffer may have an effective amount of ionic salts to control stiffness of the hydrogel. The “ionic salt” may refer to a compound which dissociates into ions in solution. The buffer may comprise between about 5 mM and 400 mM ionic salts. For example, the buffer may comprise between about 5 mM and 200 mM ionic salts, between about 50 mM and 400 mM ionic salts, between about 50 mM and 200 mM ionic salts, or between about 50 mM and 100 mM ionic salts. The ionic salt may be one that dissociates into at least one of sodium, potassium, calcium, magnesium, iron, ammonium, pyridium, quaternary ammonium, chloride, citrate, acetate, and sulfate ions. The ionic salts may comprise sodium chloride, ammonium chloride, magnesium chloride, potassium chloride, calcium chloride, ammonium sulfate, magnesium sulfate, sodium sulfate, potassium sulfate, calcium sulfate, sodium bicarbonate, and combinations thereof.

[0275] In exemplary embodiments, the buffer may comprise between about 1 mM and about 200 mM sodium chloride. For example, the buffer may comprise between about 10 mM and about 150 mM sodium chloride, for example between about 50 mM and about 100 mM sodium chloride.

[0276] The buffer may comprise counterions. The buffer may have an effective amount of counterions to render the hydrogel substantially electrically neutral. The buffer may have an effective amount of counterions to induce self-assembly of the peptide. The concentration of counterions may be dependent on the composition of the peptide preparation. Additionally, the charge of the counterions may be dependent on the charge of the peptide preparation. Thus, the counterions may be anions or cations. In general, the counterions may be cytocompatible. In certain embodiments, the counterions may be biocompatible. For instance, the counterions may comprise acetate or chloride. In other

embodiments, the biocompatible solution may be substantially free of chloride counterions.

[0277] The buffer may comprise from about 1 mM to about 150 mM of a biological buffering agent. For example, the buffer may comprise from about 1 mM to about 100 mM of a biological buffering agent, from about 1 mM to about 40 mM of a biological buffering agent, or from about 10 mM to about 20 mM of a biological buffering agent. The biological buffering agent may be selected from Bis-tris propane (BTP), 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid (HEPES), Dulbecco's Modified Eagle Medium (DMEM), tris (hydroxymethyl) aminomethane (TRIS), 2-(N-Morpholino) ethanesulfonic acid hemisodium salt, 4-Morpholineethanesulfonic acid hemisodium salt (MES), 3-(N morpholino) propanesulfonic acid (MOPS), and 3-(N-morpholino) propanesulfonic acid (MOBS), Tricine, Bicine, tris (hydroxymethyl)methylamino) propanesulfonic acid (TAPS), N-(2-Acetamido)-2-aminoethanesulfonic acid (ACES), β -Hydroxy-4-morpholinepropanesulfonic acid, 3-Morpholino-2-hydroxypropanesulfonic acid (MOPSO), (N,N-bis(2-hydroxyethyl)-2-aminoethanesulfonic acid) (BES) and combinations thereof.

[0278] Other biological buffering agents are within the scope of the disclosure.

[0279] In exemplary embodiments, the buffer may comprise from about 1 mM to about 150 mM of BTP. The buffer may comprise from about 10 mM to about 100 mM BTP, for example, from about 10 mM to about 50 mM BTP, from about 10 mM to about 40 mM, from about 20 mM to about 40 mM, or from about 20 mM to about 40 mM.

[0280] The buffer may additionally comprise at least one of water, an acid, and a base. The acid and/or base may be added in an amount effective to control pH of the buffer to be a substantially physiological pH. In other embodiments, the buffer may be acidic, alkali, or substantially neutral. The buffer may be selected to control pH of the hydrogel and maintain a desired pH at the target site. For example, to control pH of the hydrogel to be a substantially physiological pH at the target site. Thus, the properties of the buffer may be selected based on the target site. The buffer may have additional properties as selected, for example, net charge, presence or absence of additional proteins, etc. The buffer may additionally comprise one or more minerals.

[0281] The preparation may further comprise an effective amount of a mineral clay. The preparation may comprise between about 0.1% w/v to about 20% w/v of the mineral clay. For example, the preparation may comprise 0.75% w/v, 1.5% w/v, 2% w/v, 3% w/v, 4% w/v, 8% w/v, 10% w/v, or 20% w/v of the mineral clay. The amount of the mineral clay may be effective to provide desired rheological properties for the target site of application. The amount of the mineral clay may be effective to form a film. The mineral clay may be natural or synthetic. The mineral clay may comprise at least one of laponite and montmorillonite. In some embodiments, the preparation may comprise from a 1:1 to 1:2 ratio (w/v) of the peptide to mineral clay. For example, the ratio of peptide to mineral clay in the preparation may be 1:1, 3:4, 3:8, or 1:2 (w/v).

[0282] The preparation may be formulated for a target indication. For instance, the preparation may be formulated for treatment of a microbial infection or inhibition of proliferation of a microorganism, such as a pathogenic microorganism. The preparation may be formulated for treatment of a fungal infection or inhibition of proliferation of a fungal

organism. The preparation may be formulated for cell culture and/or cell delivery. The preparation may be formulated for treatment or inhibition of a wound, such as a chronic wound, or biofilm. The preparation may be formulated by engineering the peptide as described in more detail below. The preparation may be formulated by selecting the bio-compatible solution and/or additives.

[0283] In certain embodiments, the preparation may be formulated for a combination treatment. The preparation may include at least one active agent configured to provide a combination treatment. In some embodiments, the preparation may exhibit synergistic results with combination of the active agent. The active agent may be, for example, an antibacterial composition, an antiviral composition, an antifungal composition, an anti-tumor composition, an anti-inflammatory composition, a hemostat, a cell culture media, a cell culture serum, an anti-odor composition, an analgesic, local anesthetic, or a pain-relief composition. The preparation may be formulated for administration in conjunction with one of the aforementioned compositions. The preparation may be formulated for simultaneous or concurrent combination administration. The preparation may be formulated for sequential combination administration.

[0284] In some embodiments, the preparation and/or hydrogel may be designed to be thermally stable between -20°C . and 150°C ., between -20°C . and 125°C ., between -20°C . and 100°C ., between 2°C . and 125°C ., and between 37°C . and 125°C . As disclosed herein, “thermal stability” refers to the ability to withstand temperature treatment without substantial degradation, loss of biological activity, or loss of chemical activity. The graphs of FIGS. 7A-7B show peptide aggregation as measured by static light scattering (SLS) at 266 nm of exemplary peptides as a function of temperature. The exemplary peptides include arginine, lysine, valine, threonine, and proline residues. As shown in the graphs of FIGS. 7A-7B, the peptide hydrogels and peptides are thermostable as a function of temperature.

[0285] In certain embodiments, the preparation and/or peptide may be mechanically stable. For instance, the preparation may be shear thinned or sonicated. The preparation may be sonicated without substantial degradation, loss of biological or chemical activity. The preparation may be shear thinned without substantial degradation, loss of biological or chemical activity.

[0286] In certain embodiments, the preparation and/or peptide may be sterile or sterilized. The preparation and/or peptide may be sterilized by autoclave sterilization. During autoclave sterilization, the preparation and/or peptide may be heated to a temperature of between 120°C . to 150°C ., for example, up to 125°C ., up to 135°C ., or up to 150°C . The preparation and/or peptide may be held at autoclave temperature for at least about 2 minutes, for example, between about 2-20 minutes or between about 10-160 minutes. The autoclave sterilization may be sufficient to sterilize at least about 90%, 95%, 99%, 99.9%, 99.99%, 99.999%, or 100% of any pathogenic microorganism. The preparation and/or peptide may remain stable during and after autoclave sterilization. For instance, the preparation and/or peptide may remain physically, chemically, biologically, and/or functionally stable after autoclave sterilization.

[0287] In certain embodiments, the preparation and/or peptide may be pasteurized. During pasteurization, the preparation and/or peptide may be heated to a temperature of between 50°C . to 100°C ., for example, up to 60°C ., up to

70°C ., or up to 100°C . The preparation and/or peptide may be held at pasteurization temperature for at least about 15 seconds, for example, between about 1-30 minutes or between about 3-15 minutes. The pasteurization may be sufficient to sterilize at least about 90%, 95%, 99%, 99.9%, 99.99%, or 99.999% of any pathogenic microorganism.

[0288] In certain embodiments, the preparation may be sterilized by ultra-high temperature

[0289] (UHT) or high temperature/short time (HTST) sterilization. During UHT or HTST sterilization, the preparation and/or peptide may be heated to a temperature of between 100°C . to 150°C ., for example, up to 130°C ., up to 140°C ., or up to 150°C . The preparation and/or peptide may be held at UHT or HTST temperature for at least about 15 seconds, for example, between about less than 1 minute to about 6 minutes, for example, between about 2-4 minutes. The UHT or HTST sterilization may be sufficient to sterilize at least about 90%, 95%, 99%, 99.9%, 99.99%, or 99.999% of any pathogenic microorganism.

[0290] In certain embodiments, the sterilization or pasteurization may be terminal. Terminal sterilization or pasteurization may refer to treatment of the preparation in a sealed end-use package.

[0291] The preparation and/or peptide may be stable during and after heat treatment. As disclosed herein, stability during and after heat treatment, for example, autoclave sterilization, may refer to reduced or inhibited degradation, biological activity, and chemical activity. For instance, the preparation and/or peptide may be heat treated without degradation, a loss of biological activity, or a loss of chemical activity. Biological activity may refer to any bio-active property of the peptide disclosed herein. In some embodiments, biological activity may refer to antimicrobial activity. Chemical activity may refer to any chemical or physicochemical property of the peptide disclosed herein. In some embodiments, chemical activity may refer to the ability to self-assemble and or shear-thinning properties of the peptide disclosed herein. Thus, the preparation and/or peptide may be heat treated without loss of antimicrobial activity, self-assembly, or shear-thinning properties. In certain embodiments, heat treatment may enhance one or more biological activity or chemical activity of the peptide and/or preparation. For example, heat treatment may enhance antimicrobial activity, self-assembly, or shear-thinning properties of the peptide or preparation.

[0292] The preparation may be sterile. For example, the preparation may remain substantially sterile without the addition of a preservative. The preparation may be substantially sterile without gamma irradiation treatment. The preparation may be substantially sterile without electron beam treatment.

[0293] The preparation may have a predetermined shelf-life. “Shelf-life” may refer to the length of time for which the preparation may remain stable and/or maintain efficacy after storage under the given conditions. The preparation and/or hydrogel may have a shelf-life of at least about 1 year at a temperature between -20°C . and 150°C . For instance, the preparation and/or hydrogel may have a shelf-life of at least about 1 year at room temperature (between about 20°C . and 25°C .). The preparation and/or hydrogel may have a shelf-life of at least about 2 years, about 3 years, about 4 years, about 5 years, or about 6 years at room temperature. The preparation and/or hydrogel may be stable at a pressure of up to about 25 psi, for example, up to about 15 psi.

[0294] The peptide may be capable of self-assembly at a temperature between 2° C. and 40° C. For example, the peptide may be capable of self-assembly in an environment having a temperature between 2° C. and 20° C., between 20° C. and 25° C., or between 36° C. and 40° C.

[0295] The peptide may be substantially unassembled at temperatures higher than 40° C. For instance, the peptide preparation may be substantially liquid at temperatures between 40° C. and 150° C. The peptide preparation may be substantially liquid and thermally stable at temperatures between 40° C. and 125° C. or up to 150° C. Temperature may be controlled for handling of the preparation. For example, the preparation may be heated to a temperature greater than 40° C. for packaging, handling, and/or administration in a liquid state.

[0296] The preparation may be formulated for a desired route of administration. For example, the preparation may be formulated for topical or parenteral administration. In particular, the preparation may be engineered to have a viscosity appropriate for topical administration or parenteral administration. Preparations for topical administration may be formulated to withstand environmental and mechanical stressors at the site of administration or target site. Preparations for parenteral administration may be formulated to reduce migration from the site of administration or target site. In other embodiments, preparations for parenteral administration may be formulated to trigger migration from a site of administration to the target site. The preparation may be formulated for administration by a particular delivery device. For example, the preparation may be formulated for administration by spray, dropper, or syringe. The preparation may be formulated for administration by injection or catheter.

[0297] Table 1 includes the analytical characterization of three exemplary peptide preparation samples. The exemplary peptides have arginine-rich sequences comprising two lysine amino acid residues. The values were detected by conventional detection methods. Components indicated “N.D.” were below detection limit. Peptide purification, residual solvents, peptide content, and water content, may be selected to control antimicrobial activity and cell membrane disruption potential of the hydrogels.

TABLE 1

Exemplary Peptide Preparations														
#	Purity (%)	Acetate (%)	TFA (%)	Water (%)	Net Peptide (%)	Residual Solvents								
						Isopropanol (ppm)	Methanol (ppm)	Acetonitrile (ppm)	Triethylamine (ppm)	N,N-Dimethyl-formamide (ppm)	Ethyl ether (ppm)	DCM	DIC	MPS
1	97.9	19.3	N.D.	6.5	78.6	81	28	17	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.
2	93.7	16.3	N.D.	7.1	79.3	63	28	18	56	116	N.D.	N.D.	N.D.	N.D.
3	96.0	16.9	N.D.	6.2	76.1	N.D.	N.D.	4.6	2.9	N.D.	N.D.	N.D.	N.D.	N.D.

[0298] The purified peptide and hydrogel may be substantially endotoxin free without addition of a preservative or sterilization, as shown in Table 2. Thus, in some embodiments, the peptide preparation may be substantially free of a preservative.

TABLE 2

Endotoxin Levels of Different Compositions		
	OD at 410 nm	Endotoxin level (EU/mL)
Endotoxin free water	−0.000333333	−0.016148109
RO water	3.436666667	1.267607913
Tap water	3.463666667	1.2776927
Buffer solution	−0.056	−0.036940201
Peptide	0.006	−0.013782542
Peptide hydrogel	0.015	−0.010420946

[0299] The self-assembling peptide hydrogel

[0300] The preparations disclosed herein may be provided to self-assemble into a hydrogel having preselected properties. The polymeric hydrogel may have a substantially physiological pH. In general, the polymeric hydrogel may have a pH of between 4.0 and 9.0, for example, between 7.0 and 8.0, between 7.2 and 7.8, or between 7.3 and 7.5.

[0301] The polymeric hydrogel may be substantially transparent. For example, the polymeric hydrogel may be substantially free of turbidity, for example, visible turbidity. Visible turbidity may be determined by macroscopic and microscopic optical imaging. The polymeric hydrogel may be substantially free of peptide aggregates (peptide clusters), for example, visible peptide aggregates. Visible peptide aggregates may be determined by static light scattering (SLS) and UV-VIS testing. “Transparency” may refer to the hydrogel’s ability to pass visible light. The substantially transparent hydrogel may have UV-VIS light absorbance of between about 0.1 to 3.0+1.5 at a wavelength of between about 205 nm to about 300 nm.

[0302] The assembled polymeric hydrogel may have a nano-porous structure. The polymeric hydrogel may be hydrated or substantially saturated. In some embodiments, the hydrogel may have between 90% w/v and 99.9% w/v aqueous solution, for example, between 92% w/v and 99.9% w/v or between 94% w/v and 99.9% w/v. The nano-porous structure may be selected to be impermeable to a target microorganism. Thus, the hydrogel may be used to encapsulate a target microorganism or to maintain the target site free from the target microorganism. The nano-porous structure may be selected to allow gaseous exchange at the target

site. The polymeric hydrogel may have a nano-porous structure having a pore size of between 1 nm and 1000 nm, as selected (e.g., based on a target microorganism, target cell, or desired functionality). The polymeric hydrogel may have a fibril width of between 1 nm and 100 nm, as selected.

[0303] The hydrogel may generally be cationic in nature. In other embodiments, the hydrogel may be anionic in nature. In yet other embodiments, the hydrogel may be blended to contain multi-domains of cationic and/or anionic components. The hydrogel may be designed to have a preselected charge. The self-assembling peptide hydrogel disclosed herein may be tunable to biological functionality that supports the viability and function of transplanted therapeutic cells, to exhibit shear-thinning mechanical properties that allow easy and rapid administration in an intra-operative setting, to exhibit antimicrobial properties to control wound bioburden, to exhibit antiviral properties to treat or inhibit viral infection, and/or to exhibit antifungal properties to treat or inhibit fungal infection.

[0304] In particular, the peptide sequence and structure may include peptide functional groups that form nanofibers, which further self-assemble to form macromolecular structures (FIG. 1A-1B). The peptides may self-assemble in response to an environmental stimulus. The peptides may self-assemble in the presence of substantially physiological buffers, such as media or saline. The peptide hydrogels may assemble into an extracellular scaffolding matrix that is similar to native fibrillar collagen (FIG. 1A-1B). Schematics of gel matrix self-assembly and an exemplary nanostructure are shown in FIGS. 1A-1B. As shown in FIG. 1A, single peptide nanofibers self-assemble into a gel when ionic buffer is added. FIG. 1A includes a TEM image demonstrating that the nanostructure and pore size of the peptide gel look similar to native ECM (collagen). FIG. 1B includes a schematic drawing of an intra-operative mixing device for mixing a cell suspension with peptide gel matrix. A schematic SEM image in FIG. 1B of the cell-laden matrix demonstrates the exemplary concept of cells in matrix.

[0305] The peptide may be engineered by design to self-assemble into a hydrogel which is substantially biocompatible. The peptide may be engineered by design to self-assemble into a hydrogel that is cell friendly. In certain embodiments, the cell-friendly polymeric hydrogel may be non-inflammatory, and/or non-toxic. The cell-friendly polymeric hydrogel may be substantially biodegradable. The peptide may be engineered by design to be substantially antimicrobial, antiviral, and/or antifungal.

[0306] The short peptides and/or peptide functional groups may be produced synthetically. Thus, the peptides may provide ease of manufacturing, scale-up, and quality control. In general, the peptides may be manufactured without the use of plant or animal expression systems. The peptides may be substantially free of naturally occurring endotoxins and disease-transmitting pathogens. In addition, the peptide sequence and functional groups may be tuned, allowing a versatility in control and design of the assembled hydrogel, including with respect to physical and chemical properties, such as biodegradation, mechanical properties, and biological activity.

[0307] The peptide may have a functional group engineered for a target indication. For instance, the peptide may have a bioactive functional group. The target indication may be tissue engineering of a target tissue. The target indication may include, for example, cell culture, cell delivery, wound healing, and/or treatment of biofilm. Thus, the peptide may be engineered by design to self-assemble into a hydrogel which is substantially biocompatible. The peptide functional group may have between about 3 and about 30 amino acid residues. For example, the peptide functional group may have between about 3 and about 20 amino acid residues. The peptide functional group may have a sequence selected from

RGD, IKVAV, YIGSR, LKKTETQ, SNKPGVL, PKPQQFFGLM, GKLTWQELYQLKYKGI, and GGG.

[0308] In some embodiments, the peptide may include a modification selected from a linker and a spacer. Peptide “linkers” may generally refer to short amino acid sequences included to link multiple domains of the peptide. Peptide “spacers” may generally refer to amino acid sequences positioned to link and control the spatial relationship of the multiple domains of the assembled protein. The linker or spacer may be hydrophobic or hydrophilic. The linker or spacer may be rigid or flexible. Exemplary spacers include aminohexanoic acid (Ahx) (hydrophobic) and poly (ethylene) glycol (PEG) (hydrophilic). Glycine rich spacers are generally flexible.

[0309] Exemplary bioactive functional groups include laminin, bone marrow homing, collagen (e.g., I, II, and VI), bone marrow purification, and RGD/fibronectin types. Exemplary bioactive functional groups include VEGF, Substance P, Thymosin Beta, Cardiac

[0310] Homing Peptide, Bone Marrow Homing Peptide, Osteopontin, and Osteogenic peptide. Exemplary bioactive functional groups include those in Tables 3-5 below.

TABLE 3	
Exemplary Bioactive Functional Groups	
ECM Motif	ECM Type
-GGPDSGR	Laminin
-GGSDPGYIGSR	Laminin
-GGSKPPGTSS	Bone marrow homing
-GGDGEA	Collagen II
-GGPFSSTKT	Bone marrow homing
-GGFLGFPT	Bone marrow purification
-GGRGDS	RGD/fibronectin
-GGIKVAV	Laminin
-GGFPDERGVEGPGP	Collagen I
-GGPRGDSGYRGDSF	Collagen VI

TABLE 4	
Exemplary Bioactive Functional Groups	
Protein/Peptide	Bioactive Motif
VEGF	GKLTWQELYQLKYKGI
Substance P	RPKPQQFFGLM
Thymosin Beta	SNKPG
Thymosin Beta	LKKTETQG
Cardiac Homing Peptide	CSTMLKAC
Bone Marrow Homing Peptide	GCPFSSTKTE
Osteopontin	TLEGTKKGHLHLDY
Osteogenic Peptide	ALKRQGRTLYGFGG

TABLE 5

Exemplary Bioactive Functional Groups		
Peptide name	Sequence	Source
RGD	<u>KRGD</u> X	Fibronectin, osteopontin, BSP and others
pRGD/bsp-RGD	CGNGE <u>pRGD</u> TYRAY	BSP
hBSP ₂₇₈₋₂₉₃	YSENGE <u>pRGD</u> NYRAY	BSP
PHSRN	PHSRN	Fibronectin
F105	YKRSRYT	FGF-2
F119	KRTGQYKLGSKTGPQK	FGF-2
F36	PDGRVD	FGF-2
F77	KEDGRLL	FGF-2
GFOGER	GGYGGGPC (GPP) ₅ GFOGER (GPP) ₅ GPC	Collagen type I
DGEA	DGEA	Collagen type I
P15	GTPGPQGIAGQRGVV	Collagen type I
AC-100	<u>ERGD</u> NDISPFSGDGQ	MEPE
ODP (osteopontin-derived peptide)	DVDVPD <u>GRGD</u> SLAYG	Osteopontin
RRETAWA	RRETAWA	Identified by phage display
YIGSR	YIGSR	Laminin
IKVAV	IKVAV	Laminin
Ln2-P3	DLTIDDSYWYRI	Laminin
KRSR	KRSR	BSP, fibronectin, osteopontin and others
Consensus heparin binding domain	FHRIKA	BSP
HBD peptide	RKKNPNCRRH	BMP-4
BMP-2-derived peptide	NSVNSKIPKACCVPTLSAI	BMP-2
BMP-2-derived peptide	KIPKASSVPTLSAISTLYL	BMP-2
P24	S ^[PO4] KIPKASSVPTLSAIST LYLDDD	BMP-2
pBMP-9/SpBMP-9	CGGKVGKA (CC/SS) VPTKL SPISVLYK	BMP-9
BMP-9-derived peptide	RKVGKASSVPTKLSPISILYK	BMP-9
BMP-7-derived peptide	RTVPKPSSAPTQLNAISTLYF	BMP-7
BFP-1	GQGFSYPYKAVESTQ	BMP-7
OPD (osteopromotive-domain)	DWIVA	BMP-2
CBM	GLRSKSKFRRPDIQYPDA TDEDITSHM	Osteopontin
BCSP TM ₁	NGLPGPIGP	Collagen type I
OGP (osteogenic growth peptide)	ALKRQGRITLYGFGG	Naturally occurring

TABLE 5-continued

Exemplary Bioactive Functional Groups		
Peptide name	Sequence	Source
OGP ₁₀₋₁₄	YGFGG	OGP
PTH ₁₋₃₄	SVSEIQLMHNLGKHLNSME RVEWLRKKLQDVHNF	Parathyroid hormone
CB	NGVFKYRRRYLYKHAYF YPHLKRFVPVQ	BSP
CTC peptide	IAGVGGEKSGGF	Collagen-IIIa
Substance-P	RPKPQQFFGLM	Naturally occurring
Endothelin-1	CSCSSLMDEKCVYFCHLD IIW	Naturally occurring
QK	KLTWQELYQLKYKGI	VEGF
PBA2-1c	(CVRKIEIVRKK) ₂ -Ahx-Ahx- Ahx-RKRKLERIAR	PDGF
Exendin 4	HGEGTFTSDLSKQMEEEAVR LFIEWLKNGGPSSGAPPPS	Glucagon-like peptide
OPD (osteopontin-derived peptide)	SVVYGLR	Osteopontin
SPARC 113	TLEGTKKGHLHLDY	Osteonectin
SPARC 118	KKGHK	Osteonectin
TP508	AGYKPDEGKRGDACEGDS GGPFV	Thrombin
ROY	YPHIDSLGHWRR	Designed
LL-37	LLGDFFRKSEKIGKEFKRI VQRIKDFLRNLVPRTES	hCAP18

[0311] The peptide may have a functional group engineered to control or alter charge or pH of the peptide or preparation. A pre-selected charge or pH may provide bioactive properties. In some embodiments, a pre-selected charge or pH may provide antimicrobial, antifungal, and/or antiviral properties. In some embodiments, a pre-selected charge or pH may allow the preparation to be administered to a compatible target site.

[0312] The peptide may have an antimicrobial functional group. The antimicrobial functional group may include a conserved sequence of antimicrobial residues. In other embodiments, the antimicrobial functional group may overlap or partially overlap with the self-assembling functional group. In at least one embodiment, the peptide may have alternating or substantially alternating antimicrobial and self-assembling residues.

[0313] The peptide may have an antifungal functional group. The antifungal functional group may include a conserved sequence of antifungal residues. In other embodiments, the antifungal functional group may overlap or partially overlap with the self-assembling functional group. In at least one embodiment, the peptide may have alternating or substantially alternating antifungal and self-assembling residues.

[0314] The peptide may have an antiviral functional group. The antiviral functional group may include a conserved sequence of antiviral residues. In other embodiments,

the antiviral functional group may overlap or partially overlap with the self-assembling functional group.

[0315] In at least one embodiment, the peptide may have alternating or substantially alternating antiviral and self-assembling residues.

[0316] The self-assembled hydrogel may be designed to have cell protective properties at the target site. In particular, the self-assembled hydrogel may be designed to be protective against foreign microorganisms, e.g., pathogenic microorganisms. The self-assembled hydrogel may be designed to be protective against fungal organisms. The self-assembled hydrogel may be designed to be protective against immune attack from environmental immune cells. The antimicrobial, antiviral, antifungal, and/or protective properties of the hydrogel may not substantially affect the viability, growth, or function of cells at the target site.

[0317] The protective properties of the hydrogel may be engineered by altering the net charge of the peptides. In some embodiments, the net charge may be altered by controlling one or more of the presence or absence of cationic particles or peptides, the presence or absence of anionic particles or peptides, buffers, salts, peptide concentration, peptide purity, and the presence or absence of peptide counterions. The peptide may be engineered to have positively charged, negatively charged, hydrophobic, or hydrophilic amino acid residues. In an exemplary embodiment, the peptide may provide antimicrobial, antiviral, and/

or antifungal properties by inclusion amino acids which are positively charged at a substantially neutral pH level. Such amino acids may include, for example, arginine, lysine, tryptophan, and histidine.

[0318] The peptide hydrogel may additionally exhibit antimicrobial properties. In general, the antimicrobial properties may be provided by including an antimicrobial functional group. In some embodiments, the antimicrobial functional group may include a cation-rich peptide sequence. In exemplary embodiments, the antimicrobial functional group may include varying ratios of lysine (K) and arginine (R) (FIG. 4). The antimicrobial peptide hydrogel may provide antimicrobial effects against gram-positive and negative bacteria, including, for example, *E. coli* (FIG. 4), *S. aureus*, and *P. aeruginosa*. FIG. 4 is a graph showing antimicrobial activity (as percent non-viable *E. coli* remaining after 24 hours of administration) of varying concentrations of peptides having 8 arginine residues (PEP8R), 6 arginine residues (PEP6R), 4 arginine residues (PEP4R), and 2 arginine residues (PEP2R).

[0319] The peptide hydrogels may exhibit broad spectrum antimicrobial activity. In accordance with certain embodiments, the peptide hydrogels may reduce bioburden in vivo in partial thickness wounds inoculated with methicillin-resistant *S. aureus* (MRSA) (FIG. 5). FIG. 5 shows preliminary data demonstrating antimicrobial benefits of treating bioluminescent MRSA (US300) with peptide gels. Images (A) and (B) show wells plated with 100 μ l of gel and 100 μ l of US300 (1×10^8 CFU/ml) demonstrating the antimicrobial activity of peptide gels compared to controls at 1 hour and 3 hours (n=3). Image (C) shows mice with partial thickness burns inoculated with 50 μ l of 10^8 CFU/ml US300 and treated with peptide gels. As shown in image (C), the mice exhibit reduced bioburden at 3 hours after administration.

[0320] In particular, the peptide hydrogel may exhibit antimicrobial properties against foreign and/or pathogenic microorganisms, and be compatible with the administered cells. For example, such peptide hydrogels may be compatible with mammalian erythrocytes and macrophages. In one exemplary experiment, when bacteria and mammalian cells were seeded simultaneously onto the peptide hydrogels disclosed herein, the bacteria were killed while the mammalian cells remained >90% viable after 24 hours and could continue to proliferate.

[0321] In some embodiments, the peptides may include functional groups to enhance or promote biological activity compatible or synergizing with MSC function. For example, in certain embodiments, the peptide sequence may contain a functional group that mimics fibronectin and promotes adhesion and proliferation of human MSCs to a greater extent than other ECM ligands. In certain embodiments, the peptide sequence may contain a functional group comprising a neuropeptide to promote diabetic wound healing by suppressing inflammation and inducing angiogenesis. In certain embodiments, the peptide sequence may contain a functional group comprising a neuropeptide to induce the proliferation and migration of MSCs, as well as enhance the immunomodulatory function of MSCs. In certain embodiments, the peptide sequence may contain a functional group to improve wound healing by increasing angiogenesis and inducing MSC proliferation and migration. In certain embodiments, the peptide sequence may lack a functional

group that inhibits proteolytic activity. The peptide may be engineered to contain other functional groups known to one of skill in the art.

[0322] In vitro, the peptide hydrogels disclosed herein may allow cell invasion and proliferation in 3D constructs, allowing the hydrogels to serve as scaffolding matrices for tissue regeneration. The peptide hydrogels may show biocompatibility following subcutaneous implantation. Experiments show minimal cell debris or dead cells at the gel implantation site 7 days post-subcutaneous administration. Experiments further show minimal increases in cytokine concentration in the gel and surrounding tissues compared to naïve tissues, suggesting the gel has insignificant acute inflammation effects.

Kits Comprising the Peptide Preparation

[0323] Kits comprising the peptide preparation are described herein. The kit may comprise the peptide preparation and a buffer solution. The buffer may be configured to induce self-assembly of the peptide prior to or concurrently with administration of the peptide. Each of the peptide preparation and the buffer may be included in a vial or chamber. For example, the kit may comprise a pre-filled packaging containing one or more of the preparation and the buffer. The kit may comprise one or more devices for use and/or delivery of the peptide preparation. The kit may comprise a mixing device. The kit may comprise a delivery device. In certain embodiments, the delivery device and/or mixing device may be the pre-filled packaging, for example, the kit may comprise a pre-filled syringe, spray bottle, ampule, or tube. FIG. 37 is a photograph of the preparation packaged in an end-use container. The exemplary end-use container of FIG. 37 is a pre-filled syringe. The end-use container may be employed as a delivery device or a mixing device. The kit and/or any component of the kit may be sterile or sterilized. For example, the kit and/or any component may be sterilized using autoclave sterilization, optionally terminal autoclave sterilization.

[0324] Any one or more component of the kit may be autoclavable. The packaged kit may be autoclavable. Any one or more component of the kit may be sterilized or sterile. For example, any one or more component of the kit may be sterilized by autoclave. The sterilized one or more component may be packaged in a substantially air-tight container. In some embodiments, the packaged kit may be sterilized, e.g., by autoclave.

[0325] In certain embodiments, the kit may comprise the purified peptide in a dried or powder form. For example, the purified peptide may be lyophilized. The kit may comprise a biocompatible solution to be combined with the purified peptide to produce the peptide preparation. In other embodiments, the kit may comprise instructions to combine the purified peptide with a biocompatible solution to produce the preparation. The kit may additionally comprise the buffer solution.

[0326] The kit may comprise instructions for use. In particular, the kit may comprise instructions to combine the buffer with the preparation, optionally in the mixing device, to form the hydrogel. A user may be instructed to combine the preparation and the buffer at the point of use. In some embodiments, the user may be instructed to combine the preparation and the buffer prior to administration or con-

currently with administration. The user may be instructed to apply the preparation and the buffer to the target site separately.

[0327] The kit may additionally comprise instructions to store the kit under recommended storing conditions. For instance, the kit may comprise instructions to store the preparation or any component at room temperature (approximately 20-25° C.). The kit may comprise instructions to store the preparation or any component under refrigeration temperature (approximately 1-4° C.). The kit may comprise instructions to store the preparation or any component under freezer temperature (approximately 0 to -20° C.). The kit may comprise instructions to store the preparation or any component at body temperature (approximately 36-38° C.). The kit may comprise instructions to store the preparation or any component under cool and dry conditions.

[0328] The kit may additionally comprise an indication of expiration for the preparation or any component. The indication of expiration may be about 1 year after packaging. The indication of expiration may be between about 6 months and about 10 years after packaging, for example, between about 1 year and about 5 years after packaging.

[0329] The kit may comprise additional components for administration in combination with the preparation. In some embodiments, the kit may comprise instructions to combine the additional component prior to administration or concurrently with administration. The kit may comprise instructions to administer the preparation and the additional component to the target site separately. The additional component may be or comprise an antibacterial formulation, an antiviral formulation, an antifungal formulation, an anti-tumor formulation, an anti-inflammatory formulation, a cell culture media, a cell culture serum, an anti-odor formulation, an analgesic, a hemostat formulation, local anesthetic, or a pain-relief formulation. In particular embodiments, the kit may comprise a culture of cells for administration in combination with the preparation, as described herein. In some embodiments, the kit may further comprise a dressing, e.g., a topical dressing, a barrier dressing, and/or a wound dressing.

[0330] The kit may comprise one or more component configured to induce shear-thinning of the hydrogel. Mixing devices or delivery devices (described below) may be employed to induce shear-thinning of the hydrogel by mechanical agitation. The kit may comprise one or more component selected from a temperature control device, a pH control additive, an ion chelator composition, a solvent, a sound control device, a lyophilization device, and an air drying device to induce shear-thinning. For example, the kit may comprise a heater or cooler, a source of an acid or a base, a source of an ion chelator, a source of a solvent, a speaker or sound transmitter, a lyophilizer, or a compressed air dryer, or a fan.

Mixing Devices

[0331] Mixing devices for preparation of a hydrogel at a point of care are disclosed herein. The device may be a multi-chamber device. In exemplary embodiments, the device may be a two-chamber device. The devices may include a first chamber for a peptide preparation. The preparation may comprise a self-assembling peptide in a biocompatible solution. The devices may include a second chamber for a buffer solution. The first chamber and the second chamber may be separated by a barrier provided to

prevent fluid communication between the first chamber and the second chamber. The devices may, optionally, further comprise a mixing chamber. The mixing chamber may be fluidly connectable to the first chamber and the second chamber. Prior to mixing, the mixing chamber may be separated from the first chamber and/or the second chamber by a barrier. In other embodiments, the mixing device may be configured for direct mixing of the contents of the first and second chambers. In some embodiments, the devices may comprise a third chamber for an additional formulation or compound to be administered to the subject. The third chamber may be separated from the first chamber, the second chamber, and/or the mixing chamber. The third chamber may be fluidly connectable to the first chamber and/or the second chamber directly or via the mixing chamber.

[0332] The device may be a single use device. The device may be a multiple use device. In an exemplary embodiment, each of the first, second, or third chamber may be a syringe barrel. Each barrel may have an associated plunger for agitation. Each barrel may be hermetically fitted to a coupling adapter, which forms the mixing chamber. The hermetic fitting may be, for example, a luer lock or luer taper connection.

[0333] The preparation and buffer may be agitated or otherwise mixed to form a homogenous or substantially homogenous mixture, inducing hydrogelation. In some embodiments, the preparation and buffer may be agitated by transferring the mixture back and forth between the first chamber and the second chamber. In some embodiments, the hydrogel exhibits shear-thinning properties, such that during agitation the mixture is substantially liquid. Upon settling, the mixture may form a solid or gel material.

[0334] In exemplary embodiments, the device may be configured to prepare a cell graft at a point of use. In use, the first chamber may comprise the cell preparation and the second chamber may comprise the peptide preparation. The cell preparation may comprise buffer. Alternatively, a third chamber may comprise buffer. Upon actuation the cell preparation and the peptide preparation may mix or contact, i.e. in the mixing chamber. The cells may be suspended in the peptide solution, forming a cell suspension comprising self-assembling peptides.

[0335] The cell preparation and the peptide preparation may be mixed with buffer, forming a buffer suspension. The buffer suspension may be agitated as described above, inducing self-assembly of the hydrogel. The buffer suspension may be agitated to disperse the cells, forming a homogenous or substantially homogenous mixture. The homogenous or substantially homogenous suspension may self-assemble to form a hydrogel cell graft.

[0336] The mixing device may be a static mixing device. A static mixer may generally comprise a device for substantially continuous mixing of the preparation without moving components. For example, the static mixer may comprise a cylindrical or rectangular housing with one or more inlet for each component to be mixed and a single outlet for the mixture. The static mixer may comprise a plate-type mixer.

[0337] The mixing device may generally be formed or lined with an inert, thermally-stable material. In certain embodiments, the material may be opaque and/or shatter resistant.

Delivery Devices

[0338] In some embodiments, the kits may include a delivery device. For instance, the kits may include a syringe or catheter. The kits may include a dropper. The kits may include a spray, e.g. in conjunction with a bottle. The spray device may be, for example, a nasal spray.

[0339] The kits may include a tube or ampule. The kits may include a film, for example. The type of delivery device may be selected based on a target indication. Additionally, the properties of the delivery device may be selected based on a target indication. For instance, a syringe for topical delivery of the preparation may have a larger passage than a syringe for injection of the preparation.

[0340] In some embodiments, the syringe may be used for topical application of the preparation. In other embodiments, the syringe may comprise a needle for parenteral application. The needle of the syringe may have a Birmingham system gauge between 7 and 34. The catheter may be used for parenteral application. The needle of the catheter may have a Birmingham system gauge between 14 and 26. The peptide may be formulated for administration through a particular gauge needle. For instance, the peptide may be selected to have a variable viscosity that will allow application of the preparation through a particular gauge needle.

[0341] In some embodiments, the spray bottle may be used for topical application of the preparation. The spray bottle may comprise a container for the preparation and a spray nozzle. The spray nozzle may be configured for targeted delivery to a target tissue. For instance, a spray nozzle for targeted delivery to an epithelial tissue may have a substantially flat surface and a spray nozzle for targeted delivery to an intranasal tissue may have a substantially conical surface. The spray nozzle may be configured to deliver a predetermined amount of the preparation. In some embodiments, the spray nozzle may be configured to deliver the preparation in substantially unidirectional flow, optionally, regardless of orientation of the spray bottle.

[0342] The spray nozzle may be configured to reduce retrograde flow. In certain embodiments, the spray nozzle may be spring-loaded. In other embodiments, the spray nozzle may be pressure actuated. The actuation pressure may be selected based on the variable viscosity of the preparation. For instance, the actuation pressure may be selected to be sufficient to dispense the hydrogel through the spray nozzle.

[0343] The film may be used for topical application of the preparation. The film may be saturated with the preparation. The film may be used as a barrier dressing and/or hemostat. In some embodiments, the film may accompany a barrier dressing.

[0344] The delivery device may be a single use device. The delivery device may be a multiple use device. The delivery device may include a first chamber for a peptide preparation. The preparation may comprise a self-assembling peptide in a biocompatible solution. The delivery device may include a second chamber for a buffer solution. The first chamber and the second chamber may be separated by a barrier provided to prevent fluid communication between the first chamber and the second chamber. The delivery device may be constructed and arranged for administration of the peptide preparation and the buffer solution simultaneously or substantially simultaneously. In some embodiments, the delivery device may comprise a third chamber for an additional formulation or compound to be

administered to the subject. The third chamber may be separated from the first chamber and/or the second chamber.

[0345] The delivery device may generally be formed or lined with an inert, thermally-stable material. In certain embodiments, the material may be opaque and/or shatter resistant.

Coated Medical or Surgical Devices

[0346] In some embodiments, medical or surgical tools may have at least a portion of an exterior surface coated with the preparations or hydrogels disclosed herein. The coating may enable the exterior surface of the tool to exhibit antimicrobial properties, reducing incidence of infection. The coating may enable the exterior surface of the tool to be biocompatible or cytocompatible, reducing rejection and adverse reaction from contact.

[0347] The surgical tool may be a surgical instrument. For example, the tool may be a grasper, such as forceps, clamp or occluder, needle driver or needle holder, a suture or suture needle, retractor, distractor, positioner, stereotactic device, mechanical cutter, such as scalpel, lancet, drill bit, rasp, trocar, ligasure, harmonic scalpel, surgical scissors, or rongeur, dilator, specula, suction tip or tube, sealing device, such as surgical stapler, irrigation or injection needle, tip and tube, powered device, such as drill, cranial drill, or dermatome, scopes or probe, including fiber optic endoscope and tactile probe, carrier or applier for optical, electronic, and mechanical devices, ultrasound tissue disruptor, cryotome, cutting laser guide, or a measurement device, such as ruler or caliper. Other surgical tools or instruments are within the scope of the disclosure.

[0348] The medical or surgical tool may be an implantable tool. For example, the medical or surgical tool may be an implantable device, such as, implantable cardioverter defibrillator (ICD), pacemaker, intra-uterine device (IUD), stent, e.g., coronary stent, ear tube, or eye lens. Other implantable tools are within the scope of the disclosure. The implantable medical or surgical tool may be a prosthetic or a portion of a prosthetic device, for example, a prosthetic hip, knee, shoulder, or bone or a portion of a prosthetic limb. The implantable medical or surgical tool may be a mechanical tool, such as a screw, rod, pin, plate, disk, or other mechanical support. The implantable medical or surgical tool may be a cosmetic tool, such as breast implant, calf implant, buttock implant, chin implant, cheekbone implant, or other plastic or reconstructive surgery implant. Other medical or implantable tools are within the scope of the disclosure.

[0349] The formulation and/or thickness of the coating may be selected to be temporary, for example, degrading within a pre-determined period of time, for example, less than about 3 months, less than about 1 month, or less than about 2 weeks. The formulation and/or thickness of the coating may be selected to be semi-permanent, for example, degrading within a predetermined period of time of between about 3 months and 3 years, or between about 6 months and 2 years. The formulation and/or thickness of the coating may be selected to be permanent, for example, having a lifespan of more than 2 years or more than 3 years, or having a lifespan longer than the predetermined period of time that the medical or surgical tool is in contact with the subject.

Methods of Treatment by Administration of Peptide Hydrogels

[0350] In some embodiments, the preparations disclosed herein may be administered according to a predetermined

regimen. The preparations disclosed herein may be administered daily, weekly, monthly, yearly, or bi-yearly.

[0351] The preparations disclosed herein may provide immediate release effects. For example, the onset of action of the active ingredient may be less than one minute, several minutes, or less than one hour.

[0352] The preparations disclosed herein may provide delayed release effects. For example, the onset of action of the active ingredient may be more than one hour, several hours, more than one day, more than several days, or more than one week.

[0353] The preparations disclosed herein may provide extended release effects. For example, the preparations may be effective for more than one day, more than several days, more than one week, more than one month, several months, or up to about 6 months.

[0354] The preparations disclosed herein may be administered in conjunction with a medical approach that treats the relevant disease or disorder or a symptom of the relevant disease or disorder. For example, the preparations may be administered in conjunction with a medical approach that is approved to treat the relative disease or disorder or a symptom of the relevant disease or disorder. The preparations may be administered in conjunction with a medical approach that is commonly used to treat the relevant disease or disorder or a symptom of the relevant disease or disorder.

[0355] The preparations disclosed herein may be administered in combination with a surgical treatment. The preparations disclosed herein may be administered to treat wounds associated with the surgical treatment and/or to prevent or treat biofilm.

[0356] The preparations disclosed herein may be administered in combination with an anti-inflammatory agent or treatment. Anti-inflammatory agent may refer to a composition or treatment which reduces or inhibits local or systemic inflammation. The anti-inflammatory agent may comprise, e.g., non-steroidal anti-inflammatory drugs (NSAID), antileukotrienes, immune selective anti-inflammatory derivatives (ImSAID), and/or hypothermia treatment.

[0357] The preparations disclosed herein may be administered in combination with an antibacterial, antiviral, and/or antifungal agent. Such agents may refer to compositions or treatments which eliminate or inhibit proliferation of bacterial, viral, and/or fungal organisms, respectively. Exemplary antibacterial agents include antibiotics and topical antiseptics and disinfectants. The antiviral agent may be a target-specific antiviral agent or a broad-spectrum antiviral agent (e.g., remdesivir, ritonavir/lopinavir). Exemplary local antiviral agents include topical antiseptics and disinfectants. Exemplary antifungal agents include antifungal antibiotics, synthetic agents (e.g., flucytosine, azoles, allylamines, and echinocandins), and topical antiseptics and disinfectants.

[0358] The preparations disclosed herein may be administered to treat a wound, for example, an acute, a sub-acute, or a chronic wound. The wound may be a surgical wound, laceration, burn wound, bite/sting wound, vascular wound (venous, arterial or mixed), diabetic wound, neuropathic wound, pressure wound, ischemic wound, moisture-associated dermatitis, or result from a pathological process. In certain embodiments, the preparations may be administered in an amount effective to treat diabetic foot ulcers (DFU). In certain embodiments, the preparations may be administered in an amount effective to treat gastrointestinal wounds, such as anal fistulas, diverticulitis, and ulcers. In particular, the

preparations may be administered in an amount effective to promote infection free wound closures.

[0359] The preparations disclosed herein may be administered in combination with a treatment or agent to provide anesthesia and/or pain-relief, e.g., local anesthetic. "Anesthetic" may refer to a composition which provides temporary loss of sensation or awareness. The anesthetic may be a general anesthetic (e.g., GABA receptor agonists, NMDA receptor antagonists, or two-pore potassium channel activators) or a local anesthetic (e.g., ester group agents, amide group agents, and naturally derived agents).

[0360] The preparations may be administered in combination with an analgesic or pain-relief agent. "Analgesic" may refer to a composition for systemic treatment or inhibition of pain. The analgesic may comprise an anti-inflammatory agent or an opioid.

[0361] The preparations disclosed herein may be administered in combination with a hemostat agent. "Hemostat" may refer to a tool or composition to control bleeding.

[0362] Exemplary hemostat compositions include collagen-based agents, cellulose-based agents, and chitosan-based agents.

[0363] The preparations disclosed herein may be administered in combination with a treatment or agent to enhance cell or tissue graft therapy. In certain embodiments, the preparations disclosed herein may be administered in combination with a treatment or agent to prevent or inhibit cell death and/or control or reduce inflammation. The preparations disclosed herein may be administered in combination with cell culture media or cell culture serum.

[0364] The administered peptide hydrogels may have an immediate local effect. For instance, the administered preparations may provide localized wound healing or injury treatment effects by closing the wound or filling a void. In certain embodiments, the administered hydrogels may have a systemic effect. For instance, the administered hydrogels may enable cell migration or communication between cell grafts and environmental cells, resulting in a systemic effect. In other embodiments, the administered hydrogels may enable delivery of cell products or byproducts, resulting in a systemic effect. The administered peptide hydrogels may have antimicrobial, antiviral, and/or antifungal properties at a localized site of administration. In other embodiments, the administered peptide hydrogels may provide systemic antimicrobial, antiviral, and/or antifungal properties.

[0365] The administered peptide hydrogels may have long-term, sustained antimicrobial, antiviral, and/or antifungal properties at a localized site of administration. The peptide may be designed to form a hydrogel having a direct contact antimicrobial, antiviral, antifungal effect. Thus, the hydrogel may eradicate microorganisms which directly contact the hydrogel at the target site. The hydrogel may be substantially free of encapsulated antimicrobial, antiviral, and/or antifungal agents. Furthermore, the local antimicrobial, antiviral, and/or antifungal effect may persist as long as the hydrogel is in contact with the target tissue. FIG. 2 includes images which show sustained antimicrobial, antiviral, and/or antifungal effect at the target site.

[0366] To provide a systemic antimicrobial, antiviral, and/or antifungal effect, the peptide hydrogel may additionally comprise encapsulated antimicrobial, antiviral, and/or antifungal agents. Administration of such a hydrogel may generally provide: (1) local antimicrobial, antiviral, and/or antifungal treatment by direct contact as previously

described, and (2) systemic antimicrobial, antiviral, and/or antifungal treatment as a vehicle of an encapsulated therapeutic agent.

[0367] The preparations disclosed herein may be formulated as a hemostat, debridement agent, or barrier dressing (e.g., antimicrobial, antifungal, or antiviral barrier dressing). The preparations may be formulated for wound treatment. Exemplary wounds which may be treated by administration of the preparation include partial and full thickness wounds (e.g., pressure sores, leg ulcers, diabetic ulcers), first and second degree burns, tunneled/undermined wounds, surgical wounds (e.g., associated with donor sites/grafts, tissue and cell grafts, Post-Moh's surgery, post laser surgery, podiatric, sound dehiscence), trauma wounds (e.g., abrasions, lacerations, burns, skin tears), gastrointestinal wounds (e.g., anal fistulas, diverticulitis, ulcers), and draining wounds. The preparations may be formulated for administration to a predetermined target tissue, for example, mesenchymal tissue, connective tissue, muscle tissue, nervous tissue, embryonic tissue, dermal tissue, bone tissue, dental tissue, corneal tissue, cutaneous tissue, integumental tissue, soft tissue, and hard tissue, or a biological fluid.

Methods of Treatment of Microbial Infection

[0368] The preparation may be formulated to provide antimicrobial properties upon administration at a target site. For example, the self-assembled polymeric hydrogel may have antimicrobial properties. As disclosed herein, "antimicrobial" properties may refer to an effect against a microbial population, e.g., killing or inhibiting one or more microorganism from a microbial population. Thus, methods of treating a microbial infection or killing or inhibiting proliferation of a target microorganism are disclosed herein. "Proliferation" may generally refer to the metabolic or reproductive activity of an organism. Methods of management of a microbial bioburden are disclosed herein. The methods may generally comprise administering the preparation in an amount effective to promote deactivation of a target microorganism. Methods of reducing or eliminating a microbial contamination are disclosed herein. In particular, a preparation comprising about 3.0% w/v or less of the peptide, for example, 1.5% w/v or less, or 1.0% w/v or less, may provide antimicrobial properties at a target site.

[0369] The methods may comprise identifying a subject as being in need of treatment for a microbial contamination, colonization, or infection. In general, a microbial colonization or infection may be induced by proliferation of a pathogenic microorganism (disease-causing microorganism). The microbial contamination may be identified by presence of one or more microorganism. In some embodiments, the methods may be employed for prevention or treatment of a microbial colonization or infection. The microbial colonization may refer to an established colony of one or more microorganism. The microbial infection may refer to an established colony of one or more microorganism which has been diagnosed by a clinical assessment. The microbial colonization or infection may be localized or systemic. In general, a microbial contamination may develop into a microbial colonization or infection if adequate treatment is not provided.

[0370] The preparation may be administered in an amount effective to treat biofilm or a microbial infection. The methods may generally comprise administering the preparation in an amount effective to promote deactivation of a

pathogenic microorganism. In certain embodiments, the pathogenic microorganism may be a pathogenic bacterial organism. For example, the preparations and methods may be effective at promoting deactivation of broad-spectrum (gram-positive and gram-negative) bacteria. The pathogenic microorganism may be a species of a genus selected from *Bacillus*, *Bartonella*, *Bordetella*, *Borrelia*, *Brucella*, *Campylobacter*, *Chlamydia*, *Chlamydophila*, *Clostridium*, *Corynebacterium*, *Enterococcus*, *Escherichia*, *Francisella*, *Haemophilus*, *Helicobacter*, *Legionella*, *Leptospira*, *Listeria*, *Mycobacterium*, *Mycoplasma*, *Neisseria*, *Pseudomonas*, *Rickettsia*, *Salmonella*, *Shigella*, *Staphylococcus*, *Streptococcus*, *Treponema*, *Ureaplasma*, *Vibrio*, and *Yersinia*.

[0371] The preparation may be administered in combination with a surgical procedure. The methods may comprise administering the preparation in an amount effective to sterilize at least 90% of the target microorganism at the target site. For instance, the methods may comprise administering the preparation in an amount effective to sterilize at least 90%, at least 95%, at least 97%, at least 98%, at least 99%, at least 99.9%, at least 99.99%, or at least 99.999% of the target microorganism at the target site. Exemplary target sites include epithelial tissue, gastrointestinal system tissue, respiratory system tissue, cardiac system tissue, nervous system tissue, reproductive system tissue, ocular tissue, auditory tissue, and bloodstream. Epithelial tissue may include, for example, epidermis, dermis, hair, and nail.

[0372] However, additional target sites may be treated by the methods disclosed herein. As disclosed herein, "sterilize" may refer to any process that eliminates, removes, kills, or deactivates the microorganism at the target site.

Methods of Treatment of Fungal Infection

[0373] The preparation may be formulated to provide antifungal properties upon administration at a target site. For example, the self-assembled polymeric hydrogel may have antifungal properties. As disclosed herein, "antifungal" properties may refer to an effect against a fungal population, e.g., killing or inhibiting one or more organism from a fungal population. Thus, methods of treating a fungal infection or inhibiting proliferation of a fungal organism are disclosed herein. The methods may generally comprise administering the preparation in an amount effective to promote deactivation of a fungal organism. Methods of reducing or eliminating a fungal contamination are disclosed herein. In exemplary embodiments, a preparation comprising about 3.0% w/v or less of the peptide, for example, 1.5% w/v or less, or 1.0% w/v or less, may provide antifungal properties at a target site.

[0374] The methods may comprise identifying a subject as being in need of treatment for a fungal contamination, colonization, or infection. In certain embodiments, the preparation may be administered in an amount effective to treat at least one of biofilm, Tinea corporis, Candidiasis, Blastomycosis, Coccidioidomycosis, Histoplasmosis, Cryptococcosis, Paracoccidioidomycosis, Aspergillosis, Aspergilloma, Meningitis, Mucormycosis, *Pneumocystis* pneumonia (PCP), Talaromycosis, Sporotrichosis, and Eumycetoma of the subject. In some embodiment, the fungal organism may be a species of a genus selected from *Aspergillus* and *Candida*.

[0375] The preparations and methods may be effective at promoting deactivation of broad-spectrum (sporulating and non-sporulating) fungal organisms. The preparation may be

administered in an amount effective to treat a fungal infection associated with or inhibit proliferation of at least one of *Aspergillus clavatus*, *Aspergillus fischerianus*, *Aspergillus flavus*, *Aspergillus fumigatus*, *Aspergillus niger*, *Trichophyton mentagrophytes*, *Trichophyton rubrum*, *Microsporium canis*, *Candida albicans*, *Candida auris*, *Candida parapsilosis*, *Candida tropicalis*, *Blastomyces dermatitidis*, *Coccidioides immitis*, *Coccidioides posadasii*, *Cryptococcus gattii*, *Cryptococcus neoformans*, *Histoplasma capsulatum*, *Paracoccidioides brasiliensis*, *Pneumocystis jirovecii*, *Mucormycetes rhizopus*, *Mucormycetes mucor*, *Mucormycetes lichtheimia*, *Talaromyces marneffei*, *Sporothrix schenckii*, *Acremonium strictum*, *Noetastudina rosatii*, *Phaeoacremonium kraidenii*, *Pseudallescheria boydii*, *Curvularia lunata*, *Cladophilaophora bantiana*, *Exophiala jeanselmei*, *Leptosphaeria senegalensis*, *Leptosphaeria tompkinsii*, *Madurella grisea*, *Madurella mycetomatis*, *Pyrenochaeta romeroi*, *Trichosporon asahii*, *Cladosporium herbarum*, and *Fusarium sporotrichioides*.

[0376] The preparation may be administered in combination with a surgical procedure. The methods may comprise administering the preparation in an amount effective to sterilize at least 90% of the fungal organism at the target site. For instance, the methods may comprise administering the preparation in an amount effective to sterilize at least 90%, at least 95%, at least 97%, at least 98%, at least 99%, at least 99.9%, at least 99.99%, or at least 99.999% of the fungal organism at the target site. Exemplary target sites may include epithelial tissue, oral tissue, esophageal tissue, tracheal tissue, pulmonary tissue, cardiac tissue, kidney tissue, ocular tissue, and bloodstream. Epithelial tissue may include, for example, epidermis, dermis, hair, and nail. However, additional target sites may be treated by the methods disclosed herein. As disclosed herein, sterilize may refer to any process that eliminates, removes, kills, or deactivates the fungal organism at the target site.

Methods of Treatment of Biofilm

[0377] The preparation may be formulated for treatment of biofilm. Thus, the methods disclosed herein may comprise treatment of biofilm. Treatment of biofilm may generally comprise eliminating at least a portion of biofilm or inhibiting biofilm formation. Administration of the preparation may have an effect against a biofilm population, for example, killing or inhibiting one or more organism in a biofilm community. In general, the charged peptide polymer hydrogel may deconstruct the polymicrobial fungal and bacterial biofilm barrier upon contact. While not wishing to be bound by theory, it is believed the preparations disclosed herein may be selected to dismantle extracellular matrix of the biofilm population, exposing fungal, viral, and microbial organisms of the biofilm to the cationic peptide of the hydrogel. The peptide hydrogel may be effective by destroying microbes, fungi, and viral organisms within biofilms. The preparation may be administered as an antifungal, antimicrobial, and/or antiviral peptide to destroy fungi, microorganisms, and/or viral organisms, e.g., in a biofilm population, through cell lysis.

[0378] Methods of management of biofilm are also disclosed herein. For example, the methods may be employed for prevention of biofilm. The preparation may be administered to a target tissue having a population of biofilm or identified as prone to developing biofilm, e.g., a wound or

wounded tissue. The preparation may be administered in response to tissue contamination or odor.

[0379] The methods may generally comprise administering the preparation in an amount effective to promote treatment of biofilm and/or deactivation of a biofilm population. The biofilm population may comprise bacterial organisms, for example, gram-positive and/or gram-negative bacterial organisms. The biofilm population may comprise fungal organisms, for example, sporulating and/or non-sporulating fungal organisms. Thus, the preparation may provide treatment of biofilm by the antimicrobial and/or antifungal properties and methods described above. In certain embodiments, the biofilm population may comprise viral organisms. The preparation may provide treatment of biofilm by antiviral properties and methods described herein.

[0380] The preparation may be formulated as a biofilm removal agent. In some embodiments, the preparation may be administered to a target tissue for removal of biofilm. For example, the preparation may be administered for debridement of the biofilm and/or biofilm-infected tissue.

Methods of Treatment of Viral Infection

[0381] The preparation may be formulated to provide antiviral properties upon administration at a target site. For example, the self-assembled polymeric hydrogel may have antiviral properties. As disclosed herein, “antiviral” properties may refer to an effect against a viral population, e.g., killing or inhibiting one or more organism from a viral population. Thus, methods of treating a viral infection or inhibiting proliferation of a viral organism are disclosed herein. The methods may generally comprise administering the preparation in an amount effective to promote deactivation of a viral organism. Methods of reducing or eliminating a viral contamination are disclosed herein. In exemplary embodiments, a preparation comprising about 13.0% w/v or less of the peptide, for example, 1.5% w/v or less, or 1.0% w/v or less, may provide antiviral properties at a target site.

[0382] The methods may comprise identifying a subject as being in need of treatment for a viral contamination, colonization, or infection. In certain embodiments, the preparation may be administered in an amount effective to treat at least one of a respiratory viral colonization or infection (e.g., associated with rhinovirus, influenza, coronavirus, or respiratory syncytial virus), a viral skin infection (e.g., associated with molluscum contagiosum, herpes simplex virus, or varicella-zoster virus), a foodborne viral infection (e.g., associated with hepatitis A, norovirus, or rotavirus), a sexually transmitted viral infection (e.g., associated with human papilloma virus, hepatitis B, genital herpes, or human immunodeficiency virus), and other viral infections (e.g., associated with Epstein-Barr virus, West Nile virus, or viral meningitis) of the subject.

The preparation may be administered in combination with a surgical procedure. The methods may comprise administering the preparation in an amount effective to sterilize at least 90% of the viral organism at the target site. For instance, the methods may comprise administering the preparation in an amount effective to sterilize at least 90%, at least 95%, at least 97%, at least 98%, at least 99%, at least 99.9%, at least 99.99%, or at least 99.999% of the viral organism at the target site or systemically. In some embodiments, the methods may comprise administering the preparation in an amount effective to sterilize 100% of the viral organism at

the target site or systemically. In certain embodiments, the preparation may be administered in an amount effective to treat biofilm or kill or deactivate a biofilm population containing a viral organism.

Exemplary target sites may include epithelial tissue, oral tissue, esophageal tissue, tracheal tissue, pulmonary tissue, cardiac tissue, kidney tissue, ocular tissue, and bloodstream. However, additional target sites may be treated by the methods disclosed herein. As disclosed herein, sterilize may refer to any process that eliminates, removes, kills, or deactivates the viral organism at the target site.

Methods of Administration of Peptide Hydrogels

[0383] The peptide hydrogels may be administered by any mode of administration known to one of skill in the art. The method of administration may comprise selecting a target site of a subject and administering the preparation to the target site. In certain embodiments, the methods may comprise mixing the peptide with a buffer configured to induce self-assembly of the peptide to form the hydrogel. In general, the peptide may be mixed with the buffer prior to administration. However, in some embodiments, the peptide may be combined with the buffer at the target site.

[0384] The target site may be any bodily tissue or bloodstream. In some embodiments, the target site may be epithelial tissue, gastrointestinal system tissue, respiratory system tissue, cardiac system tissue, nervous system tissue, reproductive system tissue, ocular tissue, or auditory tissue. The route of administration may be selected based on the target tissue. Exemplary routes of administration are discussed in more detail below.

[0385] In some embodiments, the methods may comprise identifying a subject in need of administration of the preparation. The methods may comprise imaging a target site or monitoring at least one parameter of the subject, systemically or at the target site. Exemplary parameters which may be monitored include temperature, pH, response to optical stimulation, and response to dielectric stimulation. Thus, in some embodiments, the method may comprise providing optical stimulation or dielectric stimulation to the subject, optionally at the target site, for measuring a response. The response may be recorded, optionally in a memory storing device. In general, any parameter which may inform a user of a need or desire for administration of the preparation may be monitored and/or recorded. The methods may comprise imaging the target site or monitoring at least one parameter of the subject prior to administration of the preparation, concurrently with administration of the preparation, or subsequent to administration of the preparation. For example, the methods may comprise imaging the target site or monitoring at least one parameter of the subject after an initial dose and before a potential subsequent dose of the preparation.

[0386] In certain embodiments, the preparation may be administered responsive to the measured parameter being outside tolerance of a target value. The preparation may be administered automatically or manually in response to the measured parameter.

[0387] The preparation may be formulated for topical, parenteral, or enteral administration. The preparation may be formulated for systemic administration. Various pharmaceutically acceptable carriers and their formulations are described in standard formulation treatises, e.g., *Remington's Pharmaceutical Sciences* by E. W. Martin. See also

Wang, Y. J. and Hanson, M. A., *Journal of Parenteral Science and Technology*, Technical Report No. 10, Supp. 42:2 S, 1988; Aulton, M. and Taylor, K., *Aulton's Pharmaceutics: The Design and Manufacture of Medicines*, 5th Edition, 2017; Antoine, A., Gupta M. R., and Stagner, W. C., *Integrated Pharmaceutics: Applied Preformulation, Product Design, and Regulatory Science*, 2013; Dodou K. Exploring the Unconventional Routes—Rectal and Vaginal Dosage Formulations, *The Pharmaceutical Journal*, 29 Aug. 2012.

Parenteral Administration of Peptide Hydrogels

[0388] In certain embodiments, the hydrogels may be administered parenterally. In general, parenteral administration may include any route of administration that is not enteral. The preparation may be administered parenterally via a minimally invasive procedure. In particular embodiments, the parenteral administration may include delivery by syringe, e.g., by needle, or catheter. For instance, the parenteral administration may include delivery by injection. The parenteral administration may be intramuscular, subcutaneous, intravenous, or intradermal. The shear-thinning ability of the hydrogels may allow distribution through small lumens, while still providing minimally invasive treatment.

[0389] The method may comprise applying mechanical force to the hydrogel for parenteral administration. The hydrogel may be thinned by applied mechanical force, for example, pressure applied by a syringe to administer the preparation by injection. In particular, the pressure applied to administer the preparation through a needle or catheter may be sufficient to shear thin the hydrogel for application.

[0390] The peptide hydrogels may be administered parenterally to any internal target site in need thereof. For instance, cardiac tissue, nervous tissue, connective tissue, epithelial tissue, or muscular tissue, and others, may be the target site. The peptide hydrogels may be administered parenterally to a target site of a solid tumor. In exemplary embodiments, antifungal treatment of pulmonary tissue may be provided by parenteral administration of the peptide hydrogels described herein.

Topical Administration of Peptide Hydrogels

[0391] In certain embodiments, the hydrogels may be administered topically. In general, topical administration may include any external or transdermal administration. For instance, the target site for administration may be an epithelial tissue. In particular embodiments, the topical administration may be accompanied by a wound dressing or hemostat.

[0392] The preparation may be administered topically with a delivery device. For instance, the preparation may be administered topically by spray, aerosol, dropper, tube, ampule, film, or syringe. In particular embodiments, the preparation may be administered topically by spray. The spray may be, for example, a nasal spray. Spray parameters which may be selected for administration include droplet size, spray pattern, capacity, spray impact, spray angle, and spray diameter. Thus, the methods may comprise selecting a spray parameter to correlate with the target site or target indication. For instance, a smaller spray diameter may be selected for administration to a small wound. A specific spray angle may be selected for administration to a target

site which is difficult to reach. A denser spray pattern or larger droplet size may be selected for administration to a moist target site.

[0393] Exemplary droplet sizes may be between 65 μm to 650 μm . For instance, fine droplets having an average diameter of 65 μm to 225 μm , medium droplets having an average diameter of 225 μm to 400 μm , or coarse droplets having an average diameter of 400 μm to 650 μm may be selected. The spray pattern may range from densely packed droplets to sparse droplets. The spray diameter may range from less than 1 cm to 100 cm. For instance, spray diameter may be selected to be between less than 1 cm and 10 cm, between 10 cm and 50 cm, or between 50 cm and 100 cm. Spray angle may range from 0° to 90°. For instance, spray angle may be selected to be between 0° and 10°, between 10° and 45°, or between 45° and 90°.

[0394] In some embodiments, the preparation may be administered topically with a film. The film may be a rigid, semi-flexible, or flexible film. In certain embodiments, the flexible or semi-flexible film may be configured to adopt a topological conformation of the target site. In general, the film may be in the form of a substrate saturated with the preparation or hydrogel. The substrate may be rigid, semi-flexible, or flexible. The film may be administered as a barrier dressing and/or hemostat. The preparation may be administered topically with a film and accompanied by a barrier dressing.

[0395] The peptide formulated as a saturated film or barrier dressing may provide antimicrobial, antiviral, and/or antifungal treatment by direct contact with target population, as previously described. Conventional antimicrobial wound dressings rely on traditional antibiotics and function merely as a vehicle for antibiotic agents. However, the peptide hydrogel saturated film or barrier dressing described herein may be designed to provide a biophysical mode of cell-membrane disruption against broad-spectrum (gram-positive and gram-negative) bacterial cultures. Thus, the antimicrobial, antiviral, and/or antifungal peptide hydrogel saturated film or barrier dressing may generally avoid concerns around minimum inhibitory bacterial concentrations typical to conventional small molecule loaded polymers.

[0396] Instead, the peptide hydrogel disclosed herein may be designed to exert toxicity against gram-positive and gram-negative bacteria (including antibiotic resistant strains) while remaining cell-friendly, non-inflammatory, and non-toxic by selecting amino acid charge ratio of the peptide. Similarly, the peptide hydrogel disclosed herein may be designed to exert toxicity against fungal organisms (e.g., sporulating and non-sporulating fungal organisms) and/or viral organisms. The saturated film or barrier dressing disclosed herein may provide a temporary extracellular matrix scaffold for tissue regeneration.

[0397] The peptide hydrogels may be administered topically to any target site in need thereof. Wound healing, e.g., diabetic wound healing, is described herein as one exemplary embodiment. However, it should be understood that many other topical target sites and treatments are envisioned, for example, as previously described above. The wounds may include acute, sub-acute, and chronic wounds. The wound may be a surgical wound or ischemic wound. Chronic wounds such as venous and arterial ulcer wounds or pressure ulcer wounds, and acute wounds, such as those caused by trauma may be treated. In some embodiments, the preparation may be formulated as a film, barrier dressing,

and/or hemostat. Administration of the preparation may accompany a barrier dressing and/or hemostat.

[0398] Treatment and/or management or inhibition of biofilm is described herein as another exemplary embodiment. Moisture management and/or exudate management of wounds or tissues is described herein as another exemplary embodiment. Tissue debridement is described herein as another exemplary embodiment. The preparation may be administered topically as a prophylactic, for example, in association with a wound. The preparation may be administered topically as an analgesic, for example, to a chronic wound or site of biofilm.

Enteral Administration of Peptide Hydrogels

[0399] In certain embodiments, the hydrogels may be administered enterally. In general, enteral administration may include any oral or gastrointestinal administration. For instance, the target site for administration may be an oral tissue or a gastrointestinal tissue. In particular embodiments, the enteral administration may be accompanied by food or drink. The preparation may be administered on a substantially empty stomach. In some embodiments, water is administered to the subject after administration of the preparation. In some embodiments, several hours are waited prior to food consumption after administration.

[0400] Such enteral preparations may be formulated for oral, sublingual, sublabial, buccal, or rectal application. Oral application formulations are generally prepared specifically for ingestion through the mouth. Sublingual and sublabial formulations, e.g., tablets, strips, drops, sprays, aerosols, mists, lozenges, and effervescent tablets, may be administered orally for diffusion through the connective tissues under the tongue or lip. Specifically, formulations for sublingual administration may be placed under the tongue and formulations for sublabial administration may be placed between the lip and gingiva (gum). Sublabial administration may be beneficial when the dosage form comprises materials that may be corrosive to the sensitive tissues under the tongue. Buccal formulations may generally be topically held or applied in the buccal area to diffuse through oral mucosa tissues that line the cheek. Rectal application may be achieved by inserting the formulation in the rectal cavity, either with or without the assistance of a device. Device-assisted application may include, for example, delivery via an applicator or an insertable applicator, catheter, feeding tube, or delivery in conjunction with an endoscope or ultrasound. Suitable applicators include liquid formulation bulbs and launchers and solid formulation insertable applicators.

[0401] For any of the routes of administration disclosed herein, the methods may comprise administering a single dosage of the preparation. The site of administration may be monitored for a period of time to determine whether a booster or subsequent dosage of the preparation is to be administered. For example, the methods may comprise monitoring the site of administration. A parameter of the subject, optionally at the target site, may be monitored as previously described. The subject may be monitored hourly, every 2-3 hours, every 6-8 hours, every 10-12 hours, every 12-18 hours, or once daily. The subject may be monitored daily, every other day, once every few days, or weekly. The subject may be monitored monthly or bi-monthly. In certain embodiments, the subject may be monitored for a period of up to about 6 months. For example, the subject may be

monitored for about 1 month, about 2 months, about 3 months, about 4 months, about 5 months, or about 6 months.

[0402] The methods may comprise administering at least one booster or subsequent dosage of the preparation. For example, the methods may comprise administering a booster dosage to the target site at least about 1 day, 2 days, 3 days, 4 days, 5 days, 6 days, 7 days, 8 days, 9 days, 10 days, 11 days, 12 days, 13 days, or 14 days after the first dosage. The methods may comprise administering a booster dosage 1 week, 2 weeks, 3 weeks, 4 weeks, 5 weeks, or 6 weeks after the first dosage. The methods may comprise administering a booster dosage at least 6 months or 1 year after the first dosage. In certain embodiments, at least a portion of the hydrogel may be present at the target site at the time of the booster dosage. In other embodiments, the hydrogel may be fully metabolized or otherwise eliminated from the target site at the time of the booster dosage.

Methods of Biological Material Delivery with Peptide Hydrogels

[0403] Methods of administering biological material to a subject are disclosed herein. The methods may generally include suspending the biological material in a hydrogel. The biological material may be combined with a preparation comprising a self-assembling peptide in a biocompatible solution and a buffer comprising an effective amount of an ionic salt and a biological buffering agent to induce self-assembly of the hydrogel. The methods may comprise administering an effective amount of the biological material, the preparation, and the buffer (optionally in hydrogel form) to a target site of the subject. Suspending the biological material with the preparation or the buffer will generally produce a liquid suspension. Combining the preparation with the buffer may trigger gelation of the suspension into a hydrogel comprising the biological material, for example, as shown in FIGS. 11 and 12.

[0404] In certain embodiments, the biological material may be suspended in the preparation before inducing gelation. In other embodiments, the preparation comprising the self-assembling peptide in biocompatible solution may be combined with the buffer prior to suspending the cells in the hydrogel. In such embodiments, the biological material may be seeded on the hydrogel or encapsulated within the hydrogel to form a non-homogeneous suspension.

[0405] The non-homogeneous suspension may include clusters or spheroids of biological material (FIG. 11). The clusters or spheroids generally refer to highly concentrated biological material aggregates within the hydrogel. In certain embodiments, hydrogels comprising clusters or spheroids of biological material may exhibit greater viability, growth, or function of the biological material after administration to the target tissue. As shown in the data presented in FIG. 35E, murine MSC cluster in pre-formed hydrogels ($n=3$) showed good cell viability in self-assembling peptide matrices, as measured by relative luminescence units (RLU) using a standard luciferase viability assay at 24 hours post-incubation. Thus, it is believed clusters and spheroids may provide increased viability of biological material by reducing the effect from charge of the hydrogel.

[0406] The buffer may generally comprise an effective amount of an ionic salt and a biological buffering agent to control one or more properties of the formed hydrogel, such as stiffness, pH, and water content. The composition of the buffer may be selected to trigger a degree of gelation, a

desired pH, or a desired water content. The properties of the formed hydrogel may be selected based on a biological material type or target tissue.

[0407] Methods of treating a wound or biofilm are disclosed herein. The preparation and/or buffer may be administered in an effective amount of the biological material to treat a wound, e.g., chronic wound, acute wound. The wound may be an open wound or a closed wound. In certain embodiments, the methods may be employed to treat and/or prevent chronic wounds, including, for example, diabetic wounds, e.g., diabetic foot ulcers (DFUs), leg ulcers, first and second degree burns, tunneled/undermined wounds, surgical wounds, e.g., donor sites/grafts, tissue and cell grafts, Post-Moh's surgery, post-laser surgery, podiatric, and sound dehiscence, trauma wounds, e.g., abrasions, lacerations, burns, and skin tears, gastrointestinal wounds (e.g., anal fistulas, diverticulitis, ulcers), and draining wounds. In some embodiments, the preparation and/or buffer may be administered as a hemostat, barrier dressing (e.g., antimicrobial, antifungal, and/or antiviral barrier dressing), or an autolytic debridement agent. The methods may comprise debriding the target tissue prior to administration. The preparation and/or buffer may be administered in an effective amount of the biological material to treat an ulcer, e.g., diabetic ulcer, e.g., diabetic foot ulcer. The preparation and/or buffer may be administered in an effective amount of the biological material to treat, e.g., prevent or reduce, biofilm. The methods may additionally comprise identifying the subject as having a wound or being in need of treatment of biofilm.

[0408] Methods of treating tissue injury are disclosed herein. The preparation and/or buffer may be administered in an effective amount of the biological material to treat tissue injury, e.g., internal tissue injury. Thus, the methods disclosed herein may comprise administering the biological material, the preparation, and the buffer to internal tissue as the target tissue. The tissue injury treated may be acute, sub-acute, or chronic. The methods may comprise identifying a subject as being in need of treatment for tissue injury. The methods may comprise identifying a target tissue as being in need of treatment.

[0409] The preparations may be administered for treatment of cancer, e.g., a solid tumor. Methods of treating a tumor are disclosed herein. The preparation and/or buffer may be administered in an effective amount of the biological material to treat a tumor. In some embodiments, the methods may comprise direct administration of the preparation and/or buffer to a solid tumor. The methods may comprise identifying a subject as being in need of cancer treatment. The methods may comprise identifying a subject as being in need of treatment for a solid tumor. The methods may comprise identifying a target comprising a solid tumor.

[0410] To increase delivery efficiency, the peptide hydrogels can be designed to be sensitive to stimuli, such as tumor cell surface receptors or tumor-secreted molecules. For instance, estrogen receptors (ER) and integrin $\alpha\beta3$ are overexpressed in breast cancers and are involved in invasion, metastasis and angiogenesis. The hydrogels disclosed herein may be formed to include or express estrogen (which binds the estrogen receptor) and/or Arg-Gly-Asp (RGD) peptide (which binds $\alpha\beta3$ integrin) to target breast cancer cells. Studies performed with hydrogels expressing such modifications showed an increased necrotic and apoptotic impact of Taxol-loaded hydrogels.

[0411] The preparations disclosed herein may be administered in combination with a cancer or an anti-tumor therapy, for example, chemotherapy and radiation or surgical re-section of a tumor. The surgical excision of the tumor may be partial or complete. The preparations disclosed herein may be administered in combination with immunotherapies such as cytokine infusion therapies or other agonist/antagonists, checkpoint immunostat blockade, chimeric antigen receptor (CAR)-T cells, bi-specific T cell-engaging (BiTE) antibodies, oncolytic viruses, or cancer vaccines. The hydrogels disclosed herein may be combined with components to enhance endogenous T cell priming, expansion, survival, migration, trafficking, antigen recognition, and cytotoxic capability. The hydrogels may be modulated by incorporation of regulatory molecules and may be further adapted to deliver the biological material or efficacious molecules in a controlled, sustained manner.

[0412] The preparations disclosed herein may be administered in combination with a treatment or agent to enhance cell or tissue graft therapy. In certain embodiments, the preparations disclosed herein may be administered in combination with a treatment or agent to prevent or inhibit cell death, to control or reduce inflammation, and/or to promote nerve or other tissue regeneration such as tooth, cornea, skin, or visceral organs. In certain embodiments, the preparations disclosed herein may be administered in combination with a treatment to enhance or inhibit tissue protein expression such as to increase or decrease melanosome production, collagen and/or elastin fiber production, or histamine production.

[0413] Methods of in situ tissue regeneration are disclosed herein. The methods may comprise administering biological material in an amount effective for tissue regeneration. The hydrogel may generally provide a biocompatible biological material delivery system configured to localize the biological material at the site of tissue regeneration and improve biological material retention. The biological material may comprise cells, for example, multipotent stem cells, pluripotent stem cells, bone marrow mononuclear cells, or cells of the target tissue. The biological material may comprise tissue material or tissue-derived material, for example, of the target tissue. The biological material may comprise biological fluids, for example, natural biological fluids or synthetic biological fluids, for example, having a composition effective to provide stem cell differentiation into cells of the target tissue.

[0414] The preparations may be administered for treatment of diabetes, e.g., type 1 diabetes. Methods of treating type 1 diabetes or a symptom of type 1 diabetes are disclosed herein. The preparation and/or buffer may be administered in an effective amount of the biological material to treat diabetes or a symptom of diabetes. In particular, the methods disclosed herein may comprise administering the biological material, the preparation, and the buffer to pancreatic tissue. The biological material administered for treatment of diabetes may comprise islet cells. The methods may comprise identifying a subject as being in need of treatment for diabetes, e.g., type 1 diabetes.

[0415] The hydrogels disclosed herein may be employed as biological material delivery systems for the regeneration of many types of tissues including, but not limited to, cardiac muscle tissue (for example, in the treatment of myocardial infarction), renal tissue (for example, in the treatment of chronic kidney disease), and skin tissue (for example, in the

treatment of burn and scar victims, in particular, victims of severe burns and scars). The hydrogels may be designed to incorporate essential mechanical properties, cell-adhesion properties, or biochemical properties for each target tissue and indication. By designing the self-assembling peptide sequence, adding bioactive motifs, or altering the gel formulation, the hydrogels may be tailored for the specific needs of the selected biological material, such that tissue regeneration outcomes can be improved.

[0416] In one exemplary embodiment, the methods of tissue regeneration may comprise providing biological material therapy for transplanted tissues. The methods may comprise administering the biological material, peptide preparation, and buffer as disclosed herein in combination with a tissue transplantation procedure. The methods may be implemented with pancreatic transplantation, cardiac transplantation, lung transplantation, liver transplantation, kidney transplantation, skin transplantation, lymph node transplantation (including artificial lymph nodes), and other transplantation procedures. As previously described, the biological material and peptide may be selected and/or designed based on the target tissue and indication to be treated. In one exemplary embodiment, islet cells may be administered as the biological material to a pancreatic target tissue in combination with a pancreatic transplantation procedure. Other biological materials may be selected in combination with other transplantation procedures.

[0417] The preparation and/or buffer may be administered directly to the site for treatment, e.g., topically for a treatment of a wound, external tissue injury, or biofilm, or parenterally for treatment of an internal tissue injury or a tumor.

[0418] Methods of facilitating cell therapy in a subject are disclosed herein. The methods may comprise providing the peptide and biocompatible solution. The methods may comprise providing the buffer configured to induce self-assembly of the peptide. The methods may comprise providing instructions to suspend cells in the preparation or the buffer. The methods may comprise providing instructions to combine the cells, preparation, and the buffer to induce self-assembly of the hydrogel. The methods may comprise providing instructions to agitate the hydrogel to produce a homogeneous or non-homogeneous cell suspension. The methods may comprise providing instructions to administer an effective amount of the suspension to a target site of the subject.

[0419] In certain embodiments, the methods may comprise providing the cells for cell therapy. The cells may be collected or harvested from the subject or a donor subject. The cells may be treated prior to administration to the subject. The treatment may generally be selected based on the desired outcome of the cell therapy. In some embodiments, the cell treatment may comprise one or more of cell culture, activation, transduction, and expansion. The cell treatment may be performed in a cell engrafted hydrogel, e.g., a hydrogel cell culture, as previously described.

[0420] The methods may comprise providing cell-derived or tissue-derived material. The cell-derived or tissue-derived material may be combined with the cells. In other embodiments, the cell therapy may be provided by cell-derived or tissue-derived material. Thus, the method of facilitating cell therapy may comprise providing instructions to suspend cell-derived or tissue-derived material in the preparation or the buffer and providing instructions to combine the cell-

derived or tissue-derived material, preparation, and the buffer to induce self-assembly of the hydrogel.

[0421] The target tissue may be internal relative to the subject. The target tissue may be external relative to the subject. The methods may comprise administering the suspension and/or buffer to a target tissue selected from mesenchymal tissue, connective tissue, muscle tissue, nervous tissue, embryonic tissue, dermal tissue, bone tissue, dental tissue, corneal tissue, cutaneous tissue, integumental tissue, soft tissue, and hard tissue. In some embodiments the connective tissue may comprise cartilage. The methods may comprise administering the suspension and/or buffer to a biological fluid selected from tears, mucus, urine, menses, blood, wound exudates, and mixtures thereof.

[0422] The methods may comprise administering a first dosage of the biological material, preparation, and/or buffer. The methods may comprise administering at least one booster dosage of the biological material, preparation, and/or buffer. The methods may comprise monitoring the site of administration to determine whether to administer the at least one booster dosage, for example, responsive to one or more indication. In certain embodiments, the methods may comprise administering the at least one booster dosage after a predetermined period of time, for example, after 1 day, 1 week, 2 weeks, 1 month, 2 months, 3 months, 6 months, or 12 months.

[0423] The methods may be performed *in vitro* or *in vivo*. In certain embodiments, the biological material suspension may be administered to the target site before administration of the buffer. In such embodiments, the hydrogel may be formed *in vivo*. In other embodiments, the biological material suspension may be combined with the buffer prior to administration of either component. In such embodiments, the hydrogel may be formed *in vitro*.

[0424] In accordance with certain embodiments, the methods may be performed *in situ*. In such embodiments, the biological material may be administered to the target site and the self-assembling peptide and buffer may be administered to the target site independently. The biological material may be administered to the target site before the self-assembling peptide and the buffer. The self-assembling peptide and the buffer may be administered to the target site before the biological material. The self-assembling peptide and the buffer may be administered independently or combined, as a hydrogel.

[0425] As shown in FIG. 14, cells may be administered to the target site after administration of the self-assembling peptide and buffer. Image (A) is a schematic diagram of exemplary cells seeded on top of the hydrogel. The photograph of image (B) shows murine mesenchymal stem cells (MSCs) seeded on top of the hydrogel. The photograph of image (C) shows the seeded cells are readily absorbed onto the hydrogel without any loss of cells during delivery. Image (D) is a bright-field (BF) microscopy image of murine MSCs spreading well on top of the hydrogel. The data presented in graphs (E) and (F) shows MSCs seeded on top of self-assembling peptide (SAP) hydrogel (n=4) show greater cell viability than MSCs seeded on 2D culture and on competitor hydrogels, including competitor products containing collagen, as measured by relative luminescence units (RLU) using a standard luciferase viability assay.

[0426] In accordance with certain embodiments, methods of culturing cells *in vitro* are provided. The methods may generally include suspending the cells in the preparation

comprising the self-assembling peptide and combining the cell suspension with the buffer to induce gelation of the self-assembling peptide and produce the cell culture.

[0427] In accordance with certain embodiments, methods of preparing a tissue graft *in vitro* are provided. The methods may generally include suspending the tissue material in the preparation comprising the self-assembling peptide and combining the tissue suspension with the buffer to induce gelation of the self-assembling peptide and produce the tissue graft.

[0428] In accordance with certain embodiments, methods of preparing a cell culture or tissue graft *in situ* are provided. The methods may generally include administering the preparation comprising the peptide in a biocompatible solution to the target site. The methods may include administering the buffer configured to induce self-assembly of the peptide to the target site. The methods may comprise administering the biological material to the hydrogel *in situ* to localize the administered biological material.

[0429] The methods may comprise culturing or grafting the biological material in the preparation for a predetermined period of time prior to administration. The methods may comprise combining one or more of a cell culture media, a cell maintenance agent, a cell growth agent, and a cell culture serum with the cell culture or tissue graft. In some embodiments, the methods may comprise culturing or grafting the biological material in the hydrogel, for example, by combining the preparation with the buffer, for a predetermined period of time prior to administration. The period of time may be any period of time sufficient to grow or seed the biological material to a desired degree. Thus, the period of time may generally depend on the biological material type, the culture or graft conditions, and the desired outcome of the culture or graft prior to administration. The period of time may be, for example, at least 1 hour, 2 hours, 3 hours, 6 hours, 12 hours, 18 hours, 24 hours, 2 days, 3 days, 4 days, 5 days, 6 days, 1 week, 2 weeks, 3 weeks, or 1 month.

[0430] The biological material to be administered may include biological fluids, cells, and/or tissue material. In some embodiments, one or more biological material administered may be synthetic. For instance, the biological fluids may be or include synthetic fluids. In other embodiments, the biological material may be obtained from a donor. The biological material may be autologous (obtained from the recipient subject). The biological material may be allogeneic (obtained from a donor subject of the same species as the recipient subject) or xenogeneic (obtained from a donor subject of a different species as the recipient subject).

[0431] The cells to be administered may include whole cells, live cells, dead cells, cell-derived matter, and/or cell fragments or cell lysates. The cells may include at least one of eukaryote cells, virus, prokaryote cells, adjuvants, cytokines, and growth factors. The cells may include, for example, progenitor cells, multipotent cells, induced pluripotent cells, immune cells, specialized cells, terminally differentiated cells, bone marrow mononuclear cells, islet cells, and combinations thereof.

[0432] The cell-derived matter may include secreted material, cell fragments, and other byproducts produced by cells, by cell activity, or from cells. For example, cell-derived matter may comprise secretomes, exosomes, proteins (e.g., signaling proteins), enzymes, biological fluids, serum, plasma, spheroids, liposomes, growth factors, cytokines,

chemokines, paracrine factors, autocrine factors, genetic material (e.g., RNA or DNA), vesicles, and/or granules.

[0433] The cells may be or include mesenchymal stem/stromal cells (MSCs). Mesenchymal stem/stromal cells (MSCs) are multipotent cells. MSCs may be isolated from multiple tissues, including bone marrow and adipose tissue. Therapeutic treatment with MSCs may be employed for a variety of disease indications. MSCs typically provide therapeutic effects by, for example, secreting bioactive factors that act in a paracrine manner to recruit other cell types and enhance angiogenesis and tissue regeneration. Such therapeutic effects of MSCs may be generally employed in wound healing, e.g., chronic wound healing.

[0434] The cells may be or include stem cells. The stem cells may be pluripotent. The stem cells may be non-pluripotent. In some embodiments, the stem cells may be differentiated. For example, the stem cells may be differentiated into a target cell type. In other embodiments, the stem cells may be non-differentiated. Exemplary stem cells include neural stem cells, adipose-derived stem cells, Schwann cells, bone marrow mesenchymal stem cells, muscle-derived stem cells, hair follicle stem cells, dental pulp stem cells, skin-derived stem cells, or induced pluripotent stem cells.

[0435] The tissue to be administered may include live tissue, dead tissue, tissue-derived matter, and/or tissue fragments or tissue lysates. The tissue or tissue material may comprise, for example, bone tissue, connective tissue, neural tissue, adipose tissue, bone marrow, or combinations thereof. The biological material may comprise tissue or tissue material selected from mesenchymal tissue, connective tissue, muscle tissue, nervous tissue, embryonic tissue, dermal tissue, bone tissue, dental tissue, corneal tissue, soft tissue, and hard tissue. The tissue or tissue material may comprise a biological fluid selected from tears, mucus, urine, menses, blood, wound exudates, and mixtures thereof.

[0436] The self-assembled hydrogel may have a physical structure substantially similar to the native extracellular matrix of the biological material, e.g., target cell type, allowing the gel to serve as a temporary scaffold to promote growth, function, and/or viability. In particular, the self-assembled hydrogel may have similar properties, including, for example, pore size, density, hydration, charge, rigidity, etc., to the native extracellular matrix. The properties may be selected responsive to a target cell type.

[0437] The self-assembled hydrogel may have a selected degradation rate. The degradation rate may be selected responsive to the target site of implantation or administration. The self-assembled hydrogel properties may be selected to promote migration of the biological material, e.g., cells, to the hydrogel environment. The self-assembled hydrogel properties may be selected to promote protection in a hostile environment. The self-assembled hydrogel properties may be selected to promote anchoring of biological material within the hydrogel, for example, as with cells that will not engraft onto host tissue. The self-assembled hydrogel properties may be selected to promote migration of cell products or byproducts to the hydrogel environment, for example. In an exemplary embodiment, the self-assembled hydrogel properties may be selected to control differentiation of cells, e.g., progenitor cells or stem cells, e.g., mesenchymal stem cells.

[0438] The preparation and/or buffer may be sterile, e.g., substantially sterile. The preparation and/or buffer may be

sterilized by terminal sterilization. The preparation and/or buffer may be sterilized by autoclave sterilization. The preparation and/or buffer may be substantially free of a preservative.

[0439] The hydrogels disclosed herein have gelation kinetics which are fast enough to ensure cells and tissue material becomes substantially homogeneously incorporated within the matrix. In particular, the gelation kinetics are sufficiently fast to afford an even distribution of encapsulated cells to allow reproducible control over cell density within the matrix. In other embodiments, biofabrication may comprise controlling the gelation kinetics to form a non-homogeneous suspension, e.g., suspended cell clusters or spheroids.

[0440] Additionally, the hydrogels disclosed herein have a construct that can be introduced in vivo and remain localized at the point of administration, for example, even without a spatial cavity. The localization of the hydrogel upon administration can limit or inhibit leakage of the cell construct to neighboring tissue.

[0441] The suspension and/or peptide may be engineered to express a desired property. In certain embodiments, the suspension and/or peptide may be engineered to protect cells from hostile environments. In particular, the suspension and/or peptide may be engineered to protect the cells from environments with a high microbial burden, hostile immune cells, or environmental proteins. The suspension and/or peptide may be engineered to increase cell viability. The suspension and/or peptide may be engineered to control differentiation, control cell fate in situ, control cell fate in vivo, control cell fate ex vivo, or control cell fate in vitro.

[0442] The suspension and/or peptide may be engineered to increase cell attachment to the matrix or increase cell attachment and/or migration in the environment. The suspension and/or peptide may be engineered to decrease apoptosis, for example, by providing cell attachment and/or biological modulation.

[0443] The suspension and/or peptide may be engineered to achieve the results described above by altering the expression of protein motifs or the net charge of the peptides. The hydrogel properties may be engineered by controlling one or more of the expression of extracellular matrix or protein motifs, the presence or absence of fusion proteins, the net charge of the peptides, the presence or absence of cationic particles or peptides, the presence or absence of anionic particles or peptides, buffers, salts, peptide concentration, peptide purity, the presence or absence of peptide counterions, the presence or absence of specialized proteins, and the presence or absence of specialized small or large molecules.

[0444] The properties of the self-assembled hydrogel may be controlled by designing the peptide. For example, the peptide may include functional groups that provide one or more selected physical property. The properties may be controlled by selecting the composition of media or buffer. For example, the media may include serum or be substantially free of serum. For example, the buffer may have a net positive charge, be net neutral, or have a net negative charge. In some embodiments, the functional group may be configured to alter peptide net charge or counterions when the peptide is suspended in the solution.

[0445] The self-assembled hydrogel may be designed to have cell and/or tissue protective properties. In particular, the self-assembled hydrogel may be designed to be protective against foreign microorganisms, e.g., pathogenic micro-

organisms, as previously described. The self-assembled hydrogel may be designed to be protective against immune attack from environmental immune cells, for example, by providing a physical barrier or biochemical modulation. The antimicrobial and/or protective properties of the hydrogel may not substantially affect the viability, growth, or function of the engrafted biological material.

[0446] The protective properties of the hydrogel may be engineered by altering the net charge of the peptides. In some embodiments, the net charge may be altered by controlling one or more of the presence or absence of cationic particles or peptides, the presence or absence of anionic particles or peptides, buffers, salts, peptide concentration, peptide purity, and the presence or absence of peptide counterions.

[0447] The suspension may be designed to have a substantially physiological pH level. The suspension may have a pH level of between about 4.0 and 9.0. In some embodiments, the suspension may have a pH level of between about 6.0 and 9.0, for example between about 7.0 and 8.0. The suspension may have a pH level of between about 7.3 and 7.5, for example, about 7.4. The substantially physiological pH may allow administration of the biological material directly to a target tissue and/or at the time of preparation.

[0448] In one exemplary embodiment, murine mesenchymal stem cells (MSCs) were encapsulated into peptide hydrogels at physiological pH (about 7.4) and at pH 6 (FIG. 16). The MSCs encapsulated in hydrogel at physiological pH for 1 and 3 days showed greater cell viability than the peptide hydrogels at pH 6, as measured by absorbance at 450 nm using a standard absorbance viability assay. ($p < 0.05$). The pH value of the hydrogel may be selected based on the selected biological material. Different biological material may show improved viability with encapsulation in different pH value hydrogels.

[0449] In some embodiments, the suspension may be prepared at a point of care. In some embodiments, the hydrogel may be prepared by combining the suspension with the buffer at the point of care. For example, the suspension and/or the buffer may be prepared less than about 1 minute, less than about 2 minutes, less than about 5 minutes, or less than about 10 minutes prior to administration. In autologous administration embodiments, the methods may comprise collecting or harvesting the biological material from the subject at the point of care, prior to preparing the suspension. The material may be collected or harvested from the subject less than about 1 minute, less than about 2 minutes, less than about 5 minutes, or less than about 10 minutes prior to suspending the biological material.

[0450] The methods may comprise suspending the biological material in the preparation, optionally, agitating the suspension to provide a substantially homogeneous or non-homogeneous distribution of the biological material, and administering the suspension at a point of care. In some embodiments, the suspension may be agitated to provide a substantially homogeneous distribution of the biological material. In other embodiments, the suspension may be prepared or agitated to provide a non-homogeneous suspension, for example, comprising clusters or spheroids of the biological material.

[0451] The administration of the suspension may be topical or parenteral, as described in more detail below.

[0452] The administration of cells and cell products or byproducts at the target site may be controlled by altering

the release properties of the hydrogel. In some embodiments, the release properties may be engineered by controlling one or more of the expression of extracellular matrix or protein motifs, the presence or absence of fusion proteins, the net charge of the peptides, the presence or absence of cationic particles or peptides, the presence or absence of anionic particles or peptides, buffers, salts, peptide concentration, peptide purity, and the presence or absence of peptide counterions. The properties may be engineered to deploy cells or tissue at the target site. The properties may be engineered to deploy cell products or byproducts at the target site, for example, delivery of exosomes, growth factors, genetic material, RNA, siRNA, shRNA, miRNA, etc.

[0453] As shown in the images of FIG. 15, degradation rate of the hydrogel for targeted release of cells and/or agents may be controlled by peptide design. Briefly, peptide hydrogels of varying weight percentage were shown to deliver proteins of varying charge and varying protein size (kD) at different time points over a 30-day period. FITC-tagged proteins of varying size (20 kD, 70 kD and 150 kD) were delivered differentially from hydrogel administration over 2 weeks. Western blot shows that enzyme was released differentially from two different types of peptide gels over a 5-day period. The protein was not degraded.

[0454] The properties of the hydrogel may be selected to control a release profile of the biological material at the target site. For instance, the hydrogel may be engineered to provide rapid or immediate release of the biological material by designing a faster degradation profile. The hydrogel may be engineered to provide sustained or extended release of the biological material by designing a longer degradation profile. In certain embodiments, the administered hydrogel may comprise one or more layers of hydrogel having biological material therein. Each layer may be formed of a self-assembling peptide providing a selected degradation rate.

[0455] Exemplary layered hydrogels are shown in the schematic drawings of FIG. 11. The layered hydrogels may comprise two or more layers of hydrogel selected to have different properties, for example, degradation rate, pH, charge, etc. The layered hydrogels may comprise two or more layers of hydrogel having different additives, for example, cells, therapeutic agents, cell therapy agents, etc. In certain embodiments, cells may be substantially homogeneously suspended in one hydrogel and dispersed in clumps in a second hydrogel with different properties (as shown in image F of FIG. 11); cells may be substantially evenly dispersed in multiple hydrogels with differing degradation release properties, e.g., a first immediate release layer, a second rapid release layer, a third extended release layer, etc. (as shown in image G of FIG. 11); cells suspended in a first hydrogel having a first set of properties may be centered within a second hydrogel having a second set of properties (image H of FIG. 11), layered horizontally (image I of FIG. 11), or vertically (image J of FIG. 11).

[0456] In one exemplary embodiment, a first layer of the hydrogel, for example, an external layer in contact with the target tissue, may be designed to have the fastest degradation rate.

[0457] Thus, the first layer may be designed or configured to release the biological material into the target site of the subject after a first predetermined period of time. A second layer of the hydrogel, for example, an internal layer in contact with the first layer, may be designed to have a second

fastest degradation rate. Thus, the second layer may be designed or configured to release the biological material into the target site of the subject after a second predetermined period of time. The hydrogel may be designed to include additional consecutive layers, for example, third, fourth, fifth, sixth, seventh, eighth, ninth, tenth layers and so forth. Each layer having a different degradation rate and being designed or configured to release biological material into the target site of the subject at different predetermined periods of time. The biological material released by each hydrogel layer may be the same or different. In some embodiments, one or more hydrogel layers may be designed to release a combination therapy or other drug, with or without biological material. Such a gradient design may allow temporal regulation over the release of biological material to the target tissue.

[0458] Methods of producing a layered hydrogel having one or more classes of biological material are disclosed herein. The methods may comprise producing a first hydrogel layer having a first biological material suspended therein as previously described. The methods may comprise producing a second hydrogel layer adjacent the first hydrogel layer having a second biological material suspended therein. Each hydrogel layer may be formed by combining a self-assembling peptide having a selected degradation rate with a buffer and the biological material, as previously described. Additional hydrogel layers may be produced adjacent the first or the second hydrogel layer. In some embodiments, each hydrogel layer is produced to encapsulate a previous hydrogel layer.

[0459] In other embodiments, hydrogels having temporal regulation over the release of biological material may be designed by including a shell formed of a material having a slower degradation rate encapsulating each hydrogel layer. The degradation rate of the shell material may be selected to regulate timing of release of the biological material within each inner hydrogel layer. The methods of producing the layered hydrogel may comprise forming a shell layer adjacent to one or more hydrogel layer, for example, encapsulating one or more hydrogel layer. Exemplary shell layers may be formed of liposomes and/or Janus-faced molecules to confer bifunctionality and tunability to the layered hydrogel.

[0460] The hydrogel may be engineered to increase cytocompatibility with the biological material, while maintaining antimicrobial properties. In some embodiments, the biological material may be suspended in a hydrogel as previously described. The biological material hydrogel may be cytocompatible. The cytocompatible biological material hydrogel may be encapsulated with an antimicrobial hydrogel. For instance, a preparation comprising a self-assembling peptide configured to form an antimicrobial hydrogel may be combined with a buffer to form the antimicrobial hydrogel. The antimicrobial self-assembling peptide may have a net positive charge. For example, the antimicrobial peptide hydrogel may have a charge of +2 or greater, for example, up to +11, a charge of +2 to +9, or a charge of +5 to +9. The antimicrobial peptide preparation may be combined with the buffer on a surface of the cytocompatible hydrogel comprising biological material. Thus, the hydrogel may be formed of an interior hydrogel layer comprising biological material and an exterior hydrogel layer being antimicrobial. FIG. 11 includes schematic diagrams of multi-layered hydrogels.

Biofabrication of Cell Grafts with Peptide Hydrogels

[0461] Methods of preparing biological material grafts in vitro for administration in vivo are disclosed herein. The methods may include self-assembly of a preparation comprising biological material and a peptide into a hydrogel scaffold matrix in vitro. The self-assembled higher order structure may be administered to a target site of the subject.

[0462] The methods may comprise obtaining a biological material from a donor subject. The donor subject may be the recipient subject for autologous treatment or another subject for allogeneic or xenogeneic treatment. The methods may comprise suspending the biological material in the preparation comprising the peptide in a biocompatible solution. The methods may comprise combining the preparation with a buffer to induce self-assembly of the hydrogel, biofabricating the implantable sample.

[0463] In some embodiments, the methods may further comprise administering the hydrogel to a target tissue of the recipient subject.

[0464] In some embodiments, the biological material may be treated before administration to the recipient subject. Treatment may include, for example, one or more of, culturing, growing, activating, transducing, and expanding the biological material or cells for a period of time prior to administration. Treatment may include washing the biological material. Treatment may include combining the biological material with one or more additives or agents disclosed herein.

[0465] In some embodiments, the methods may include biofabrication of the biological material graft at a point of care. Thus, one or more of the steps for biofabrication may be performed at the point of care. In some embodiments, all of the steps may be performed at the point of care. For example, the biological material may be administered immediately after suspending or engrafting.

[0466] FIGS. 32A-32C are schematic diagrams of exemplary methods of biofabricating the self-assembling peptide hydrogel. FIG. 32A shows an exemplary method comprising performing all steps of biofabrication and administration of biological material at point of care. In the exemplary methods of FIG. 32A-32B the biological material is autologous. However, the biological material encapsulated and administered may be allogeneic or xenogeneic. FIG. 32B shows an exemplary method comprising harvesting the biological material from the subject at the point of care, engineering the biological material at an external facility, and biofabricating the biological material encapsulated in hydrogel at the point of care. FIG. 32C shows an exemplary method comprising harvesting the biological material from a donor subject and administering the biological material encapsulated in hydrogel to a recipient subject. FIGS. 32B-32C show engineering the biological material at an external facility. However, the biological material may be engineered or treated at the point of care. Additionally, the point of care for harvesting the biological material from a donor subject need not be the same point of care for encapsulating the biological material in the hydrogel and administering the biological material encapsulated in hydrogel to the recipient subject.

[0467] Mixing devices for biofabrication of cell grafts at a point of care are disclosed herein.

[0468] The devices may include a first chamber for a cell preparation. The cell preparation may comprise cells suspended in water, media, or buffer. The devices may include

a second chamber for a peptide preparation, and optionally a third chamber for a buffer, as previously described.

EXAMPLES

[0469] The function and advantages of these and other embodiments can be better understood from the following examples. These examples are intended to be illustrative in nature and are not considered to be limiting the scope of the invention.

Example 1: Treating Pathogen Contaminated Diabetic Wounds with Engrafted Peptide Hydrogels at Varying Concentrations of 0.5%, 0.75%, and 1.5% w/v

[0470] Cell therapy may be a viable approach for treating chronic wounds such as diabetic foot ulcers by inducing the expression of pro-regenerative growth factors and cytokines.

[0471] Several different cell types may be employed for such treatment. In an exemplary embodiment, mesenchymal stem/stromal cells (MSCs) may be used to provide cell therapy to diabetic wounds. Cell therapy treatment with MSCs may improve wound healing in diabetic subjects, for example, by accelerating wound closure, increasing granulation tissue formation, augmenting angiogenesis, and/or improving biomechanical strength of regenerated tissues. The preparations and methods disclosed herein may be employed to deliver therapeutic cells at a point of care and substantially protect cells from the wound microenvironment, which is vulnerable to inflammation and pathogenic colonization. The cells may be progenitor cells, multipotent cells, induced pluripotent cells, immune cells, specialized cells, terminally differentiated cells, or combinations thereof.

[0472] In accordance with certain embodiments, the methods disclosed herein may comprise injecting cells intradermally into or in the periphery of the wound area, both in preclinical and clinical settings. The therapeutic efficacy of MSCs may be enhanced by improving cell viability, retention, and function following administration. The preparations may allow easy cell seeding and void filling. The preparations may be administered in combination with application of a wound dressing. The preparations may generally not be occlusive in nature, and thus may not require removal because they have the ability to substantially biodegrade in situ. The preparations may be amenable to cell attachment, without requiring pre-seeding in vitro, and without requiring cell culture for extended time periods, while providing cell attachment and retention. In certain embodiments, pre-seeding and cell culture prior to administration may be employed.

[0473] The disclosed cell scaffolding matrix technology may allow simple and rapid cell encapsulation at the point of care, as well as conformal filling of wounds. The preparations may withstand rapid degradation in the protease-rich diabetic wound environment. As such, the preparations may provide a balance between ECM formation and degradation, preventing, reducing, or inhibiting the development of chronic ulcers or fibrosis, decreasing the incidence of pathogenic colonization of the wound matrix and tissue, and preventing, reducing, or inhibiting the incidence of wound stalling in an inflammatory phase. Thus, the preparations and methods may be used for promoting wound closure.

[0474] The preparations and methods disclosed herein may additionally prevent, reduce, or inhibit the susceptibility to infections. Several gram-positive pathogens commonly found in diabetic foot ulcers, including *S. aureus*, express adhesins that bind to collagen, a conventionally used cell graft protein. The antimicrobial properties of the proposed technology may enable treatment of chronic wounds that are at risk for pathogenic colonization, for example, by preventing, reducing, or inhibiting pathogen binding to the cell matrix.

[0475] The self-assembling preparations disclosed herein may be employed as a cell delivery vehicle that supports the viability and function of transplanted cells to provide therapeutic benefit, while remaining simple to manufacture and easy to administer in an intra-operative setting. In some embodiments, the preparations may be used in combination with a surgical wound debridement procedure to deliver therapeutic cells into chronic diabetic wounds that are at a high risk of pathogenic colonization. The cell suspension may be prepared ex situ by a medical provider at a point of care prior to administration.

[0476] The efficacy of the peptide hydrogels described herein for delivering therapeutic cells to accelerate tissue regeneration in both clean and contaminated diabetic wounds will be examined. A splinted, full-thickness excisional wound model in db/db mice will be used. It has been shown that application of a silicone splint to wounds in mice minimizes skin contraction during healing, resulting in a model that better approximates human wound healing and allows new treatment strategies to be evaluated for their ability to improve wound re-epithelialization and granulation tissue formation. Primary allogeneic mouse MSCs will be delivered into immunocompetent diabetic mice. However, autologous or xenogeneic MSCs may be used. A subsequent study will use a swine model.

[0477] In particular, the study will confirm in vivo viability of MSCs encapsulated in the peptide hydrogels. The study will demonstrate that the peptide hydrogels may be used to rapidly and gently encapsulate primary MSCs, as well as provide a scaffold that supports their viability in vivo. The study will demonstrate that the peptide hydrogels are biocompatible in vivo and that the gels can encapsulate primary MSCs with high cell retention and viability following transplantation.

[0478] Manufacturing specifications of peptides have been already defined and will be confirmed through certificate of analysis. The self-assembling peptide hydrogel with antimicrobial motif will be used. Additional peptide hydrogels including various biofunctional motifs will also be used (e.g., enhance cell adhesion and function). Aseptic formulation (0.2 μ m filtering) of pre-clinical product will be performed.

[0479] In vitro viability and proliferation of cells following encapsulation in the antimicrobial extracellular matrices will be demonstrated. Specific peptides will be tested for high cell retention and viability of MSCs post-encapsulation in vitro. Primary MSCs from C57BL/6 mice expressing GFP (Cyagen), which have been used in mouse wound healing models, will be used throughout the study to facilitate detection of transplanted cells in vivo. GFP+ MSCs will be encapsulated within the peptide matrices with peptide contents of 0.5%, 0.75%, and 1.5% w/v. Following mixing, gels will be dispensed into cell culture well plates through a syringe to simulate application into a wound bed. MSCs

seeded onto tissue culture polystyrene (TCPS) and onto collagen scaffolds (Integra® wound matrix or other similar porous collagen scaffold) will serve as controls.

[0480] Following cell encapsulation or seeding, MSCs will be allowed to adhere for 30 min. Media will then be added to the different conditions in order to culture the cells and evaluate their viability and proliferation at 1, 3, and 7 days after initial cell encapsulation/seeding. The cell matrices will be dissociated by gentle pipetting and dilution in media in order to disrupt peptide networks. For the control conditions, cells will be enzymatically released using trypsin-EDTA. Cells will be evaluated for total cell number, viability, and proliferation by staining with a viability stain and counting using a hemocytometer. The study will confirm that the peptide hydrogels encapsulate primary MSCs in a rapid, safe, and gentle manner, leading to high retention and viability of encapsulated cells.

[0481] Preliminary results using the MG-63 osteoblast progenitor cell line have demonstrated that cells exhibit high viability following encapsulation and shear thinning within gels. The study will extend these findings using primary MSCs, a cell type that has therapeutic potential. It is expected that cell encapsulation within the peptide hydrogels will be rapid and gentle, as a result of the peptide self-assembly mechanism and the flowable properties of the gels. High (>95%) initial cell retention within the peptide matrices, as well as high (>80%) cell viability in the days following encapsulation are expected. Peptides with bioactive motifs are expected to further enhance cell viability and/or proliferation. Peptide formulations that lead to >90% cell viability of MSCs at days 1, 3, and 7 will be carried forward into in vivo testing.

[0482] The study will demonstrate in vivo biocompatibility of the implanted matrices and viability of encapsulated cells. The study will demonstrate that implanted matrices are biocompatible in vivo, and that they support the survival of MSCs encapsulated within the gels at days 3 and 14 after transplantation. GFP+MSCs will be used to allow detection of the cells following in vivo delivery. 40 female CD1 mice will receive 100 μ l subcutaneous injections of the following treatments: 1) PBS, 2) 0.5×10^6 MSCs only (control), 3) collagen scaffold+ 0.5×10^6 MSCs (comparator product control), 4) self-assembling peptide hydrogels+ 0.5×10^6 MSCs, and 5) Bioactive self-assembling peptide hydrogels+ 0.5×10^6 MSCs. A 0.75% w/v concentration of the self-assembling peptides will be used and each mouse will receive two subcutaneous implants. Mice will be euthanized at day 3 and day 14, and the gel implants will be excised together with surrounding tissue for analysis of implant biocompatibility, as well as MSC viability and functional activity.

[0483] At each timepoint, 4 implants per condition will be processed for histology, and another 4 implants per condition will be stored to analyze the expression of paracrine factors associated with tissue regeneration. Histology samples will be stained with hematoxylin and eosin (H&E), and the different conditions will be evaluated for biocompatibility by assessing tissue morphology, necrosis, and fibrotic tissue thickness around the implant. In order to identify conditions that promote survival of MSCs following in vivo delivery, tissue sections will be analyzed to enumerate GFP+MSCs in the gels (cells/cm²) and exclude cells undergoing apoptosis (TUNEL assay). The study will con-

firm that the peptide formulations are safe and biocompatible in vivo and promote MSC survival following implantation.

[0484] Preliminary in vitro results have shown that the peptide hydrogel is cytocompatible with mammalian cells (FIG. 5). Thus, it is anticipated that the peptide hydrogels will be safe and biocompatible with MSCs throughout the implantation period. Fewer than 10% necrotic cells within the gels are expected at all time points (quantified by cell/cm²) and minimal fibrotic tissue surrounding the gels (quantified by thickness/cm²). Gels on day 14 will show some degradation compared to day 3, but significant macrophage or giant body cell responses (quantified by cell/cm²) in response to gel degradation products are not expected. In addition, it is expected that the peptide hydrogels will support the viability of the MSCs delivered in the gels, confirmed by quantifying viable GFP+MSCs in tissue sections. The bioactive peptide formulation may promote greater survival of the MSCs within the matrices and increased infiltration of endogenous cells into the implants.

[0485] Sixteen additional animals are provided to account for dropouts and test the bioactive formulations, which are endowed with biological functionalities that can act on MSCs. Expression of pro-regenerative paracrine factors, such as VEGF, Ang-1, EGF, and KGF, will be analyzed by ELISA (quantified as pg protein/mg of tissue) using stored samples, as well as by immunohistochemical staining of tissue sections (image analysis to identify protein spatial and temporal localization at days 3 and 14).

Example 2: Antimicrobial Properties of MSC Engrafted Peptide Hydrogels

[0486] The study will demonstrate that MSCs encapsulated in antimicrobial peptide matrices will enhance tissue regeneration of full-thickness wounds in db+/db+ diabetic mice. The db/db mouse model of diabetes (db+/db+; BKS. Cg-Dock7m+/+Leprdb/J) is a commonly used model of diabetic wound healing that exhibits vulnerability to infections, altered host response, and impaired healing. Tissue morphology between the different treatment groups at day 14 (partial wound closure) and day 28 (complete wound closure) for clean diabetic wounds and at day 28 for pathogen-contaminated wounds, which are expected to exhibit delayed tissue regeneration will be examined. The study will demonstrate that the peptide matrices delivering MSCs lead to accelerated rate of wound closure and improved quality of regenerated tissue as compared to controls.

[0487] The study will demonstrate accelerated tissue regeneration in clean diabetic wounds. In this study, it will be demonstrated that the hydrogel matrices can improve healing of non-infected wounds in db+/db+ mice. A total of 30 female 10-week old db+/db+ mice will each receive two full thickness skin wounds, following which they will be randomized into the following treatment groups: 1) PBS (control), 2) 0.5×10^6 MSCs only (control), 3) collagen scaffold+ 0.5×10^6 MSCs (comparator product control), 4) self-assembling peptide hydrogel, and 5) self-assembling peptide hydrogel+ 0.5×10^6 MSCs.

[0488] Animal surgeries will be performed according to previously established protocols. Briefly, mice will be placed under anesthesia and two 5-mm full thickness excisional wounds will be created on the dorsum of each mouse, on either side of the midline. Donut-shaped silicone splints will be placed around the wounds and affixed using liquid

adhesive (Krazy® Glue, Elmer's Products) and interrupted sutures. 100 μ L of treatment will be applied to the wounds before covering with Tegaderm™ (3M), a non-adherent wound dressing. For the MSC only control, MSCs suspended in 100 μ L of PBS will be injected intradermally at 4 sites in the periphery of the wound. For the comparator product control, MSCs will be pre-seeded onto collagen scaffolds (Integra® wound matrix or other similar porous collagen scaffold) for 30 min at 37° C. prior to placement of the scaffold in the wound bed. Wounds will be photographed at days 0, 3, 7, 14, 21, and 28 in order to measure the wound surface area and quantify the percent wound closure over time by image analysis.

[0489] When the wound dressings are removed, wounds will be scored (wound scoring, Draize scoring for skin irritation) and qualitatively assessed for hydration, crusting, exudate, and manageability. Mice will be euthanized at days 14 and 28, and the wounds and surrounding tissue will be excised using 10 mm biopsy punches. 6 wounds per condition will be processed for H&E staining, and histological sections will be evaluated for re-epithelialization, granulation tissue formation, edema, erythema, fibrotic tissue, and giant body cell response (quantified by cell/cm²). The study will establish accelerated rate of wound closure and improved quality of regenerated tissue (increased re-epithelialization and granulation tissue formation) in treated groups compared to controls.

[0490] Preliminary in vitro results have shown that the peptide hydrogels are cytocompatible, suggesting that the gels can both support the survival and function of exogenously delivered MSCs, as well as allow invasion and proliferation of endogenous cells for tissue regeneration. It is anticipated that the peptide hydrogels delivering MSCs will show improved healing of diabetic wounds compared to controls, as measured by the rate of wound healing (by image analysis) and histopathology assessments by a board-certified pathologist.

[0491] The peptide hydrogels can mediate cell attachment (FIG. 6) to support the viability and function of encapsulated MSCs, as well as serve as a scaffolding matrix for infiltration of endogenous cells during the wound healing process. FIG. 6 includes images demonstrating selective toxicity against pathogens. In particular, FIG. 6 shows live cells (green) and dead cells (red) on an assay of co-cultured MRSA and C3H10t1/2 mesenchymal stem cells on the peptide hydrogels. As shown in the right image of FIG. 6, MRSA is killed while mammalian cells remain healthy. Higher magnification shows C3H10t1/2 cells are able to adhere and spread on the hydrogel. Co-culture of MRSA with bone-marrow derived mesenchymal stem cells (BMSC) shows that hydrogel selectively kills up to 10⁶ CFU MRSA/mL. Bright-field microscopy images show lysis of BMSCs in untreated co-cultures and BMSC cell adhesion and spreading in hydrogel-treated co-cultures.

[0492] Bioactive peptide hydrogel effects on enhanced cell attachment capacity and wound healing will be examined. In addition, bioactive peptide matrices have the potential to synergize with MSCs through their biological activity to accelerate wound healing.

[0493] The peptide hydrogels may accelerate tissue regeneration in pathogen contaminated diabetic wounds. The inherent antimicrobial properties of the peptide hydrogels will be determined to protect therapeutic MSCs encapsulated within the matrix following delivery into infected

diabetic wounds, enabling improved wound healing. Wounds will be created in 20 db+/db+ mice as described above. After wound creation and splint application, 105 CFU of *S. aureus* (ATCC 25923) in 10 μ L of PBS will be placed on the wound bed and allowed to sit undisturbed for 15 minutes. Following inoculation, 100 μ L of treatment will be applied before covering with Tegaderm wound dressing.

[0494] Treatment groups will consist of: 1) PBS (control), 2) 0.5 $\times$ 10⁶ MSCs only (control), 3) collagen scaffold+0.5 $\times$ 10⁶ MSCs (comparator product control), 4) self-assembling peptide hydrogels, and 5) self-assembling peptide hydrogels+0.5 $\times$ 10⁶ MSCs. The rate of wound closure will be monitored at days 3, 7, 14, 21, and 28 by digital photographs, and wounds will be scored following identical methods as described above. Following euthanasia at day 28, wounds will be excised using 10 mm biopsy punches. 8 wounds per treatment group will be processed for H&E staining and subjected to histopathology assessments. The study will establish accelerated rate of wound closure and improved quality of regenerated tissue (increased re-epithelialization and granulation tissue formation) in the groups treated with the peptide hydrogels compared to controls.

[0495] Preliminary in vitro results have shown that the peptide hydrogels exhibit potent antimicrobial effects, while simultaneously remaining cytocompatible with mammalian cells. Therefore, it is anticipated that the peptide hydrogels delivering MSCs will improve the healing of infected wounds in diabetic mice as compared to control treatments, as measured by the rate of wound healing (by image analysis) and histopathology assessments by a board certified pathologist. The parallel treatment groups examined as described above will serve as uninfected wound healing controls to compare against the infected wounds studied here.

[0496] Bioburden in wounds may be measured using tissue biopsies (vs. swabs) between days 1-7. Healing phenotype may also be assessed. Wound bioburden in the treatment groups will be compared at days 3 and 7 by taking wound biopsies and quantifying the bacterial burden (CFUs/g of tissue). Complementary and ongoing studies will test a larger set of clinically relevant pathogens.

[0497] Feasibility of peptide hydrogels will be demonstrated by confirming in vivo viability of transplanted MSCs encapsulated in peptide hydrogels, and improved tissue regeneration in vivo in clean and contaminated diabetic wounds following treatment with MSCs encapsulated in the peptide hydrogels. Following, the efficacy of the peptide hydrogels for promoting tissue regeneration in a diabetic swine model will be determined. The study will consist of topically applied therapeutic cells encapsulated in synthetic matrices.

Example 3: Cancer Treatment by Administration of the Self-Assembling Peptide Hydrogel

[0498] The hydrogels disclosed herein exhibit inherent anti-tumor cell activity. In general, tumor cell membranes differ from healthy cell membranes by possessing proportionally more phosphatidylserine, conferring greater electronegativity to the tumor cell membrane outer leaflet. The self-assembling peptides disclosed herein may adopt a β -hairpin conformation upon contact with the higher electron negatively charged tumor cell membrane. In hairpin shape, hydrophilic peptide residues may interact with the hydrophilic portion of the membrane. While not wishing to

be bound by theory, it is believed the hydrophobic portion of the peptide may insert into the lipid portion of the cell bilayer, disrupting the structural integrity of the membrane and forming pores with lytic activity on the cells. Thus, the hydrogel may confer anti-tumor properties upon contact with tumor cells.

[0499] FIG. 20 is a schematic diagram of hydrogel assembly and dissolution with encapsulated therapeutics such as anti-cancer agents. The hydrogel may be assembled or disassembled in a controlled response to external stimuli. Exemplary external stimuli include change in temperature, pH, ionic strength, optics, magnetic force, ultrasonic frequency, laser, and antibody-bound enzymes. Therapeutics encapsulated or suspended in the hydrogel may be injected at or near a solid tumor site. Targeted, controlled hydrogel dissolution may be used to control release of therapeutics to the target site.

[0500] Cancer treatment may comprise combination administration of anti-cancer agents with the preparations disclosed herein. The anti-cancer agent may be suspended in the preparation and/or encapsulated by the hydrogel at the target site. Administration may comprise direct administration to a tumor site, for example, by injection to a solid tumor. Localized drug delivery to solid tumors by administration of the hydrogels disclosed herein may provide additional advantages over mainstream treatments. For instance, targeted drugs may be delivered in lower doses or concentrations, limiting the frequency and severity of off-target effects such as tissue toxicities or auto-immune responses.

[0501] Thus, in certain embodiments, the hydrogels disclosed herein may be employed as therapeutic delivery vehicles with flexible, tunable properties such as porosity, viscosity, stability or biodegradability, and cytotoxicity or cytocompatibility with cells or cell-derived biological materials. The hydrogel may be disassembled by stimuli, such as changes in temperature, pH, or ionic strength, providing controlled release of therapeutics after administration to a solid tumor site. Sustained release of therapeutics can provide benefits to patients at the point of care, reducing the need for additional office visits.

[0502] Exemplary anti-cancer agents which may be encapsulated in the hydrogel or otherwise administered in combination with the preparations disclosed herein include Doxorubicin (DOX), metformin (ME), and 5-fluorouracil (5-FU), Taxol, and Matrix Metalloproteinase 8 (MMP8). The anti-cancer agents may be encapsulated and/or delivered in conjunction with agents that bind cell surface receptors, such as estrogen receptor and integrin $\alpha\beta3$. Exemplary agents that bind estrogen receptor and integrin $\alpha\beta3$ which may be employed include estrogen and RGD peptide, respectively.

[0503] In one exemplary trial, thermosensitive, biodegradable hydrogels were used to deliver Doxorubicin (DOX) for cancer treatment. At body temperature (36.5-37.5° C.), hydrogels of varying concentration differentially released DOX and permitted endocytic uptake at the tumor cell level. The gradual release of DOX and local delivery of DOX to the tumor site improved treatment outcomes compared to systemic treatment modalities. However, the tested thermosensitive polymers can undergo gelation prematurely in the syringe needle, obstructing delivery of therapeutics. It is believed the hydrogels disclosed herein can be employed to deliver DOX through syringe needles with improved results.

[0504] In another exemplary trial, pH sensitive hydrogels were used for therapeutic delivery of anti-cancer agents. Due to increased metabolism in tumor cells, the pH of the tumor microenvironment (TME) is lower than non-tumor tissue. Agents such as metformin (ME) and 5-fluorouracil (5-FU) delivered with pH sensitive hydrogels resulted in greater synergistic inhibitory effect on carcinomas, resulting in a significant decrease in tumor size compared to systemic delivery. Encapsulation of the agents in hydrogel protected normal tissues from toxic drug exposure.

[0505] The hydrogels described herein, which are tunable, have an advantage over conventional static delivery systems because the in vivo environment is in a dynamic state of flux. In certain embodiments, the peptide hydrogel may be modulated to increase delivery efficiency by improving sensitivity to multiple stimuli, such as the recognition of cell surface receptors or tumor-secreted molecules. In an exemplary trial, hydrogels with molecular recognition capability showed more effective delivery of therapeutics. Estrogen receptors (ER) and integrin $\alpha\beta3$, are highly overexpressed in breast cancers and are involved in invasion, metastasis and angiogenesis. Hydrogels incorporating estrogen, which binds the estrogen receptor, and Arg-Gly-Asp (RGD) peptide, which binds $\alpha\beta3$ integrin, were able to deliver Taxol payloads specifically to breast cancer cells. Dual modification of the hydrogel promoted the necrotic and apoptotic impact of Taxol-loaded hydrogels.

Immunomodulation Using Hydrogel-Based Drug Delivery Systems in Cancer Treatment

[0506] Low tumor antigen presentation can result in low numbers of activated T cells. Cancer therapies, such as cancer vaccines and cytokine infusion therapies (TNF-alpha, IL-1, IL-2, IL-12, IFN-alpha, anti-CD40L/CD40, anti-CD137, anti-CD27, anti-OX40/OX40L, TLR agonists), can increase antigen presentation and T cell priming, and thereby increase the numbers of activated T cells peripheral to the tumor. However, activated T cells can still be blocked by a host of suppressors in the tumor microenvironment. Tumor suppression of activated T cells might include blocking proliferation, trafficking and infiltration of T cells, blocking recognition of cancer cells, and blocking cytotoxic activity of T cells. To combat the hostile tumor microenvironment, tumor agents may include any one or more of a host of regulatory molecules to increase T cell trafficking (CX3CL1, CXCL9, CXCL10, CCL5), increase tumor infiltration (anti-VEGF, LFA-1, selectins), increase T cell recognition (CAR-T cells), or T cell cytotoxic activity (IFN-gamma). Checkpoint blockade agents (anti PD-L1/PD-1, IDO inhibitors, LAG-3, VISTA, A2AR, TIM-3) can enhance the activity of cytotoxic T cells in the tumor microenvironment.

[0507] Combination therapies as described herein may employ one or more combination agents to enhance host immune system function. Exemplary combination therapies include antigen presentation combined with checkpoint inhibitors, or the use of multiple checkpoint inhibitors simultaneously. Other exemplary agents to increase cytotoxic T cell recognition of tumor cells or increase efficacy of checkpoint blockade therapies include bi-specific T-cell-engaging (BITE) antibody structures with receptors for more than one antigen. BITE antibodies allow dual recognition, for instance, of cytotoxic T cells and tumor cells, to increase

T cell engagement. Oncolytic viruses, trained to recognize tumor cells and loaded with drugs, may also be employed.

[0508] The hydrogels disclosed herein may be employed for the enhancement of endogenous immunity using Chimeric Antigen Receptor-T cells (CAR-T cells). CAR-T cells are cytotoxic T lymphocytes that express T-cell receptors trained to recognize and bind specific proteins on tumor cells and target tumor cells for destruction. Trained CAR-T cells can be treated, e.g., amplified, ex vivo and infused in high numbers to treat tumor cells. CAR-T cell therapy has produced promising clinical results in hematological malignancies such as lymphomas and leukemias, but the application of CAR-T cell therapy against solid tumors has faced challenges.

[0509] Challenges for the application of CAR-T cell therapy in solid tumor treatment include a low survival rate of infused CAR-T cells, the inability of CAR-T cells to penetrate the fibrotic tumor extracellular matrix (ECM), the de-activation of CAR-T cells in the tumor microenvironment, and safety concerns related to off-target effects. The self-assembling peptide (SAP) hydrogels disclosed herein may overcome these challenges.

[0510] The hydrogel may be used as a delivery vehicle for encapsulated T cells. The hydrogel may provide a scaffold for the concentration and expansion of CAR-T cells and the localization of those cells specifically to the tumor site, reducing losses associated with delivery. To address CAR-T cell penetration of the tumor, hydrogels may also be infused with enzymes designed to break down the fibrotic extracellular matrix (ECM) proteins surrounding the tumor. The hydrogel may be designed to break down the fibrotic ECM prior to the release of CAR-T cells from the hydrogel.

[0511] T-cell trafficking into the tumor is particularly reduced in breast, lung, and pancreatic carcinomas. Studies have shown that Matrix Metalloproteinase 8 (MMP8), an endopeptidase protease, is capable of lysing most tumor extracellular matrix proteins. MMP-8 has also been shown to prevent tumor metastasis, one of the indicators of increased mortality rates associated with solid tumors. In some embodiments, the hydrogel may be infused with MMP8 to enhance infiltration of CAR-T cells to the tumor.

[0512] The hydrogel may provide protective effects to address concerns about survival of infused CAR-T cells in the hostile tumor microenvironment. The tumor microenvironment is characterized by oxidative stress and hypoxia and contains suppressors to T cell recognition and cytotoxic activity. In some embodiments, the hydrogels may be loaded with catalase or hydrogen peroxidase to “neutralize” reactive oxygen species in the tumor microenvironment. Hydrogels may be loaded one or more regulatory factors to improve endogenous cytotoxic T cell activity. Furthermore, the one or more regulatory factors may be employed to increase CAR-T cell activity. These tunable modifications to the hydrogel may permit sustained dosing of CAR-T cells and reduce CAR-T cell exhaustion after administration.

[0513] Conventional CAR-T cell therapy also suffers from off-target toxicity. Systemic delivery of CAR-T cells may result in the development of autoimmunity to detrimental sequelae of host tissues and/or organs. Thus, CAR-T cell therapies can be life-threatening. However, localized delivery of CAR-T cells using hydrogels may minimize the risk of off-target toxicity by delivering cells directly to the tumor.

[0514] The hydrogels disclosed herein with inherent anti-tumor beta hairpin properties can be combined with existing

modalities to enhance endogenous T cell priming, expansion, survival, migration, trafficking, antigen recognition, and cytotoxic capability. The hydrogels may also confer the benefits to infused CAR-T cells. The hydrogels may be easily modulated by incorporation of regulatory molecules and can be further adapted to deliver cells or efficacious molecules in a controlled, sustained manner. These qualities are expected to reduce off-target effects, decrease required dosages, decrease the need for office and hospital visits, and improve patient outcomes.

Example 4: Hydrogel-Based Therapies for Organ Transplantation or Replacement

[0515] The hydrogels disclosed herein may be employed as biodegradable materials to serve as provisional scaffolds for transplanted cells to adhere, proliferate and differentiate, as well as promote tissue regeneration. In vitro, the peptide hydrogels disclosed herein may allow cell invasion and proliferation in 3D constructs, allowing the hydrogels to serve as scaffolding matrices for tissue regeneration. The peptide hydrogels may show biocompatibility following subcutaneous implantation. Experiments show minimal cell debris or dead cells at the gel implantation site 7 days post-subcutaneous administration. Experiments further show minimal increases in cytokine concentration in the gel and surrounding tissues compared to naïve tissues, suggesting the gel has insignificant acute inflammation effects.

[0516] The methods of administering cells disclosed herein may be employed to supplement the function of simple organs. In one exemplary embodiment, the hydrogels may provide a scaffolding environment supporting an “artificial lymph node.” The hydrogel may allow efficient interaction of professional antigen-presenting cells with T and B lymphocytes. For instance, in tumor immunotherapy, the artificial lymph node may be used to enhance tumor-specific clonal T-cell activation and anti-tumor immune responses. The hydrogel may be administered by local injection to allow peritumor/intratumor residence, which may further enhance efficacy by increasing transport into solid tumors that are poorly reached via systemic vascular access. In the case of microbial (bacterial, fungal, and/or viral) infection, the artificial lymph node may be used to enhance the immune response from infections, especially in immunocompromised patients.

[0517] In another exemplary embodiment, injectable hydrogels may be used as biomaterials for therapeutic delivery of cells and bioactive molecules for tissue regeneration. In one study, intramyocardial injection of autologous bone marrow mononuclear cells in hyaluronan hydrogel was shown to improve cardiac performance in a mini pig model of myocardial infarction. The study showed that although the intramyocardial injection of hyaluronan hydrogel or bone marrow mononuclear cells alone only slightly elevated left ventricular ejection fraction, the injection of hydrogel loaded with bone marrow mononuclear cells showed increased bone marrow mononuclear cell retention as well as significant improvement in left ventricular ejection.

[0518] Successful implementation of hydrogel-based cell delivery systems for tissue regeneration generally depends on factors such as the biocompatibility of hydrogel, the cell-hydrogel interaction, and the controlled release of cells and growth factors. Many conventional hydrogel materials are not biocompatible and may trigger inflammatory and

immune responses in the body. These materials are not suitable for use as delivery systems for tissue regeneration. Some conventional materials are biocompatible but lack other characteristics for optimal tissue regeneration. For instance, extracellular matrix (ECM)-based hydrogels typically suffer from poor mechanical properties and rapid degradation rates.

[0519] These materials are unsuitable for the controlled release of cells and their regulatory growth factors. Other conventional materials, such as Matrigel, are potentially tumorigenic, and thus, must be used with caution. Another conventional natural hydrogel, alginate, is relatively biocompatible and bio-inert, but does not have cell-binding properties which are essential for a successful outcome in tissue regeneration. Polyethylene glycol (PEG) is a conventional synthetic hydrogel which is bio-inert and biocompatible but is non-injectable, non-degradable, and also has low cell adhesion properties.

[0520] The self-assembling peptide hydrogels disclosed herein have the ability to overcome many of the challenges faced by competitor hydrogels. The hydrogels have been shown to be biocompatible and cytocompatible. In one study, the hydrogels disclosed herein were shown to be more cytocompatible with mammalian cells than conventional gel products (see FIG. 21). Control products showed antimicrobial efficacy but were significantly less cytocompatible with green fluorescent protein-expressing mesenchymal stem cells (GFP-MSCs) than the self-assembling peptide hydrogel. Image (A) of FIG. 46 shows methicillin-resistant *Staphylococcus aureus* (MRSA) cultured on BHI agar plates with no treatment (control), the self-assembling peptide hydrogel cultured with and without GFP-MSCs, and competitor products cultured with GFP-MSCs. Fluorescence microscopy images show poor GFP-MSC viability with competitor products, but good cell attachment and spreading on the self-assembling peptide hydrogel. Iodosorb iodine gel (distributed by Smith and Nephew, London, UK) showed high autofluorescence.

[0521] Cell attachment to the self-assembling peptide hydrogels was compared to polyethylene glycol (PEG) polymer. Bright-field images of the cell encapsulating materials are shown in FIG. 33. The self-assembled peptide hydrogel showed rapid cell adhesion when compared to PEG, leading to increased cell survival.

[0522] The peptide hydrogel was shown to be biocompatible and safe in vivo. Test articles (100u L per site) were administered as bilateral subcutaneous injections in the dorsum of female CD-1 mice. At indicated timepoints (day 3 and day 14), implants were excised and processed for hematoxylin and eosin (H&E) staining (FIG. 22). Stained tissue sections were evaluated and scored. The biocompatibility of competitor gels (Collagen, Medi-honey, and Silver-sept) were also studied. The peptide hydrogel showed similar early inflammation (within expected range for a degradable biomaterial) with other commercial products (Collagen and Silver-sept) which have met the Good Manufacturing Practices standards. Medi-honey showed severe inflammation and 2 out of 5 mice injected with Medi-honey died after administration.

[0523] It was shown that the peptide hydrogel does not trigger inflammation. The data is presented in FIG. 24. Briefly, seven days post-implantation of a lysine-rich hydrogel, hematoxylin and eosin (H&E)-stained tissue sections show no signs of inflammation. Tissue cytokine arrays from

hydrogel-treated and naïve mice show no significant differences in expression of cytokines commonly associated with inflammation (IL-1a/b, G-CSF, GM-CSF, MCP-1, MIP-1a, SCF, and Rantes) on Days 3, 7 and 30.

[0524] In one study, almost no endotoxins were detected in the hydrogel. The data is presented in Tables 6-7 and the graph of FIG. 29. Briefly, the test for endotoxin levels showed the peptide gel meets USP ch. 161 criteria for non-pyrogenic devices (generic endotoxin limit of 20 EU/device). The test for endotoxin levels showed the peptide hydrogel has an endotoxin level less than 0.1 EU/mL.

TABLE 6

O.D. values of the standard solutions after subtracting the blank (endotoxin-free water) value, EU/mL values were back-calculated to check the validity of the equation.						
Standard	O.D. at 410 nm				Ave. O.D.	Calculated EU/mL
	n1	n2	n3	n4		
0.1 EU/mL	0.051	0.064	0.07	0.065	0.062	0.097
0.25 EU/mL	1.186	1.221	1.206	0.911	1.131	0.254
0.5 EU/mL	2.904	3.024	2.811	2.439	2.794	0.498

TABLE 7

O.D. values of the test samples after subtracting the blank (endotoxin-free water)** value. Endotoxin levels (EU/mL) values were calculated.						
Sample	O.D. at 410 nm				Ave O.D.	Calculated EU/mL
	n1	n2	n3	n4		
Hydrogel	0.039	0.044	0.085	0.009	0.044	0.095

[0525] As previously described, stiffness of the hydrogel may be controlled, which allows control over cell fate and function. For instance, tests have shown that hydrogel stiffness can be increased with increasing peptide concentration, buffer concentration and composition, salt concentration, and as a function of time, or any combination thereof (FIGS. 30A-30C). One or more of post-formulation time (FIG. 30A), salt concentration (FIG. 30B), and peptide concentration (FIG. 30C) may be modified as shown in FIGS. 30A-30C to control hydrogel stiffness. In the graphs, a greater storage modulus (G') corresponds with a greater hydrogel stiffness.

[0526] Antimicrobial properties and stiffness of the self-assembling peptide hydrogels are not significantly affected by encapsulation of cells. As shown in the images of FIG. 34, the hydrogel encapsulated with cells retained antimicrobial properties. Antimicrobial efficacy of the peptide hydrogel against Methicillin-resistant *Staphylococcus aureus* (MRSA ATCC: 33591) showed complete clearance of 4 log CFU of MRSA. In the samples, 10-100 µl of D1 mesenchymal stem cells in DMEM mixed with 1 ml of preparation demonstrated complete elimination of 4 log CFU of MRSA. As shown in the graphs of FIG. 34, the hydrogel encapsulated with cell media retained mechanical properties. The graphs show storage modulus (G') and loss modulus (G'') of self-assembling peptide hydrogel versus strain sweep of self-assembling peptide hydrogel only (top) and self-assembling peptide hydrogel with cell media (bottom). The results

indicate that the self-assembling peptide hydrogel retains its shear-thinning kinetics even when cell media is added into the gel.

[0527] In some embodiments, the peptide sequence may be modified to include extracellular matrix motifs and bioactive domains to enhance tissue regeneration. The motifs may include the tripeptide Arg-Gly-Asp (RGD) which mediates cell attachment, an important property for any cell delivery system. The physical cross-linking of the hydrogels may be disrupted by localized stimuli, such as changes in temperature, pH, or ionic strength, providing controllable temporal release of cells and agents, for example, growth factors. The tunable nature of the hydrogels may permit control over the spatial organization of cells and allow flexibility to accommodate the needs of a variety of cell types.

[0528] Multi-layering may allow for the spatial control of different cell types encapsulated in differently tuned hydrogels (see FIG. 11). In an exemplary embodiment, hydrogels with two-faced Janus motifs or other physical properties may be encapsulated within hydrogels with different types of cells and targeted to different transplantation sites. Cells or cell-derived materials suspended within one hydrogel type may also be dispersed in pockets within a second hydrogel with other properties. Cells and biomolecules suspended in one type of hydrogel may be layered with another hydrogel loaded with other cell types and biomolecules. Photographs of exemplary multi-layered hydrogels are shown in FIG. 31. Photograph A of FIG. 31 shows a first hydrogel containing cells encapsulated by a second hydrogel. Photograph B of FIG. 31 shows alternating horizontal layers of a first hydrogel and a second hydrogel.

Example 5: Hydrogel-Based Therapies for Pancreatic Transplantation

[0529] The hydrogels disclosed herein may be used as cell delivery systems for the regeneration of many types of tissues including, but limited to, cardiac muscle tissue for myocardial infarction patients, renal tissue for chronic kidney disease patients, and skin regeneration for severely-burned or scarred victims. The hydrogels may be controlled to incorporate selected mechanical properties, cell-adhesion properties, or biochemical properties. By changing the self-assembling peptide sequence, adding bioactive motifs, or altering the formulation, hydrogels may be controlled to conform with specific needs of cells, improving tissue regeneration outcomes.

[0530] In one exemplary embodiment, the hydrogels may be administered in connection with pancreatic transplantation therapy. Subjects with diabetes, for example, Type 1 diabetes, require insulin because the immune system attacks and destroys its own pancreatic beta cells. Pancreatic islet cell transplantation has shown some promise as a therapeutic strategy for treating Type 1 diabetes. Briefly, pancreatic islets harvested from donors have been injected into immunologically compatible diabetic patients, permitting about half of the transplant recipients to stop requiring insulin treatments for one year, and in some instances 3-5 years.

[0531] For the treatment, sufficient (e.g., surplus) islets must be harvested because functional beta cells have been notoriously difficult to expand in culture and cultured human fetal cells lose the ability to release insulin over time. Harvested islets may be treated prior to administration. Generally, harvested islets are transfected with PDX-1 to

increase their expression of insulin. Even so, subjects receiving the transplanted islets require immunosuppressant drugs and the islets do not express insulin as efficiently as endogenous islets in healthy patients.

[0532] Encapsulation of the transplanted islets with biopolymers may overcome the challenges of graft rejection. In some embodiments, encapsulation of the transplanted islets may reduce or eliminate the lifetime need to remain on immunosuppressant drugs. Alginate, a biocompatible material, has been used in tests with success as an encapsulation polymer.

[0533] Alginate forms a gel under mild conditions and can be paired with a semi-permeable membrane that permits transplanted islets to regulate blood glucose levels but prevent host antibodies from destroying encapsulated cells. Addition of a second external alginate layer perfused with fibroblast growth factor (FGF) permitted the release of FGF for up to 30 days. Because islet cells consume a disproportionate amount of oxygen, FGF-triggered angiogenesis improves graft viability within the alginate bead. Even so, FGF-dependent angiogenesis is too slow to prevent significant hypoxia-related death. To overcome hypoxia-related islet cell death, transplants were deposited into the portal vein in the liver, a process that permits faster re-vascularization.

[0534] However, translation of microencapsulation techniques to the clinic are still fraught with problems due to the difficulties of scaling-up the production of alginate microbeads to produce sufficient quantities for transplantation into humans. A human requiring as many as 1 million islets would require approximately 200 hours for nanotechnology encapsulation in alginate microbeads. In addition, humans transplanted with alginate microspheres have experienced a strong foreign-body response (FBR) following pericapsular fibrotic overgrowth (PFO), as well as sedimentation of the microspheres at the site of transplant. The development of fibrotic tissue surrounding the capsule leads to hypoxia poor transplant survival. Attempts to reduce PFO formation have led to increased manufacturing time for alginate microcapsules, including layer-by-layer coating and nanofiber, thin-film nanoporous devices.

[0535] The self-assembling peptide hydrogels disclosed herein may overcome the challenges of microencapsulation by alginate microbeads and provide improvements to alginate-based technologies. The self-assembling peptide hydrogels may be designed to not invoke a foreign-body response or pericapsular fibrotic overgrowth. The self-assembling peptide hydrogels may not demand laborious processes for scaling-up of manufacturing. Furthermore, unlike alginate, the hydrogels may be designed to permit cell attachment. Cells seeded on hydrogels have shown higher viability than cells grown on 2D untreated tissue culture plates. Cells seeded on hydrogels have been shown to adhere well and proliferate on the surface. Cells encapsulated within the hydrogels can be uniformly distributed within the matrix. The shear-thinning properties of the hydrogel may permit the hydrogel to uniformly fill spaces without sedimentation. Additionally, the hydrogels are tunable for greater functionality and could be infused with FGF or any other agents to enhance graft survival, as previously described. The self-assembling peptide hydrogels may be designed to be infused with neutralizing antibodies against host antibodies, offering protection for islet grafts and reduc-

ing or eliminating the need for the additional manufacturing processes of adding a semi-permeable membrane.

[0536] In other studies, islet cells have been conventionally administered via tubular or planar diffusion chambers made from polyacrylonitrile and polyvinylchloride. The tubular chambers have good biocompatibility but tend to rupture, leading to low transplantation survival. Planar diffusion chambers in the form of sheets have better graft survival than tubular chambers but still do not provide adequate oxygen to implanted cells. In an attempt to overcome the challenge of hypoxia, islets have been conventionally transplanted with macro-devices that supply oxygen. Although pilot studies have shown increased graft functionality and morphology, Phase I clinical studies are ongoing to assess the safety of implanting oxygen macro-devices and the longevity of these devices when implanted in humans.

[0537] The self-assembling peptide hydrogels disclosed herein may overcome some of the drawbacks of tubular and planar diffusion chambers. For instance, the hydrogels may be designed to not rupture. Additionally, the hydrogels may be employed with or without an oxygen macro-device.

[0538] In other studies, pancreatic ductal cells (PDCs) were induced to form pluripotent cells that could then be differentiated into islet-like clusters using a cocktail of growth factors including epidermal growth factor (EGF), hepatocyte growth factor (HGF), and nicotinamide. The PDC-derived islets were transplanted into the kidney for partial restoration of glucose regulation in mice. The use of PDCs overcomes ethical concerns associated with the use of embryonic stem cells (ESCs) and overcomes complications of spontaneous tumor formation associated with the use of pluripotent stem cells (PSCs). Additional studies are being performed on the expansion of PDCs to achieve insulin-producing cells within several weeks. Thus, the methods disclosed herein may comprise administration of PDCs with the self-assembling peptide hydrogels.

[0539] Any of the biological materials disclosed herein may be administered in combination with one or more growth factor such as FGF, EGF, HGF, and nicotinamide.

Example 6: In Vitro Biofabrication of Cell Grafts with Peptide Hydrogels for Administration In Vivo

[0540] In the absence of a delivery vehicle, administration of the biological material in vivo can result in poor survival or diffusion from the target site. Biomaterials that encapsulate cells serve as delivery vehicles, protecting cells from destructive environmental factors, and providing a substrate for cell anchoring and attachment.

[0541] The biological material and the self-assembling peptide preparation may be packaged separately or together and delivered to the point of care. The biological material and/or preparation may be packaged in syringes, vials, or any other appropriate containers.

[0542] The method of administering the biological material may vary. The most suitable method of administration may be chosen depending on the therapy and situation. The biological material may be encapsulated in the hydrogel, seeded on the hydrogel, or delivered separately to the same target site as the hydrogel. The amount of biological material to be delivered to the target site may vary depending on the therapy. The cells to be delivered may be single cells or aggregates (including, e.g., spheroids and/or highly concen-

trated cell suspensions). The volume of self-assembling peptide preparation administered may vary depending on the therapy.

[0543] An exemplary method of encapsulation of the biological material in the self-assembling peptide hydrogel is described (see FIG. 26). Briefly, the biological material and self-assembling peptide hydrogel are packaged separately and delivered to the point of care (FIG. 26). The self-assembling peptide hydrogel may be provided in a pre-filled sterile Luer Lock syringe barrel. At the point of care just prior to use, a sterile Luer Lock connector may be fastened onto the barrel. The plunger may be depressed until the peptide solution is almost extruded from the barrel. A second sterile 1 mL Luer Lock syringe may be loaded with about 40 million cells suspended in about 200 μ L of culture medium or 1 $\times$ phosphate buffered saline (PBS). The plunger may be depressed until the cell suspension is almost extruded, releasing any air bubbles from the barrel. The two syringes may be joined by the Luer Lock connector. The cell-gel suspension may be mixed by gently depressing the plunger of one syringe barrel at a time in alternating fashion for about ten times or until the desired mixing is achieved (FIG. 26), at which point the entire volume of mixed biological material suspension may be loaded into either one of the two syringe barrels. The Luer Lock connector and the empty syringe barrel may be detached and discarded. If desired, an applicator tip such as a syringe needle may be used to administer the hydrogel comprising the biological material to the target site of the subject (FIG. 26). Other optional tips may include nozzle-shaped tips for filling deep, wide puncture wounds or fan-shaped tips for spreading the hydrogel across large surface areas, and paddles for spreading the hydrogel into a thin layer.

[0544] In some embodiments, cellular aggregates (for example, cell spheroids and high concentrated cell suspensions) may be encapsulated in the hydrogel. An exemplary method of encapsulation of cellular aggregates into the hydrogel is provided. The self-assembling peptide hydrogel may be pre-filled in a sterile Luer Lock syringe barrel. The hydrogel may be extruded out of the syringe and into the target site of the subject (see FIG. 27). The cellular aggregates may be provided in a container. The cellular aggregates may be taken out of the container and encapsulated into a hydrogel at the target site of the patient (FIG. 27).

[0545] In some embodiments, cells may be seeded on a top of the hydrogel, on a bottom of the hydrogel, or both. An exemplary method of seeding cells on the hydrogel is provided. Cells and/or cell-derived materials and the self-assembling peptide preparation may be packaged separately and delivered to the point of care (see FIG. 28). The self-assembling peptide hydrogel may be pre-filled in a sterile Luer Lock syringe barrel. The hydrogel may be extruded out of the syringe and into the target site of the subject (FIG. 28). The cell and or cell-derived material suspension may be dropped gently on top of the hydrogel on the target site of the subject (FIG. 28).

[0546] Data is shown in the graphs of FIG. 12. Briefly, image (A) is a schematic drawing of the Luer Lock syringe mixing device. The cells can be distributed substantially homogeneously by connecting the two Luer Lock syringes as previously described. In image (C), fluorescence microscopy of green fluorescence protein (GFP)-expressing mesenchymal stem cells (MSCs) shows substantially homogeneous distribution of the cells within the hydrogel. The data

presented in the graph of image (D) shows the substantially homogeneously distributed MSCs in 1.5% w/v and 0.75% w/v peptide in hydrogel have good cell viability at 24 hours post incubation, as measured in relative luminescence units (RLU) using a standard luciferase viability assay. The data presented in the graph of image (E) shows the substantially homogeneously distributed MSCs in hydrogels also have good cell viability at 72 hours post incubation.

Example 7: Properties of Modified Peptide Hydrogels

[0547] The data presented below show the effect of peptide and preparation design on hydrogel cytocompatibility, antimicrobial efficacy, and cell viability. The methods disclosed herein comprise selection of one or more aspects of the preparation to control cytocompatibility, antimicrobial efficacy, and/or cell viability of the hydrogel for a selected cell type and/or target tissue.

[0548] As shown in the data presented in FIG. 17, modifications of peptide hydrogels with functional groups RGD, VEGF, and/or Thym1/2 showed an increase in cytocompatibility of the hydrogel with mesenchymal stem cells (MSCs) seeded on top of gel over 2D cell culture (n=4) (p<0.0001). The modifications also showed an increase in viability of human retinal pigment epithelial (RPE) cells (n=3) compared to unmodified peptide hydrogels (p<0.05), as measured by relative luminescence units (RLU) using a standard luciferase viability assay.

[0549] Images of methicillin-resistant *Staphylococcus aureus* (MRSA, ATCC 33591) (n=6) (A-C) or *Pseudomonas aeruginosa* (PA01) (N=3) (E-G) cultured on BHI agar plates with SAP hydrogels of differing charge: +9 hydrogel, +7 hydrogel, or +5 hydrogel are shown in FIG. 18. As shown by the data presented in FIG. 18, the hydrogels with +9 and +7 charge had full clearance of 6-log MRSA or PA01 after 24 hours, but +5 hydrogel showed limited anti-microbial efficacy against the tested microorganisms.

[0550] An exemplary method of testing the hydrogel for antimicrobial efficacy is shown in FIG. 23. Briefly, microbe may be made to contact an exterior surface or interior surface of the hydrogel. The hydrogel may be applied on top of the bacterial inoculum spotted on the surface of an agar plate or bacterial broth culture can be swirled uniformly into a bottom layer of hydrogel and covered with a top layer of hydrogel and nutrient broth, thoroughly mixed and plated in serial dilutions. As shown in the data presented in FIG. 23, the hydrogel demonstrated complete clearance of 8×10^6 CFU methicillin-resistant *Staphylococcus aureus* (MRSA)/mL and complete clearance of 1×10^6 CFU *Pseudomonas aeruginosa* (PA01) spotted on a BHI agar plate. When the bacterial culture was swirled uniformly into the hydrogel, the hydrogel demonstrated complete clearance of 1×10^6 CFU MRSA and PA01.

[0551] As shown in the data presented in FIG. 19, murine mesenchymal stem cells (MSCs) seeded on top of peptide hydrogels for 24 hours (n=4) show strong cell viability as measured by relative luminescence units (RLU) using a standard luciferase viability assay. Hydrogels were made using BIS-TRIS propane or TRIS buffer at different buffer concentrations (33 mM, 50 mM or 100 mM). All gels showed significantly higher viability and proliferation than the control (MSCs grown in non-tissue culture-treated wells) (p<0.001).

[0552] The images of FIG. 25 show that modified hydrogels retain antimicrobial efficacy and demonstrate increased biotherapeutic effect. Antimicrobial efficacy of modified peptide gel against Methicillin-resistant *Staphylococcus aureus* (MRSA, ATCC: 33591) inoculated on BHI agar plates was tested. The hydrogels formed from peptides with a VEGF modification or a Substance-P modification showed complete elimination of 10^6 CFU MRSA. Skin section from swine were inoculated with MRSA. FIG. 25 includes an image of day 22 post full-thickness cutaneous wound bed with MRSA. After treatment of the wound with VEGF-modified hydrogel, there was a complete elimination of MRSA and evidence of tissue angiogenesis (alpha-SMA staining).

Example 8: Non-Homogeneous Cell Suspension in the Self-Assembling Peptide Hydrogel

[0553] Non-homogeneous cell suspensions in the peptide hydrogel may comprise cell clusters and/or cell spheroids. Non-homogeneous cell suspensions were found to have high cell viability for certain cell types.

[0554] Cell viability of highly concentrated cells in hydrogel was examined. The results are shown in FIG. 35. Briefly, a highly concentrated cell suspension was injected into a formed hydrogel and cell culture media was added on top. Cell viability assays were carried out after 24 hours of incubation in the hydrogel. Murine mesenchymal stem cells (MSCs) encapsulated in high cell concentrations showed good cell viability (p<0.0001) in hydrogels (n=3) compared to tissue culture plates (treated and non-treated), as measured by relative luminescence units (RLU) using a standard luciferase viability assay at 24 hours post incubation. Delivery of highly concentrated cell suspensions increased cell viability in positively charged hydrogels (+5, +7, and +9).

[0555] Cell viability of cells encapsulated in the hydrogel in the form of cell spheroids was examined. The results are shown in FIG. 36. Briefly, the cell spheroids formed after MSCs were seeded in low attachment/suspension well plates and incubated for 24 hours at 37° C. Murine mesenchymal stem cells (MSCs) encapsulated in spheroid form showed good cell viability at day 1 and day 3 (p<0.0001) in hydrogels (n=3), as compared to single cells mixed homogeneously in the hydrogel, as measured by relative luminescence units (RLU) using a standard luciferase viability assay at 24 hours post incubation.

[0556] As shown in the data presented in FIGS. 35-36, non-homogeneous cell suspensions showed good cell viability, especially in positively charged self-assembling peptide hydrogels (+5 to +9).

Example 9: Cell Viability of In Vivo Self-Assembling Peptide Hydrogel Cell Suspensions

[0557] Cell viability in vivo was examined. RFP expressing bone marrow derived mouse mesenchymal stem cells (MSC) were encapsulated in the self-assembling peptide hydrogel and implanted in subcutaneous pockets of animal models. The implants were extracted after 4 days and directly placed under microscope for imaging of fluorescence RFP signal. The images are shown in FIGS. 38A-38B and FIGS. 39A-39B.

[0558] FIG. 38A includes 20× magnified images of the extracted MSC peptide hydrogel. As shown in the images of

FIG. 38A, prominent clusters of RFP-fluorescing cells are present with intact nuclei and spreading morphology. FIG. 38B includes 20× magnified images of the extracted MSC peptide hydrogel, which utilized an RGD-modified peptide sequence. As shown in the images of FIG. 38B, prominent clusters of RFP-fluorescing cells are present with intact nuclei and spreading morphology. Accordingly, cell survival in vivo was shown for up to 4 days of implantation in both the peptide hydrogel and RGD-modified peptide hydrogel.

[0559] FIG. 39A is a 20× magnified image of the extracted MSC peptide hydrogel, which utilized a non-antimicrobial RGD-modified peptide hydrogel. FIG. 39B is a 20× magnified image of the extracted MSC peptide hydrogel, which utilized an antimicrobial peptide hydrogel. As shown in the images of FIGS. 39A-39B, cells are viable in non-antimicrobial and antimicrobial peptide hydrogels at 4 days after implantation. The RGD-modified peptide hydrogels show a higher RFP signal, demonstrating that cells attach to the peptide and proliferate.

[0560] Other extracted cell-containing hydrogel samples were digested by trypsin to remove the cells from the hydrogels. The removed cells were cultured onto tissue culture plates for 24 hours and imaged. The images are shown in FIG. 40. FIG. 40 includes RFP channel and bright field 20× magnified images of the cells recovered from an extracted non-antimicrobial RGD modified peptide hydrogel and an extracted antimicrobial peptide hydrogel. As shown in FIG. 40, the recovered cells are viable and proliferate as evidenced by the RFP signal.

[0561] Accordingly, cell viability and ability to proliferate were demonstrated in vivo in different formulations of the peptide hydrogel.

Example 10: Cell Delivery Wound Healing of Non-Contaminated Diabetic Wound

[0562] Mesenchymal stem cells (MSC) were delivered in the self-assembling peptide hydrogel to non-contaminated diabetic wound sites. Diabetic splinted wound healing mouse models were treated topically with the MSCs encapsulated in a lysine rich antimicrobial peptide hydrogel (G4D29-MSC), an arginine rich antimicrobial peptide hydrogel (6R-MSC), and PBS (PBS-MSC). Wound images were taken on day 0, day 10, and day 21 post-treatment. The images are shown in FIG. 41.

[0563] Wound closure rate (percentage of wound area by time) of the treated wounds was determined from percentage of wound closure measured at day 0, day 10, and day 21

post-treatment. The data is shown in the graph of FIG. 42. As shown in FIG. 42, the MSCs delivered in the self-assembling peptide hydrogels showed good wound closure rate by 21 days after treatment.

[0564] Accordingly, the MSCs delivered in the self-assembling peptide hydrogel show good wound closure rate.

[0565] The phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. As used herein, the term “plurality” refers to two or more items or components. The terms “comprising,” “including,” “carrying,” “having,” “containing,” and “involving,” whether in the written description or the claims and the like, are open-ended terms, i.e., to mean “including but not limited to.” Thus, the use of such terms is meant to encompass the items listed thereafter, and equivalents thereof, as well as additional items. Only the transitional phrases “consisting of” and “consisting essentially of,” are closed or semi-closed transitional phrases, respectively, with respect to the claims. Use of ordinal terms such as “first,” “second,” “third,” and the like in the claims to modify a claim element does not by itself connote any priority, precedence, or order of one claim element over another or the temporal order in which acts of a method are performed, but are used merely as labels to distinguish one claim element having a certain name from another element having a same name (but for use of the ordinal term) to distinguish the claim elements.

[0566] Having thus described several aspects of at least one embodiment, it is to be appreciated various alterations, modifications, and improvements will readily occur to those skilled in the art. Any feature described in any embodiment may be included in or substituted for any feature of any other embodiment. Such alterations, modifications, and improvements are intended to be part of this disclosure and are intended to be within the scope of the invention. Accordingly, the foregoing description and drawings are by way of example only.

[0567] Those skilled in the art should appreciate that the parameters and configurations described herein are exemplary and that actual parameters and/or configurations will depend on the specific application in which the disclosed methods and materials are used. Those skilled in the art should also recognize or be able to ascertain, using no more than routine experimentation, equivalents to the specific embodiments disclosed.

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Val Leu Thr Lys Val Lys Thr Lys Val Pro Pro Thr Lys Val Glu Val
1 5 10 15

Lys Val Leu Val
20

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Val Leu Thr Lys Val Lys Thr Lys Val Pro Pro Thr Lys Val Glu Val
1 5 10 15

Lys Val Leu Val Gly Gly Gly Gly Arg Gly Asp Val
20 25

<210> SEQ ID NO 18
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Arg Gly Asp Ser Val Lys Val Arg Val Arg Val Arg Val Pro Pro Thr
1 5 10 15

Arg Val Arg Val Arg Val Lys Val
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<223> OTHER INFORMATION: D-amino acid

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Ile Lys Val Ala Val Gly Val Lys Val Arg Val Arg Val Arg Val Pro
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Pro Thr Arg Val Arg Val Arg Val Lys Val
20 25

-continued

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Tyr Ile Gly Ser Arg Gly Val Lys Val Arg Val Arg Val Pro
1 5 10 15

Pro Thr Arg Val Arg Val Arg Val Lys Asn
20 25

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Val Leu Thr Lys Val Lys Thr Val Pro Pro Thr Lys Val Glu Val Lys
1 5 10 15

Val Leu

<210> SEQ ID NO 22
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<223> OTHER INFORMATION: D-amino acid

<400> SEQUENCE: 22

Val Glu Val Ser Val Ser Val Glu Val Pro Pro Thr Glu Val Ser Val
1 5 10 15

Glu Val Glu Val Gly Gly Gly Gly Arg Gly Asp Val
20 25

<210> SEQ ID NO 23
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<223> OTHER INFORMATION: D-amino acid

<400> SEQUENCE: 23

-continued

Val	Glu	Val	Ser	Val	Ser	Val	Glu	Val	Pro	Pro	Thr	Glu	Val	Ser	Val
1				5					10					15	
Glu	Val	Glu	Val												
				20											

1. A method of administering biological material to a subject, comprising:

combining the biological material with:

a preparation comprising a purified amphiphilic peptide in an aqueous biocompatible solution, the peptide comprising a folding group having a plurality of charged amino acid residues and hydrophobic amino acid residues arranged in a substantially alternating pattern and a turn sequence, the peptide being configured to self-assemble into a hydrogel, the hydrogel being sterile, and

a buffer configured to induce self-assembly of the hydrogel; and

administering the hydrogel comprising the biological material to a target tissue of the subject.

2. An in vitro method of preparing a culture of biological material, comprising:

combining the biological material with:

a preparation comprising a purified amphiphilic peptide in an aqueous biocompatible solution, the peptide comprising a folding group having a plurality of charged amino acid residues and hydrophobic amino acid residues arranged in a substantially alternating pattern and a turn sequence, the peptide being at least 80% purified and configured to self-assemble into a hydrogel, and

a buffer configured to induce self-assembly of the hydrogel and prepare the culture of biological material.

3. A method of grafting biological material in a subject, comprising:

administering to a target site of the subject a hydrogel formed from a preparation comprising a purified amphiphilic peptide in an aqueous biocompatible solution, the peptide comprising a folding group having a plurality of charged amino acid residues and hydrophobic amino acid residues arranged in a substantially alternating pattern and a turn sequence, the peptide being configured to self-assemble into the hydrogel, the hydrogel being sterile;

and

administering to the target site an effective amount of the biological material to graft the biological material.

4. The method of claim 1, comprising combining the biological material with the preparation to produce a biological material suspension prior to combining the biological material suspension with the buffer.

5. The method of claim 1, comprising combining the biological material with the buffer to produce a biological material buffer suspension prior to combining the biological material buffer suspension with the preparation.

6.-11. (canceled)

12. The method of claim 3, comprising administering to the target site an effective amount of the biological material

to treat a wound, biofilm, tissue injury, tissue regeneration, microbial contamination, fungal contamination, viral contamination, or a tumor.

13. The method of claim 1, wherein the biological material comprises at least one of cells, tissue material, cell-derived material, tissue-derived material, and biological fluids.

14. The method of claim 13, wherein the biological material comprises active or inactive forms of at least one of eukaryote cells, virus, prokaryote cells, adjuvants, cytokines, and growth factors, or wherein the cells comprise progenitor cells, multipotent cells, induced pluripotent cells, immune cells, specialized cells, terminally-differentiated cells, bone marrow mononuclear cells, islet cells, or combinations thereof, or wherein the tissue material comprises bone tissue, connective tissue, neural tissue, adipose tissue, cartilage, epithelial tissue, muscle tissue, bone marrow, or combinations thereof.

15.-16. (canceled)

17. The method of claim 1, wherein the biological material is autologous, allogeneic, xenogeneic, or synthetic.

18.-19. (canceled)

20. The method of claim 1, comprising collecting the biological material from the subject or from a donor subject.

21. The method of claim 1, further comprising culturing the biological material in the hydrogel for a predetermined period of time prior to administration.

22. The method of claim 1, wherein the hydrogel comprises a non-homogeneous suspension of the biological material, e.g., comprising clusters or spheroids.

23. (canceled)

24. The method of claim 1, comprising combining the preparation and the buffer at a point of care less than about 10 minutes prior to administration.

25. (canceled)

26. The method of claim 1, comprising administering the hydrogel or the biological material topically or parenterally to the subject.

27.-28. (canceled)

29. The method of claim 1, wherein the peptide comprises an effective amount of counterions to form the hydrogel, the peptide being free of chloride counterions and the counterions selected from acetate and citrate counterions.

30.-31. (canceled)

32. The method of claim 1, wherein the buffer comprises between about 10 mM and 150 mM sodium chloride and between about 10 mM and 100 mM Bis-tris propane (BTP).

33. The method of claim 1, further comprising combining the biological material with a cell culture media, cell maintenance agent, cell growth agent, cell culture serum, or combination thereof.

34. The method of claim 1, wherein administration to the target tissue or target site comprises administration to a tissue selected from mesenchymal tissue, connective tissue, muscle tissue, nervous tissue, embryonic tissue, dermal tissue, bone tissue, dental tissue, corneal tissue, cutaneous

tissue, integumental tissue, soft tissue, and hard tissue, or a biological fluid selected from tears, mucus, urine, menses, blood, wound exudates, and mixtures thereof.

35.-38. (canceled)

39. The method of claim 1, wherein:

the folding group has a sequence comprising $Y[XY]_M[T][YX]_MY$, where X is 1-3 charged amino acids, Y is 1-3 hydrophobic amino acids, T is 2-8 turn sequence amino acids, and N and M are each independently between 2 and 10;

the hydrophobic amino acid residues are independently selected from glycine, alanine, valine, leucine, isoleucine, proline, phenylalanine, methionine, threonine, tryptophan, and combinations thereof;

the charged amino acid residues are independently selected from arginine, lysine, histidine, and combinations thereof; and

the turn sequence amino acids are independently selected from a D-proline, an L-proline, aspartic acid, threonine, asparagine, and combinations thereof.

40.-44. (canceled)

45. The method of claim 1, wherein the peptide is at least 80% purified.

46.-48. (canceled)

49. The method of claim 1, wherein the peptide includes a functional group having between 3 and 30 amino acids, the functional group being engineered to express a bioactive property.

50.-62. (canceled)

63. The method of claim 1, wherein the preparation comprises between 0.5% w/v and 3.0% w/v of the peptide.

64.-66. (canceled)

67. The method of claim 1, wherein the peptide is anionic or cationic, the cationic peptide having a net charge between +2 and +11.

68.-69. (canceled)

70. The method of claim 2, wherein the preparation is sterile.

71.-77. (canceled)

78. The method of claim 1, wherein the target site is associated with a wound or biofilm, tissue injury or providing tissue regeneration, or a tumor in a subject, the method comprising administering an effective amount of the hydrogel and the biological material to the target site of the wound or biofilm, the tissue injury or tissue regeneration, or the tumor of the subject.

79.-80. (canceled)

81. A method of biofabricating a hydrogel having suspended biological material, the method comprising:

obtaining the biological material from a donor subject; combining, at a point of care, the biological material with:

a preparation comprising a purified amphiphilic peptide in an aqueous biocompatible solution, the peptide comprising a folding group having a plurality of charged amino acid residues and hydrophobic amino acid residues arranged in a substantially alternating pattern and a turn sequence, the peptide being configured to self-assemble into a hydrogel, and

a buffer configured to induce self-assembly of the hydrogel and biofabricate the hydrogel having suspended biological material.

82.-85. (canceled)

86. A method of facilitating cell therapy in a subject, comprising:

providing a preparation comprising a purified amphiphilic peptide in an aqueous biocompatible solution, the peptide comprising a folding group having a plurality of charged amino acid residues and hydrophobic amino acid residues arranged in a substantially alternating pattern and a turn sequence, the peptide being configured to self-assemble into a hydrogel;

providing instructions to combine a biological material comprising cells with the preparation and a buffer configured to induce self-assembly of the hydrogel, the biological material being autologous or allogeneic;

providing instructions to agitate the hydrogel comprising the biological material to produce a non-homogeneous cell suspension hydrogel;

providing instructions to administer an effective amount of the non-homogeneous cell suspension hydrogel to a target site of the subject to provide cell therapy to the subject.

87.-88. (canceled)

89. The method of claim 86, further comprising providing at least one of a mixing device configured to agitate the hydrogel and a delivery device configured to administer the non-homogeneous cell suspension hydrogel.

90. The method of claim 1, comprising combining the preparation and the biological material at a point of care less than about 10 minutes prior to administration.

91. The method of claim 1, comprising combining the buffer and the biological material at a point of care less than about 10 minutes prior to administration.

92. The method of claim 1, comprising combining the preparation, the buffer, and the biological material at a point of care less than about 10 minutes prior to administration.

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